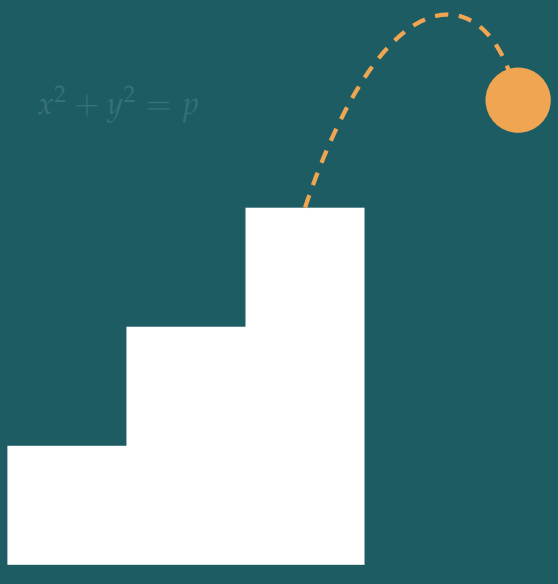
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2004 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2004 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2004 Worked Solutions

Worked solutions to all 42 questions of the 2004 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2004, Question 1

Pure • Surds and Cube Roots

STEP I 2004, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q1

The key idea

Numbers of the form $a + b\sqrt{n}$ with integer a, b behave like a little number system: multiplying two of them gives another of the same form, and two such numbers are equal exactly when their rational parts agree and their $\sqrt{n}$ parts agree. Part (i) is a warm-up in cubing such a number; part (ii) runs the process backwards by comparing parts. Part (iii) is a quadratic in x^3, and its roots turn out to be exactly the cubes found in (ii).

Solution

Part (i)

By the binomial theorem, with $(2\sqrt{5})^2 = 20$ and $(2\sqrt{5})^3 = 40\sqrt{5}$,

$$\begin{aligned}\left(3+2\sqrt{5}\right)^3 &= 3^3 + 3\cdot 3^2\cdot 2\sqrt{5} + 3\cdot 3\cdot 20 + 40\sqrt{5}\\ &= 27 + 54\sqrt{5} + 180 + 40\sqrt{5}\\ &= 207 + 94\sqrt{5}.\end{aligned}$$

So $a = 207$ and $b = 94$.

Part (ii)

The same expansion with $\sqrt{2}$ in place of $\sqrt{5}$ gives, for any c and d,

$$\begin{aligned}\left(c - d\sqrt{2}\right)^3 &= c^3 - 3c^2d\sqrt{2} + 3c\cdot 2d^2 - 2d^3\sqrt{2}\\ &= \left(c^3 + 6cd^2\right) - \left(3c^2d + 2d^3\right)\sqrt{2}.\end{aligned}$$

Since the real cube root is unique, we need $(c - d\sqrt{2})^3 = 99 - 70\sqrt{2}$.

Ask yourself Why may we compare the rational parts and the $\sqrt{2}$ parts separately?

If $p+q\sqrt{2}=r+s\sqrt{2}$ with p,q,r,s integers and $q\neq s$, then $\sqrt{2}=(p-r)/(s-q)$ would be rational, which it isn't. So $q=s$, and then $p=r$. Hence we need

$$c^3+6cd^2=99, \qquad 3c^2d+2d^3=70.$$

Since c and d are positive, the first equation forces $c^3<99$, so $c\leq 4$. Trying each value in $6cd^2=99-c^3$:

- $c=1$: $6d^2=98$, no integer solution;
- $c=2$: $12d^2=91$, no;
- $c=3$: $18d^2=72$, so $d=2$;
- $c=4$: $24d^2=35$, no.

The only candidate is $c=3$, $d=2$, and it satisfies the second equation too: $3\cdot 9\cdot 2+2\cdot 8=54+16=70$. Therefore

$$\sqrt[3]{99-70\sqrt{2}}=3-2\sqrt{2}, \qquad c=3,\ d=2.$$

(As a sanity check on signs, $2\sqrt{2}\approx 2.83<3$, so $3-2\sqrt{2}$ is positive, matching the fact that $99-70\sqrt{2}\approx 0.005$ is positive.)

Part (iii)

Put $y=x^3$. The equation becomes $y^2-198y+1=0$, so

$$y=\frac{198\pm\sqrt{198^2-4}}{2}=99\pm\sqrt{99^2-1}=99\pm\sqrt{9800}=99\pm 70\sqrt{2},$$

using $9800=4900\cdot 2$. Both values are positive and real.

From part (ii), $99-70\sqrt{2}=(3-2\sqrt{2})^3$. For the other root, notice that the product of the two roots of the quadratic is 1, so

$$99+70\sqrt{2}=\frac{1}{99-70\sqrt{2}}=\frac{1}{(3-2\sqrt{2})^3}=\left(3+2\sqrt{2}\right)^3,$$

since $(3-2\sqrt{2})(3+2\sqrt{2})=9-8=1$. (Equally, changing the sign of d in the expansion of part (ii) gives $(3+2\sqrt{2})^3=99+70\sqrt{2}$ directly.)

Each real number has exactly one real cube root, so each value of y gives exactly one real x. The two real solutions are

$$x=3-2\sqrt{2} \qquad \text{and} \qquad x=3+2\sqrt{2}.$$

The other four solutions of the sextic are complex: the remaining cube roots of each y.

Checks

- Numerically, $(3+2\sqrt{5})^3 \approx 417.19$ and $207+94\sqrt{5} \approx 417.19$.
- $3+2\sqrt{2} \approx 5.8284$, and $x^6 - 198x^3 + 1$ at this value is zero to rounding error; the same holds at $3-2\sqrt{2} \approx 0.1716$.
- The two answers to (iii) multiply to 1, as they should: if x is a root then so is $1/x$, because dividing the equation by x^6 gives the same equation in $1/x$.

Common slips

- Forgetting that $(d\sqrt{2})^3 = 2\sqrt{2}\,d^3$, not $\sqrt{2}\,d^3$, when expanding.
- Finding $c=3$, $d=2$ from the first equation but not checking the second.
- Giving six "solutions" in (iii): only one cube root of each positive number is real.

Beyond STEP

The set of numbers $a+b\sqrt{2}$ with a, b integers is a ring, written $\mathbb{Z}[\sqrt{2}]$, and $3+2\sqrt{2}$ is a *unit* in it: it has an inverse, $3-2\sqrt{2}$, in the same ring. Units like this are tied to Pell's equation $a^2 - 2b^2 = \pm 1$ (note $99^2 - 2\cdot 70^2 = 1$), and every unit of $\mathbb{Z}[\sqrt{2}]$ is $\pm(1+\sqrt{2})^n$. For more, see Davenport, *The Higher Arithmetic*, or Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP I 2004, Question 2

Pure • Floor Function Integrals

STEP I 2004, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q2

The key idea

The function $[x]$ is constant on each interval $r \leq x < r+1$, so anything built from it is a step function. Integrating a step function is just adding up areas of rectangles: split $[0, a]$ at the integers, and on each unit piece the integrand is a constant. When a is not an integer, the only new feature is a final, narrower rectangle.

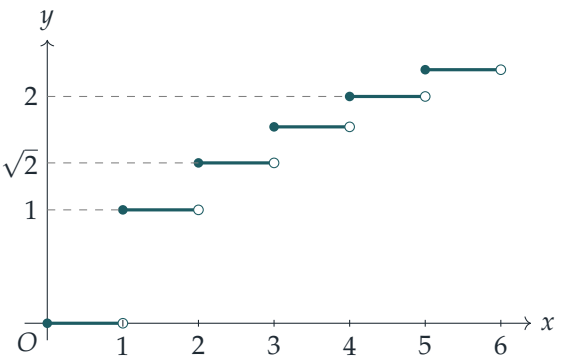
Solution

Part (i)

For $x \geq 0$, write $r = [x]$. Then r is a non-negative integer and $r \leq x < r+1$, so on that interval

$$y = \sqrt{[x]} = \sqrt{r}.$$

So the graph is a staircase: $y = 0$ on $[0,1)$, $y = 1$ on $[1,2)$, $y = \sqrt{2}$ on $[2,3)$, and so on. Each step includes its left-hand endpoint (solid dot) but not its right-hand one (open dot), since $[x]$ jumps up exactly at each integer. The steps rise by ever smaller amounts, as $\sqrt{r+1} - \sqrt{r} = 1/(\sqrt{r+1} + \sqrt{r})$ decreases. For $x < 0$ we have $[x] \leq -1$, so $\sqrt{[x]}$ is not defined there and the graph is only for $x \geq 0$.



Now let a be a positive integer. Split the range of integration at the integers $1, 2, \ldots, a-1$:

$$\int_0^a \sqrt{[x]}\,\mathrm{d}x = \sum_{r=0}^{a-1} \int_r^{r+1} \sqrt{[x]}\,\mathrm{d}x.$$

On $r \le x < r+1$ the integrand equals the constant $\sqrt{r}$. (At the single point $x = r+1$ it takes a different value, but changing a function at one point does not change its integral.) So each piece is the area of a rectangle of width 1 and height $\sqrt{r}$:

$$\int_r^{r+1} \sqrt{[x]}\,\mathrm{d}x = \sqrt{r}, \qquad \text{and hence} \qquad \int_0^a \sqrt{[x]}\,\mathrm{d}x = \sum_{r=0}^{a-1} \sqrt{r},$$

as required.

Part (ii)

Exactly the same argument works for $2^{[x]}$, which equals 2^r on $r \le x < r+1$. For a positive integer a,

$$\int_0^a 2^{[x]}\,\mathrm{d}x = \sum_{r=0}^{a-1} \int_r^{r+1} 2^r\,\mathrm{d}x = \sum_{r=0}^{a-1} 2^r = 1 + 2 + 4 + \cdots + 2^{a-1}.$$

This is a geometric series with first term 1, common ratio 2 and a terms, so its sum is

$$\frac{2^a - 1}{2 - 1} = 2^a - 1,$$

which is the required result.

Part (iii)

Now let $a > 0$ not be an integer, and put $n = [a]$, so n is an integer with $n < a < n+1$ and $n \ge 0$.

> **Ask yourself** Where does the integral stop being a sum of whole unit rectangles?

Up to $x = n$ everything is as in part (ii); the leftover piece is $n \le x \le a$, a strip of width $a - n$ lying inside a single step, where $[x] = n$. (If $n = 0$ the first piece is empty and contributes $2^0 - 1 = 0$, so the same formula works.) Therefore

$$\begin{aligned}\int_0^a 2^{[x]}\,\mathrm{d}x &= \int_0^n 2^{[x]}\,\mathrm{d}x + \int_n^a 2^n\,\mathrm{d}x \\ &= \left(2^n - 1\right) + (a - n)\,2^n \\ &= 2^{[a]}\left(1 + a - [a]\right) - 1.\end{aligned}$$

Checks

- If $0 < a < 1$, then $[a] = 0$ and the formula gives $1 \cdot (1 + a) - 1 = a$: correct, since the integrand is 1 on $[0, a]$.

- As a approaches an integer m from below, $[a] = m - 1$ and the formula tends to $2^{m-1} \cdot 2 - 1 = 2^m - 1$, matching part (ii). So the integral is a continuous function of a, as any integral of a bounded function with respect to its upper limit must be.
- Numerically, with $a = 2.5$ the formula gives $4 \times 1.5 - 1 = 5$, and a midpoint-rule evaluation of the integral also gives 5.000. For part (i) with $a = 5$, both sides are $0 + 1 + \sqrt{2} + \sqrt{3} + 2 \approx 6.1463$.

Common slips

- Summing from $r = 1$ to a instead of $r = 0$ to $a - 1$: on $[r, r + 1)$ the height is $\sqrt{r}$, the value at the *left* end.
- In part (iii), forgetting the partial strip, or giving it width 1 rather than $a - [a]$.
- Drawing the steps joined by vertical lines, or with closed dots at both ends: the floor function is not continuous at the integers.

Beyond STEP

Integrating a step function by summing rectangles is exactly how the Riemann integral is built at university: one defines the integral for step functions first, then squeezes a general function between step functions from above and below. Part (i) is also the first move in comparing sums with integrals, which gives estimates such as $\sum_{r=1}^{n} \sqrt{r} \approx \frac{2}{3} n^{3/2}$. Abbott's *Understanding Analysis* develops the Riemann integral in this way.

WORKED SOLUTION

STEP I 2004, Question 3

Pure • Factorising Two-Variable Polynomials

STEP I 2004, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q3

The key idea

A polynomial in two variables can be treated as a polynomial in *one* of them, with coefficients that involve the other. Viewed as a polynomial in x, the factor theorem still works, long division still works, and a quadratic in x still factorises when its discriminant is a perfect square. In part (ii) the leading coefficient in x is itself linear in y, which hints that one factor involves y alone.

Solution

Part (i)

Collect the expression $(*)$ by powers of x:

$$P(x) = x^3 + (2y-5)x^2 + (y^2 - 8y + 6)x + (6y - 3y^2).$$

Substituting $x = 3$,

$$\begin{aligned} P(3) &= 27 + 9(2y-5) + 3(y^2 - 8y + 6) + 6y - 3y^2 \\ &= (27 - 45 + 18) + (18y - 24y + 6y) + (3y^2 - 3y^2) = 0 \end{aligned}$$

for every y, so by the factor theorem $x - 3$ is a factor, as required.

To find the other factor, write $P(x) = (x-3)(x^2 + px + q)$, where p and q may depend on y. Expanding the right-hand side gives $x^3 + (p-3)x^2 + (q-3p)x - 3q$, and comparing coefficients:

$$\begin{aligned} x^2: &\quad p - 3 = 2y - 5, && \text{so } p = 2y - 2, \\ x^0: &\quad -3q = 6y - 3y^2, && \text{so } q = y^2 - 2y, \end{aligned}$$

and the x coefficient checks out: $q - 3p = y^2 - 8y + 6$, as it should. So the quadratic factor is

$$x^2 + (2y-2)x + y^2 - 2y = (x^2 + 2xy + y^2) - 2(x + y) = (x+y)(x+y-2).$$

Ask yourself How did we spot that the quadratic factorises?

The terms $x^2+2xy+y^2$ are a perfect square $(x+y)^2$, and the remaining terms $-2x-2y$ are a multiple of $x+y$. (If that grouping doesn't jump out, the discriminant in x is $(2y-2)^2-4(y^2-2y)=4$, a perfect square, so the roots are $x=-y+1\pm1$, that is $x=-y$ and $x=2-y$.) Therefore

$$(*)=(x-3)(x+y)(x+y-2),$$

so $a=1$, $b=0$, $c=1$, $d=-2$ (or the same with the two factors swapped: $a=1$, $b=-2$, $c=1$, $d=0$).

Part (ii)

Call the expression Q and again collect by powers of x:

$$Q=(y+2)x^2+(12-4y-5y^2)x+(6y^3-y^2-21y+10).$$

The coefficient of x^2 is $y+2$. If Q is a product of three linear factors, the x^2 term comes from the x-parts of two of them, so it's natural to ask whether $y+2$ divides the whole thing. Set $y=-2$:

$$12+8-20=0,\qquad -48-4+42+10=0,$$

so both remaining coefficients vanish at $y=-2$, and by the factor theorem (in y) each is divisible by $y+2$:

$$\begin{aligned}12-4y-5y^2&=-(5y^2+4y-12)=-(y+2)(5y-6),\\ 6y^3-y^2-21y+10&=(y+2)(6y^2-13y+5).\end{aligned}$$

Also $6y^2-13y+5=(2y-1)(3y-5)$. Hence

$$Q=(y+2)\left[x^2-(5y-6)x+(2y-1)(3y-5)\right].$$

Ask yourself Does the bracket factorise, and how can we tell?

Its discriminant in x is

$$\begin{aligned}(5y-6)^2-4(2y-1)(3y-5)&=25y^2-60y+36-(24y^2-52y+20)\\ &=y^2-8y+16=(y-4)^2,\end{aligned}$$

a perfect square, so it does. More quickly, $(2y-1)+(3y-5)=5y-6$, so the two numbers with product $(2y-1)(3y-5)$ and sum $5y-6$ are $2y-1$ and $3y-5$, giving roots $x=2y-1$ and $x=3y-5$. Therefore

$$Q=(y+2)(x-2y+1)(x-3y+5).$$

Checks

- Put $x = y = 1$ in part (i): $(*)$ gives $1-5+2+1-8-3+6+6 = 0$, and $(1-3)(2)(0) = 0$. With $x = 2$, $y = 1$: $(*)$ gives $8-20+8+2-16-3+12+6 = -3$, and $(-1)(3)(1) = -3$.
- Put $x = y = 0$ in part (ii): $Q = 10$ and $(2)(1)(5) = 10$. With $x = 1$, $y = 1$: $Q = 6-1-21+2+12-4+1-5+10 = 0$, and indeed $x - 2y + 1 = 0$.
- We checked both factorisations with exact rational arithmetic at 2000 random points (x, y); they agreed every time.

Common slips

- Treating the expression as a polynomial in x but forgetting that the "constant" term is everything without an x, here $6y - 3y^2$ in part (i).
- In part (ii), trying to find a root $x = k$ with k a number: no factor of the form $x - k$ exists. The factor with no x in it, $y + 2$, is the one to find first.
- Sign errors when dividing $-(5y^2 + 4y - 12)$ by $y + 2$; check by expanding.

Beyond STEP

Polynomials in several variables form a ring, and $\mathbb{Z}[x, y]$ can be viewed as $(\mathbb{Z}[y])[x]$: polynomials in x whose coefficients are polynomials in y. That is exactly what we did, and it is why the factor theorem and division still work. These rings have unique factorisation (Gauss's lemma), so the factorisations we found are the only ones up to order and signs. Allenby's *Rings, Fields and Groups* develops these ideas.

WORKED SOLUTION

STEP I 2004, Question 4

Pure • Integration by Substitution

STEP I 2004, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q4

The key idea

Whenever you see $\sqrt{x^2-1}$, think of the identity $\sec^2 t - 1 = \tan^2 t$: putting $x = \sec t$ turns the square root into $\tan t$, and since $\mathrm{d}x = \sec t \tan t \, \mathrm{d}t$, the $\tan t$ cancels too. Parts (ii) and (iii) look different, but completing the square inside the root turns each into the same shape, $\sqrt{u^2 - a^2}$ with $u = x + 2$, and the factor $x + 2$ outside the root is then exactly what the substitution needs.

Solution

The derivative

Writing $\sec t = (\cos t)^{-1}$ and using the chain rule,

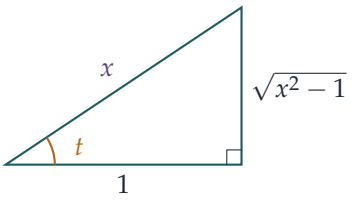
$$\frac{\mathrm{d}}{\mathrm{d}t} \sec t = -(\cos t)^{-2} \cdot (-\sin t) = \frac{\sin t}{\cos^2 t} = \sec t \tan t.$$

Part (i)

Put $x = \sec t$, so $\mathrm{d}x = \sec t \tan t \, \mathrm{d}t$. When $x = \sqrt{2}$, $\cos t = 1/\sqrt{2}$ and $t = \frac{\pi}{4}$; when $x = 2$, $\cos t = \frac{1}{2}$ and $t = \frac{\pi}{3}$. For t in $[\frac{\pi}{4}, \frac{\pi}{3}]$ we have $\tan t > 0$, so

$$\sqrt{x^2-1} = \sqrt{\sec^2 t - 1} = \sqrt{\tan^2 t} = \tan t$$

(with no modulus sign needed). The triangle below shows the same thing: hypotenuse x, adjacent side 1, so the opposite side is $\sqrt{x^2-1}$.



The integral becomes

$$\begin{aligned}\int_{\sqrt{2}}^{2} \frac{1}{x^3\sqrt{x^2-1}}\,\mathrm{d}x &= \int_{\pi/4}^{\pi/3} \frac{\sec t\tan t}{\sec^3 t\,\tan t}\,\mathrm{d}t = \int_{\pi/4}^{\pi/3} \cos^2 t\,\mathrm{d}t \\ &= \int_{\pi/4}^{\pi/3} \tfrac{1}{2}(1+\cos 2t)\,\mathrm{d}t = \left[\tfrac{1}{2}t + \tfrac{1}{4}\sin 2t\right]_{\pi/4}^{\pi/3} \\ &= \left(\frac{\pi}{6}+\frac{\sqrt{3}}{8}\right) - \left(\frac{\pi}{8}+\frac{1}{4}\right) = \frac{\sqrt{3}-2}{8} + \frac{\pi}{24},\end{aligned}$$

using $\cos^2 t = \frac{1}{2}(1+\cos 2t)$, $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$ and $\sin\frac{\pi}{2} = 1$. This is the required result.

Part (ii)

Ask yourself Part (ii) has no $\sqrt{x^2-1}$ in it. How can part (i) help?

Multiply out under the root and complete the square:

$$(x+1)(x+3) = x^2+4x+3 = (x+2)^2 - 1.$$

So with $u = x+2$ (and $\mathrm{d}u = \mathrm{d}x$) the integral is

$$\int \frac{1}{u\sqrt{u^2-1}}\,\mathrm{d}u.$$

The integrand is only defined when $u^2 > 1$. Take first $u > 1$ (that is, $x > -1$), and put $u = \sec t$ with $0 < t < \frac{\pi}{2}$, so that $\tan t > 0$ and $\sqrt{u^2-1} = \tan t$. Then

$$\int \frac{\sec t\tan t}{\sec t\,\tan t}\,\mathrm{d}t = \int 1\,\mathrm{d}t = t + C,$$

and since $\sec t = u$ we have $t = \operatorname{arcsec} u$ (equivalently $\arccos(1/u)$). Hence, for $x > -1$,

$$\int \frac{1}{(x+2)\sqrt{(x+1)(x+3)}}\,\mathrm{d}x = \operatorname{arcsec}(x+2) + C.$$

For completeness, the integrand is also defined for $x < -3$, where $u < -1$. There, put $u = -v$ with $v > 1$: the integrand $\dfrac{1}{u\sqrt{u^2-1}}\,\mathrm{d}u$ becomes $\dfrac{1}{(-v)\sqrt{v^2-1}}(-\mathrm{d}v) = \dfrac{1}{v\sqrt{v^2-1}}\,\mathrm{d}v$, which integrates to $\operatorname{arcsec} v$. Both cases are covered by

$$\operatorname{arcsec}|x+2| + C = \arccos\frac{1}{|x+2|} + C,$$

with possibly different constants on the two intervals.

Part (iii)

Again complete the square:

$$x^2+4x-5 = (x+2)^2 - 9,$$

so with $u = x+2$ we need $\displaystyle\int \frac{1}{u\sqrt{u^2-9}}\,\mathrm{d}u$. This time the natural substitution is $u = 3\sec t$, chosen so that $u^2 - 9 = 9\tan^2 t$. For $u > 3$ (that is, $x > 1$) take $0 < t < \frac{\pi}{2}$, so $\sqrt{u^2-9} = 3\tan t$ and $\mathrm{d}u = 3\sec t\tan t\,\mathrm{d}t$. Then

$$\int \frac{3\sec t\tan t}{3\sec t\cdot 3\tan t}\,\mathrm{d}t = \int \frac{1}{3}\,\mathrm{d}t = \frac{t}{3} + C,$$

and $\sec t = u/3$. Hence, for $x > 1$,

$$\int \frac{1}{(x+2)\sqrt{x^2+4x-5}}\,\mathrm{d}x = \frac{1}{3}\operatorname{arcsec}\frac{x+2}{3} + C.$$

Exactly as in part (ii), the case $x < -5$ (where $u < -3$) is handled by $u = -v$, and both intervals are covered by $\frac{1}{3}\operatorname{arcsec}\left(\frac{1}{3}|x+2|\right) + C$.

Checks

- Simpson's rule on the integral in part (i) gives $0.0974060\ldots$, matching $\frac{1}{8}(\sqrt{3}-2) + \frac{\pi}{24}$.
- Differentiate the answers: $\displaystyle\frac{\mathrm{d}}{\mathrm{d}u}\operatorname{arcsec} u = \frac{1}{u\sqrt{u^2-1}}$ for $u > 1$ (put $u = \sec t$ and use the derivative from the start). For example, $\int_0^3$ of the part (ii) integrand is numerically 0.322241, and $\operatorname{arcsec} 5 - \operatorname{arcsec} 2 = \arccos\frac{1}{5} - \frac{\pi}{3}$ agrees.
- Part (iii) should be part (ii) rescaled: with $u = 3w$, $\int \frac{\mathrm{d}u}{u\sqrt{u^2-9}} = \frac{1}{3}\int \frac{\mathrm{d}w}{w\sqrt{w^2-1}}$, which explains the factor $\frac{1}{3}$.

Common slips

- Forgetting to change the limits in part (i), or getting them the wrong way round: $\sec t = \sqrt{2}$ gives $t = \frac{\pi}{4}$, not $\frac{\pi}{3}$.
- Writing $\sqrt{\tan^2 t} = \tan t$ without checking the sign of $\tan t$ on the range used.
- In part (iii), substituting $u = \sec t$ rather than $u = 3\sec t$, which leaves $\sqrt{\sec^2 t - 9}$ with no identity to simplify it.

Beyond STEP

The inverse secant is one member of a family: substitutions $x = a\sin t$, $a\tan t$, $a\sec t$ (or their hyperbolic cousins $a\sinh t$, $a\cosh t$) tame $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$ and $\sqrt{x^2-a^2}$, and completing the square reduces any $\sqrt{\text{quadratic}}$ to one of these. The care needed over signs and over which interval an antiderivative lives on is a first taste of how analysis courses make integration rigorous. Bartle and Sherbert, *Introduction to Real Analysis*, treats the substitution theorem properly, and Siklos, *Advanced Problems in Mathematics*, has more STEP integrals in this spirit.

WORKED SOLUTION

STEP I 2004, Question 5

Pure • Residues Modulo Five

STEP I 2004, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q5

The key idea

The five progressions sort the positive integers by their remainder on division by 5. Once each class has a general term like $5n-3$, sums and squares can be tracked by expanding and pulling out a factor of 5. That gives a complete picture of where squares can land, and both impossibility proofs follow from it: the right-hand side lands in a class that the left-hand side can never reach.

Solution

The general terms

Each progression has common difference 5, so for $n = 1, 2, 3, \ldots$ the nth terms are

$$A:\ 5n-4, \qquad B:\ 5n-3, \qquad C:\ 5n-2, \qquad D:\ 5n-1, \qquad E:\ 5n.$$

(Put $n = 1$ to check that each gives the first listed term.) So a positive integer is in A, B, C, D or E exactly when it leaves remainder 1, 2, 3, 4 or 0 on division by 5.

Sum of a term in B and a term in C

Take any term $5m-3$ of B and any term $5n-2$ of C, where $m, n \geq 1$. Then

$$(5m-3)+(5n-2) = 5m+5n-5 = 5(m+n-1).$$

Since $m+n-1 \geq 1$ is a positive integer, this is the $(m+n-1)$th term of E, as required.

Squares of terms in B and C

For a term $5m-3$ of B,

$$\begin{aligned}(5m-3)^2 &= 25m^2-30m+9\\ &= 5(5m^2-6m+2)-1.\end{aligned}$$

Here $k = 5m^2 - 6m + 2$ is an integer, and $k = 5m^2 - 6m + 2 = m(5m - 6) + 2 \geq 1$ for every $m \geq 1$ (at $m = 1$ it is 1, and for $m \geq 2$ both m and $5m - 6$ are positive). So $(5m - 3)^2 = 5k - 1$ is the kth term of D. Every square of a term in B is in D.

The similar claim for C: *the square of every term in C is also a term in D.* Indeed, for a term $5m - 2$ of C,

$$\begin{aligned}(5m - 2)^2 &= 25m^2 - 20m + 4 \\ &= 5(5m^2 - 4m + 1) - 1,\end{aligned}$$

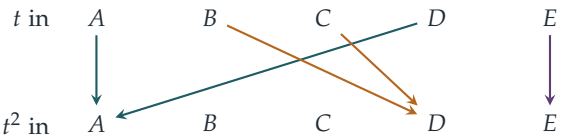
and $k = 5m^2 - 4m + 1 = m(5m - 4) + 1 \geq 2$ for $m \geq 1$. So $(5m - 2)^2$ is the kth term of D.

? **Ask yourself** Where do the squares of terms in A, D and E go?

We need this for parts (i) and (ii), so it is worth doing the other three classes in the same way:

$$\begin{aligned}(5m - 4)^2 &= 25m^2 - 40m + 16 = 5(5m^2 - 8m + 4) - 4 && \text{(in } A\text{)}, \\ (5m - 1)^2 &= 25m^2 - 10m + 1 = 5(5m^2 - 2m + 1) - 4 && \text{(in } A\text{)}, \\ (5m)^2 &= 5(5m^2) && \text{(in } E\text{)}.\end{aligned}$$

The brackets are positive integers for $m \geq 1$ (for instance $5m^2 - 8m + 4 = m(5m - 8) + 4 \geq 1$, and $5m^2 - 2m + 1 \geq 4$). So squaring sends the classes as follows:



In particular, *no perfect square lies in B or C*: every positive square lies in A, D or E.

Part (i)

Suppose that positive integers x and y satisfy $x^2 + 5y = 243\,723$. Since $243\,723 = 5 \times 48\,744 + 3$, the right-hand side is the term $5 \times 48\,745 - 2$ of C. Then

$$x^2 = 243\,723 - 5y = 5(48\,745 - y) - 2.$$

This is positive (it is x^2), so $48\,745 - y$ is a positive integer, and x^2 is a term of C. But we showed that no square lies in C. This contradiction shows that there are no such positive integers x and y.

Part (ii)

A fourth power is the square of a square. The square x^2 lies in A, D or E, and squaring a term of A or D gives a term of A, while squaring a term of E gives a term of E. So for

any positive integer x,

$$x^4 \text{ is in } A \text{ or } E, \quad \text{that is,} \quad x^4 \text{ leaves remainder 1 or 0 on division by 5.}$$

The same holds for y^4, so $2y^4$ leaves remainder 2 or 0. Adding, the remainder of $x^4 + 2y^4$ on division by 5 is the remainder of one of

$$1+2=3, \qquad 1+0=1, \qquad 0+2=2, \qquad 0+0=0,$$

so $x^4 + 2y^4$ is in C, A, B or E, and *never* in D.

Ask yourself Why can we just add the remainders?

If $x^4 = 5a + r$ and $2y^4 = 5b + s$, then $x^4 + 2y^4 = 5(a+b) + (r+s)$, so the sum differs from $r+s$ by a multiple of 5 and is in the same class as $r+s$. This is exactly the calculation we did for $B + C$ above, in general form.

Now $26\,081\,974 = 5 \times 5\,216\,395 - 1$ ends in 4, so it is a term of D. Hence it cannot equal $x^4 + 2y^4$ for any positive integers x and y, as required.

Checks

- Small cases: $2^2 = 4$, $7^2 = 49$ and $12^2 = 144$ all end in 4 or 9, so lie in D; $3^2 = 9$, $8^2 = 64$, $13^2 = 169$ likewise.
- Fourth powers: $1, 16, 81, 256, 625, 1296$ all end in 1, 6 or 5, so lie in A or E.
- A computer search over all $x \le \sqrt{243\,723} \approx 493.7$ finds no x with $243\,723 - x^2$ a positive multiple of 5, and a search over all x with $x^4 < 26\,081\,974$ (so $x \le 71$) finds no y with $2y^4 = 26\,081\,974 - x^4$.

Common slips

- Writing the general term of E as $5n+5$ and of A as $5n+1$ is fine, but then n starts at 0; mixing the two conventions gives terms that are off by one.
- In part (i), forgetting to use that $x^2 > 0$ so $48\,745 - y$ is a *positive* integer; without this, x^2 need not be a term of C as the progressions only contain positive integers.
- In part (ii), handling only x^4 and forgetting that the factor 2 moves y^4 from remainder 1 to remainder 2.

Beyond STEP

The five progressions are the residue classes modulo 5, and the whole question is arithmetic in $\mathbb{Z}/5\mathbb{Z}$: the squares there are $0, 1, 4$ (the *quadratic residues*) and the fourth powers are $0, 1$, an instance of Fermat's little theorem $a^4 \equiv 1 \pmod 5$ for $5 \nmid a$. Reducing a Diophantine equation modulo a well-chosen number is the first tool for proving it has

no solutions. Davenport's *The Higher Arithmetic* develops congruences and quadratic residues beautifully.

WORKED SOLUTION

STEP I 2004, Question 6

Pure • Vector Geometry of Triangles

STEP I 2004, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q6

The key idea

Work with position vectors rather than separate x- and y-coordinates: write $\mathbf{a} = (p_1, q_1)$, $\mathbf{b} = (p_2, q_2)$, $\mathbf{c} = (p_3, q_3)$, so that H has position vector $\mathbf{a} + \mathbf{b} + \mathbf{c}$. Every line in the question then has a short vector description, and "at right angles" becomes "the scalar product is zero". The scalar product that turns up is a difference of two squares, $(\mathbf{c} + \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b})$, and that is the whole second half.

Solution

We assume A, B, C are not collinear, so that they form a genuine triangle (otherwise the lines in the first part can coincide and there is no single point of intersection).

The medians meet

Let L, M, N be the midpoints of BC, CA, AB, with position vectors

$$\mathbf{l} = \tfrac{1}{2}(\mathbf{b} + \mathbf{c}), \qquad \mathbf{m} = \tfrac{1}{2}(\mathbf{a} + \mathbf{c}), \qquad \mathbf{n} = \tfrac{1}{2}(\mathbf{a} + \mathbf{b}).$$

A general point of the line AL is

$$\mathbf{a} + s(\mathbf{l} - \mathbf{a}) = (1 - s)\,\mathbf{a} + \tfrac{s}{2}\,\mathbf{b} + \tfrac{s}{2}\,\mathbf{c},$$

and a general point of the line BM is

$$\mathbf{b} + t(\mathbf{m} - \mathbf{b}) = \tfrac{t}{2}\,\mathbf{a} + (1 - t)\,\mathbf{b} + \tfrac{t}{2}\,\mathbf{c}.$$

These two expressions agree if the coefficients of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ match: $1 - s = \frac{t}{2}$, $\frac{s}{2} = 1 - t$ and $\frac{s}{2} = \frac{t}{2}$. The last gives $s = t$, and then $1 - s = \frac{s}{2}$ gives $s = t = \frac{2}{3}$, which satisfies all three. So both lines pass through the point G with position vector

$$\mathbf{g} = \tfrac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c}), \qquad \text{that is,} \qquad G = \left(\tfrac{p_1+p_2+p_3}{3},\ \tfrac{q_1+q_2+q_3}{3}\right).$$

Ask yourself Matching coefficients found *a* common point. How do we know it is *the* point of intersection?

Two distinct lines meet in at most one point, so we only need AL and BM to be different lines. If they were the same line, it would contain A, B and L, hence the whole line BL, which is the line BC; then A would lie on BC, contradicting our assumption. So G is the unique point of intersection.

For the third median CN, take the point $\mathbf{c} + u(\mathbf{n} - \mathbf{c})$ with $u = \frac{2}{3}$:

$$\mathbf{c} + \tfrac{2}{3}\left(\tfrac{1}{2}\mathbf{a} + \tfrac{1}{2}\mathbf{b} - \mathbf{c}\right) = \tfrac{1}{3}\mathbf{a} + \tfrac{1}{3}\mathbf{b} + \tfrac{1}{3}\mathbf{c} = \mathbf{g}.$$

So G lies on the line joining C to the midpoint of AB, as required. (The value $\frac{2}{3}$ each time says that G, the centroid, is two-thirds of the way along each median from the vertex.)

Perpendicularity at H

The point H has position vector $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$, so

$$\overrightarrow{AH} = \mathbf{h} - \mathbf{a} = \mathbf{b} + \mathbf{c}, \qquad \overrightarrow{BC} = \mathbf{c} - \mathbf{b}.$$

If the line AH meets BC at right angles then these direction vectors have zero scalar product:

$$0 = (\mathbf{c} + \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b}) = \mathbf{c} \cdot \mathbf{c} - \mathbf{b} \cdot \mathbf{b} = (p_3^2 + q_3^2) - (p_2^2 + q_2^2),$$

where the cross terms $\mathbf{c} \cdot \mathbf{b}$ and $\mathbf{b} \cdot \mathbf{c}$ cancel because the scalar product is commutative. Hence

$$p_2^2 + q_2^2 = p_3^2 + q_3^2,$$

which is the required result. In words, B and C are the same distance from the origin O.

In exactly the same way, $\overrightarrow{BH} = \mathbf{a} + \mathbf{c}$ and $\overrightarrow{AC} = \mathbf{c} - \mathbf{a}$, so if BH meets AC at right angles then $(\mathbf{c} + \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a}) = 0$, giving

$$p_1^2 + q_1^2 = p_3^2 + q_3^2.$$

The third altitude

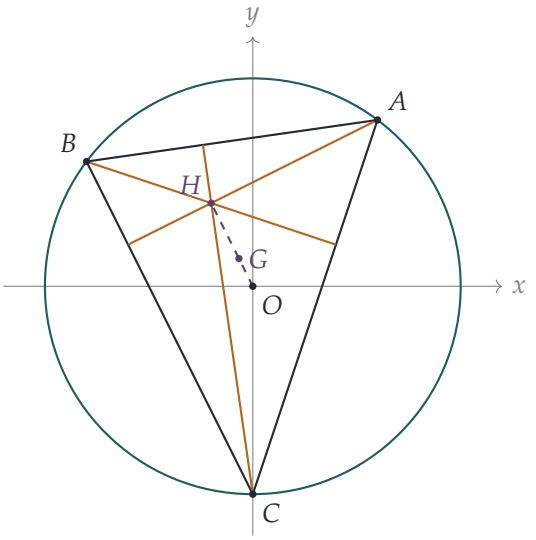
Suppose both perpendicularity conditions hold. Combining the two results above,

$$p_1^2 + q_1^2 = p_3^2 + q_3^2 = p_2^2 + q_2^2, \qquad \text{so} \qquad \mathbf{a} \cdot \mathbf{a} = \mathbf{b} \cdot \mathbf{b}.$$

Now $\overrightarrow{CH} = \mathbf{a} + \mathbf{b}$ and $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$, and

$$\overrightarrow{CH} \cdot \overrightarrow{AB} = (\mathbf{b} + \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{b} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{a} = 0.$$

So CH is perpendicular to AB (provided $H \neq C$, so that CH is a genuine line; if $H = C$ the statement is empty). This completes the deduction.



The figure shows the situation in the final part: A, B, C lie on a circle centred at the origin O, the three altitudes (amber) all pass through H, and O, G, H are collinear with $\mathbf{h} = 3\mathbf{g}$.

Checks

- Take $A = (3,4)$, $B = (-4,3)$, $C = (0,-5)$, all at distance 5 from O. Then $H = (-1,2)$, and $\overrightarrow{AH} \cdot \overrightarrow{BC} = (-4)(4) + (-2)(-8) = 0$; the other two scalar products are also 0. The centroid is $G = (-\frac{1}{3}, \frac{2}{3})$, one third of H.
- Numerically, for about 2000 random triangles with integer vertices, the intersection of two medians computed exactly with fractions was always $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$ and lay on the third median; and for 1000 random triangles inscribed in a circle centred at O, all three scalar products for H vanished.

Common slips

- Showing the two medians pass through G without saying why they cannot be the same line. One sentence about A, B, C not being collinear settles it.
- Using $\overrightarrow{OH} = \mathbf{a}+\mathbf{b}+\mathbf{c}$ as the direction of AH. The direction is $\mathbf{h}-\mathbf{a} = \mathbf{b}+\mathbf{c}$.
- Thinking the result holds for every triangle. The hypothesis forces O to be equidistant from A, B, C; for a general origin, H is not the orthocentre.

Beyond STEP

The final part proves that the three altitudes of a triangle are concurrent: place the origin at the circumcentre O (equidistant from the vertices), and the point $\mathbf{a}+\mathbf{b}+\mathbf{c}$ lies on all three. Since $\mathbf{h}=3\mathbf{g}$, it also proves that the circumcentre, centroid and orthocentre lie on a line, the *Euler line*, with $OG : GH = 1 : 2$. This vector style of geometry, where coordinate-free identities do the work, is how geometry is done in first-year linear algebra; Strang's *Introduction to Linear Algebra* develops it, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice.

WORKED SOLUTION

STEP I 2004, Question 7

Pure • Generating Functions

STEP I 2004, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q7

The key idea

A linear recurrence and a power series are two descriptions of the same object. Multiplying the series by $1 - 7x + 10x^2$ lines up each coefficient a_n with a_{n-1} and a_{n-2}, so the recurrence makes almost every coefficient of the product vanish. What survives is a polynomial, so f is a rational function. Partial fractions and geometric series then give a_n directly. Part (ii) is the same machine, once we have found p and q from the given values.

Solution

Part (i)

For $|x| < \frac{1}{5}$ the series converges, so we may multiply by powers of x and add series term by term:

$$\begin{aligned} \mathrm{f}(x) &= a_0 + a_1 x + \sum_{n=2}^{\infty} a_n x^n, \\ -7x\,\mathrm{f}(x) &= -7a_0 x - \sum_{n=2}^{\infty} 7a_{n-1} x^n, \\ 10x^2 \mathrm{f}(x) &= \sum_{n=2}^{\infty} 10a_{n-2} x^n. \end{aligned}$$

(In the second and third lines we have relabelled the index so that every sum is over x^n with $n \geq 2$.) Adding,

$$\mathrm{f}(x) - 7x\,\mathrm{f}(x) + 10x^2 \mathrm{f}(x) = a_0 + (a_1 - 7a_0)x + \sum_{n=2}^{\infty} \left(a_n - 7a_{n-1} + 10a_{n-2}\right) x^n.$$

Every bracket in the sum is zero by the recurrence, and $a_0 = 2$, $a_1 - 7a_0 = 7 - 14 = -7$. So

$$\left(1 - 7x + 10x^2\right) \mathrm{f}(x) = 2 - 7x.$$

Now $1-7x+10x^2=(1-2x)(1-5x)$, which is non-zero for $|x|<\frac{1}{5}$, so we may divide:

$$f(x)=\frac{2-7x}{(1-2x)(1-5x)}.$$

The numerator is the sum of the two factors, $(1-5x)+(1-2x)=2-7x$, so

$$f(x)=\frac{(1-5x)+(1-2x)}{(1-2x)(1-5x)}=\frac{1}{1-2x}+\frac{1}{1-5x},$$

as required.

For $|x|<\frac{1}{5}$ we have $|2x|<1$ and $|5x|<1$, so both fractions are sums of convergent geometric series:

$$f(x)=\sum_{n=0}^{\infty}(2x)^n+\sum_{n=0}^{\infty}(5x)^n=\sum_{n=0}^{\infty}\left(2^n+5^n\right)x^n.$$

Ask yourself Why can we now read off a_n?

We have two power series that agree for every x in the interval $|x|<\frac{1}{5}$. A power series is determined by its values near 0: its coefficients can be recovered as $a_n=f^{(n)}(0)/n!$. So the coefficients must match, and

$$a_n=2^n+5^n \quad \text{for all } n\geq 0.$$

(As a quick check, $a_0=1+1=2$ and $a_1=2+5=7$.)

Part (ii)

First we need p and q. Putting $n=2$ and $n=3$ into the recurrence:

$$\begin{aligned} b_2=pb_1+qb_0: &\quad 40=10p+5q, \quad \text{so } 2p+q=8;\\ b_3=pb_2+qb_1: &\quad 100=40p+10q, \quad \text{so } 4p+q=10. \end{aligned}$$

Subtracting, $2p=2$, so $p=1$ and $q=6$. The recurrence is $b_n=b_{n-1}+6b_{n-2}$.

Exactly as in part (i), multiplying by $1-x-6x^2$ kills every coefficient from x^2 onwards:

$$(1-x-6x^2)\,g(x)=b_0+(b_1-b_0)x=5+5x.$$

Since $1-x-6x^2=(1-3x)(1+2x)$, which is non-zero for $|x|<\frac{1}{3}$,

$$g(x)=\frac{5+5x}{(1-3x)(1+2x)}=\frac{A}{1-3x}+\frac{B}{1+2x},$$

where $5+5x=A(1+2x)+B(1-3x)$. Putting $x=\frac{1}{3}$ gives $\frac{20}{3}=\frac{5}{3}A$, so $A=4$; putting $x=-\frac{1}{2}$ gives $\frac{5}{2}=\frac{5}{2}B$, so $B=1$. Hence

$$g(x)=\frac{4}{1-3x}+\frac{1}{1+2x}.$$

For $|x| < \frac{1}{3}$ we have $|3x| < 1$ and $|2x| < 1$, so expanding both as geometric series,

$$g(x) = \sum_{n=0}^{\infty} \left(4 \cdot 3^n + (-2)^n\right) x^n,$$

and comparing coefficients as before,

$$b_n = 4 \cdot 3^n + (-2)^n.$$

Checks

The formulae reproduce the given values: $4 + 1 = 5$, $12 - 2 = 10$, $36 + 4 = 40$, $108 - 8 = 100$. Computing a_n and b_n from the recurrences up to $n = 39$ agrees exactly with $2^n + 5^n$ and $4 \cdot 3^n + (-2)^n$. Numerically, at $x = 0.1$ the partial sums of the series give 3.25 for f and 6.547619… for g, matching $\frac{1}{0.8} + \frac{1}{0.5} = 3.25$ and $\frac{4}{0.7} + \frac{1}{1.2} = \frac{275}{42}$.

Common slips

- Forgetting that the recurrence only holds from $n = 2$: the x^0 and x^1 terms of the product must be worked out separately, and they give the numerator.
- In part (ii), using the recurrence with p and q still unknown, or writing the numerator as $b_0 + b_1 x$ rather than $b_0 + (b_1 - pb_0)x$.
- Factorising $1 - x - 6x^2$ with the wrong signs, which leads to 3^n and 2^n instead of 3^n and $(-2)^n$.

Beyond STEP

The series f is called the *generating function* of the sequence (a_n), and the method works for any linear recurrence with constant coefficients: the denominator is always built from the recurrence, and its factors give the "characteristic roots" (here 2 and 5). Generating functions are a central tool in combinatorics and probability, where probability generating functions turn sums of random variables into products. The step "equal power series have equal coefficients" is justified properly in a first analysis course; see Abbott, *Understanding Analysis*, for power series and their uniqueness.

WORKED SOLUTION

STEP I 2004, Question 8

Pure • Monotone Sequences and Induction

STEP I 2004, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q8

The key idea

To show a sequence is increasing, look at the difference $t_{n+1} - t_n$ as a function of t_n. In both parts it factorises, and the factors are exactly the fixed points 2 and 3 of the recurrence (the values where $t_{n+1} = t_n$). The sign of the difference then depends only on where t_n lies relative to 2 and 3, so the real work is an induction showing that every term stays in the right region.

Solution

Part (i)

For any n,

$$x_{n+1} - x_n = \frac{x_n^2 + 6}{5} - x_n = \frac{x_n^2 - 5x_n + 6}{5} = \frac{(x_n - 2)(x_n - 3)}{5}.$$

If $x_n > 3$ then both factors are positive, so $x_{n+1} - x_n > 0$, that is, $x_{n+1} > x_n$.

It remains to check that *every* term exceeds 3, which we do by induction. The statement $x_n > 3$ holds for $n = 0$ by hypothesis. Suppose $x_n > 3$. By the calculation above, $x_{n+1} > x_n > 3$, so $x_{n+1} > 3$ too. Hence $x_n > 3$ for all $n \geq 0$, and therefore $x_{n+1} > x_n$ for all $n \geq 0$: the sequence is strictly increasing.

? **Ask yourself** Why isn't it enough to show $x_1 > x_0$?

The inequality $x_{n+1} > x_n$ was deduced from $x_n > 3$, so we need that condition at every step, not just the first. Here it comes for free: an increasing step from above 3 stays above 3. (Note that starting at exactly $x_0 = 3$ gives the constant sequence, so the strict inequality matters.)

Part (ii)

First note that if $2 < y_n < 3$ then $y_n > 0$, so $y_{n+1} = 5 - 6/y_n$ is defined. For such y_n,

$$y_{n+1} - y_n = 5 - \frac{6}{y_n} - y_n = -\frac{y_n^2 - 5y_n + 6}{y_n} = \frac{(y_n - 2)(3 - y_n)}{y_n},$$

which is positive, since $y_n - 2 > 0$, $3 - y_n > 0$ and $y_n > 0$. So $y_{n+1} > y_n$ *provided* $2 < y_n < 3$.

As in part (i), we must show that every term lies in this interval. This time it is not automatic, because an increasing sequence could overshoot 3. So we look at the distance from 3:

$$3 - y_{n+1} = 3 - 5 + \frac{6}{y_n} = \frac{6 - 2y_n}{y_n} = \frac{2(3 - y_n)}{y_n}.$$

? **Ask yourself** How do we know the sequence never jumps past 3?

The formula above says that $3 - y_{n+1}$ has the same sign as $3 - y_n$ whenever $y_n > 0$. So a term below 3 is always followed by a term below 3.

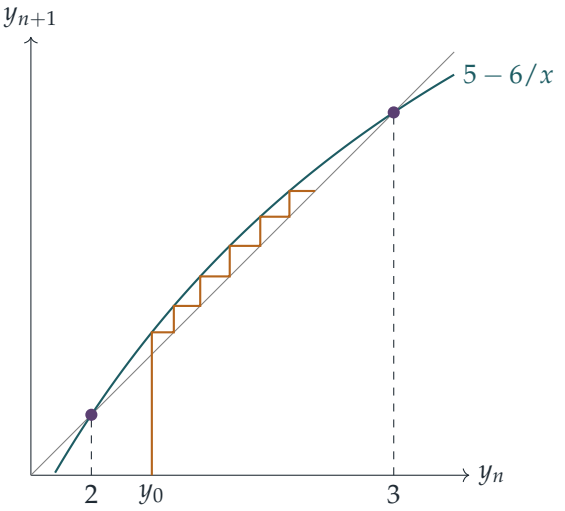
Now the induction. Claim: $2 < y_n < 3$ for all $n \geq 0$. This holds for $n = 0$ by hypothesis. Suppose $2 < y_n < 3$. Then:

- $y_{n+1} > y_n > 2$, by the difference calculation;
- $3 - y_{n+1} = 2(3 - y_n)/y_n > 0$, since $3 - y_n > 0$ and $y_n > 0$, so $y_{n+1} < 3$.

Thus $2 < y_{n+1} < 3$, completing the induction.

Since $2 < y_n < 3$ for every n, the difference calculation applies at every step, so $y_{n+1} > y_n$ for all $n \geq 0$. The sequence is strictly increasing, and yet $y_n < 3$ for all n, as required.

The cobweb diagram shows what is going on: the curve $y = 5 - 6/x$ crosses the line $y = x$ at $x = 2$ and $x = 3$, lies above the line between them, and the iterates climb towards 3 without reaching it.



Checks

- At the fixed points the recurrences give constant sequences: $(3^2 + 6)/5 = 3$ and $5 - 6/3 = 3$, $5 - 6/2 = 2$, matching the factors $(t - 2)(t - 3)$.
- Starting from $y_0 = 2.5$: $y_1 = 2.6$, $y_2 \approx 2.692$, $y_3 \approx 2.771$, increasing and below 3; after 200 steps y_n is within 10^{-15} of 3. Starting from $x_0 = 3.1$: $x_1 = 3.122$, $x_2 \approx 3.149$, increasing.
- A computer tested both inductions with exact fractions for hundreds of random starting values, and confirmed that x_0 just below 3 gives $x_1 < x_0$, so the condition $x_0 > 3$ is needed.

Common slips

- Showing $x_{n+1} > x_n$ under the assumption $x_n > 3$ and stopping there, without proving by induction that every term exceeds 3.
- In part (ii), proving $y_{n+1} < 3$ but forgetting that $y_{n+1} > 2$ is also needed to continue the induction (it follows from $y_{n+1} > y_n$).
- Multiplying an inequality through by y_n without noting that $y_n > 0$.

Beyond STEP

Part (ii) is a bounded increasing sequence, and the *monotone convergence theorem* of analysis says such a sequence must converge; any limit L satisfies $L = 5 - 6/L$, so here $y_n \to 3$. In part (i), the sequence increases without bound, since a finite limit would be a fixed point greater than 3, and there is none. Whether iterates are attracted to or repelled from a fixed point depends on the slope of the map there, which is the start

of the theory of discrete dynamical systems. Abbott's *Understanding Analysis* treats monotone convergence carefully.

WORKED SOLUTION

STEP I 2004, Question 9

Mechanics • Projectile Motion

STEP I 2004, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q9

The key idea

The first part is the standard "vertical velocity is zero at the top" calculation. In the tunnel, the speed u isn't given, but it isn't free either: once θ is chosen, requiring the path to pass through P fixes u. So we eliminate u, write the greatest height as a function of θ alone (best in terms of $t = \tan\theta$), and ask when it stays below the roof. The answer comes out as a quadratic inequality in t.

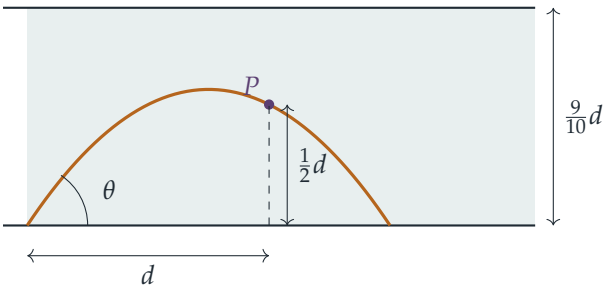
Solution

Greatest height

Model the particle as a point moving freely under gravity, with no air resistance. Take axes with origin at the point of projection, x horizontal and y vertically up. The vertical component of velocity starts at $u\sin\theta$ and has constant acceleration $-g$. At the highest point the vertical velocity is zero, so by $v^2 = v_0^2 + 2as$ in the vertical direction,

$$0 = u^2\sin^2\theta - 2gH, \qquad \text{so} \qquad H = \frac{u^2\sin^2\theta}{2g}.$$

The tunnel



The path of the particle comes from eliminating the time T between $x = uT\cos\theta$ and $y = uT\sin\theta - \frac{1}{2}gT^2$:

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = xt - \frac{gx^2(1+t^2)}{2u^2},$$

where $t = \tan\theta$ and we have used $\sec^2\theta = 1 + t^2$.

Passing through P. Putting $x = d$, $y = \frac{1}{2}d$ and dividing by d:

$$\tfrac{1}{2} = t - \frac{gd(1+t^2)}{2u^2}, \qquad \text{so} \qquad \frac{gd(1+t^2)}{2u^2} = t - \tfrac{1}{2}.$$

The left-hand side is positive, so we need $t > \frac{1}{2}$. (This makes sense: the particle starts heading in direction θ and gravity only bends the path downwards, so it can only reach P if it is aimed above the line to P, which has gradient $\frac{1}{2}$.) For each such θ there is exactly one speed that works:

$$u^2 = \frac{gd(1+t^2)}{2t-1}.$$

Ask yourself The question gives no value of u. How can the condition be on θ alone?

Because the speed is not ours to choose: the condition "passes through P" pins u down once θ is fixed. So the greatest height is really a function of θ only.

Clearing the roof. Using $\sin^2\theta = \dfrac{t^2}{1+t^2}$ in the formula for H,

$$H = \frac{u^2}{2g}\cdot\frac{t^2}{1+t^2} = \frac{d(1+t^2)}{2(2t-1)}\cdot\frac{t^2}{1+t^2} = \frac{dt^2}{2(2t-1)}.$$

The particle misses the roof exactly when $H < \frac{9}{10}d$. Since $2t - 1 > 0$ we may multiply through by $10(2t-1)/d$ without reversing the inequality:

$$\begin{aligned}\frac{dt^2}{2(2t-1)} < \frac{9d}{10} &\iff 5t^2 < 9(2t-1)\\ &\iff 5t^2 - 18t + 9 < 0\\ &\iff (5t-3)(t-3) < 0\\ &\iff \tfrac{3}{5} < t < 3.\end{aligned}$$

Every t in this interval satisfies $t > \frac{1}{2}$, so the earlier condition adds nothing. If $t = \frac{3}{5}$ or $t = 3$ the particle just touches the roof, which is excluded. Because the roof is never hit, the particle comes back down to the floor and lands inside the tunnel.

Finally, θ is between 0 and $\frac{\pi}{2}$, where tan is increasing, so

$$\arctan\tfrac{3}{5} < \theta < \arctan 3,$$

as required.

Checks

- At $t = 3$: $H = \dfrac{9d}{2 \cdot 5} = \frac{9}{10}d$, and at $t = \frac{3}{5}$: $H = \dfrac{(9/25)d}{2 \cdot (1/5)} = \frac{9}{10}d$. Both endpoints graze the roof, as they should.
- As $t \to \frac{1}{2}^{+}$ the required speed $u \to \infty$ and $H \to \infty$: the particle travels almost along the straight line to P, so it must be very fast and climbs very high afterwards.
- Numerically, with $d = 3$ and $g = 9.8$ we tested 10^5 random angles. For each we found u, confirmed that the path passes through P, and found that $H < \frac{9}{10}d$ exactly when $0.5404 < \theta < 1.2490$ (that is, $\arctan 0.6$ and $\arctan 3$ in radians). A step-by-step simulation also confirmed $H = u^2 \sin^2\theta/(2g)$.

Common slips

- Treating u as a separate unknown and getting stuck: it must be eliminated using the condition at P.
- Forgetting to check the sign of $2t - 1$ before multiplying the inequality through by it.
- Checking only that P is below the roof. The highest point of the path usually comes after P, so it is the greatest height that matters.

Beyond STEP

Writing the trajectory in terms of $\tan\theta$ turns projectile questions into quadratics in t. The same trick finds the "envelope" (bounding parabola) of all trajectories with a given speed, and gives the two possible angles for hitting a target. Problems like this, where some parameters are fixed by constraints and the rest form an allowed region, lead into optimisation and the calculus of variations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is an excellent next step for this style of mechanics.

WORKED SOLUTION

STEP I 2004, Question 10

Mechanics • Constant Acceleration Kinematics

STEP I 2004, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

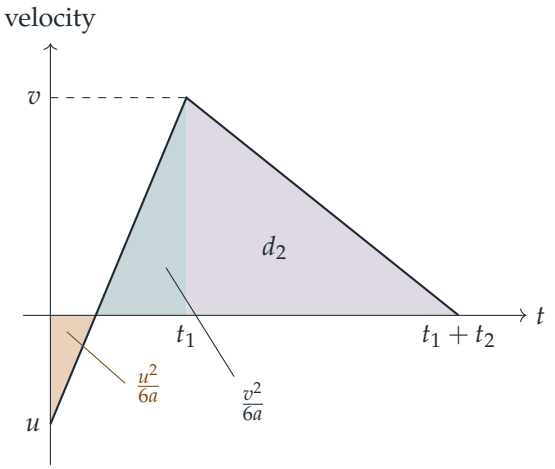
step.maths.org/questions/04-s1-q10

The key idea

Everything here is constant acceleration, so $v^2 = u^2 + 2as$ and $v = u + at$ do all the work. The one trap is the word *distance*. When $u < 0$ the particle starts by moving backwards, stops, and only then speeds up forwards, so the distance d_1 is the sum of two pieces rather than a single displacement. A velocity–time graph makes this vivid: distance is the total area between the graph and the time axis, counting areas below the axis as positive.

Solution

Take the direction of the final velocity v as positive. In the first stage the acceleration is $+3a$ (it takes the velocity from u up to $v > u$); in the second it is $-a$, until the velocity is 0. Throughout, $a > 0$.



The graph shows the case $u < 0$: the small triangle below the axis is the backward

journey.

The second stage (both cases). From speed v to rest with deceleration a, the particle never reverses, so

$$0 = v^2 - 2ad_2 \quad \Longrightarrow \quad d_2 = \frac{v^2}{2a}.$$

Part (i)

If $u > 0$ the velocity rises from u to v and stays positive, so distance equals displacement:

$$v^2 = u^2 + 2(3a)d_1 \quad \Longrightarrow \quad d_1 = \frac{v^2 - u^2}{6a}.$$

Hence

$$3d_1 = \frac{v^2 - u^2}{2a} < \frac{v^2}{2a} = d_2,$$

since $u^2 > 0$. So $3d_1 < d_2$, as required.

Part (ii)

Now $u < 0$, and the first stage splits in two.

Ask yourself Why can't we just use $v^2 = u^2 + 6ad_1$ again?

That formula gives the *displacement*, $\frac{v^2-u^2}{6a}$. But the particle moves backwards first. With acceleration $3a$ it slows from speed $|u|$ to rest, covering $\frac{u^2}{6a}$ (from $0 = u^2 - 2(3a)s$). It then speeds up from rest to v, covering $\frac{v^2}{6a}$. The distance is the sum:

$$d_1 = \frac{u^2 + v^2}{6a}, \qquad \text{so} \qquad 3d_1 = \frac{u^2 + v^2}{2a}.$$

Comparing with $d_2 = \frac{v^2}{2a}$ and $2d_2 = \frac{2v^2}{2a}$:

- $3d_1 - d_2 = \dfrac{u^2}{2a} > 0$, since $u \neq 0$;
- $2d_2 - 3d_1 = \dfrac{v^2 - u^2}{2a} > 0$, since $v > |u|$.

Hence $d_2 < 3d_1 < 2d_2$, as required.

Average speed

The times do not care about direction. In stage one the velocity changes by $v - u$ at rate $3a$, and in stage two it changes by v at rate a, so

$$t_1 = \frac{v - u}{3a}, \qquad t_2 = \frac{v}{a}, \qquad t_1 + t_2 = \frac{4v - u}{3a},$$

whichever the sign of u.

Case $u > 0$. The total distance is $\frac{v^2-u^2}{6a} + \frac{3v^2}{6a} = \frac{4v^2-u^2}{6a}$, so the average speed is

$$S_+ = \frac{4v^2-u^2}{6a} \cdot \frac{3a}{4v-u} = \frac{4v^2-u^2}{2(4v-u)}.$$

Case $u < 0$. Write $u = -w$ with $0 < w < v$. The total distance is $\frac{v^2+w^2}{6a} + \frac{3v^2}{6a} = \frac{4v^2+w^2}{6a}$ and the time is $\frac{4v+w}{3a}$, so

$$S_- = \frac{4v^2+w^2}{2(4v+w)}.$$

To compare these, measure each against $\frac{1}{2}v$:

$$S_+ - \tfrac{1}{2}v = \frac{4v^2-u^2-v(4v-u)}{2(4v-u)} = \frac{u(v-u)}{2(4v-u)},$$

$$S_- - \tfrac{1}{2}v = \frac{4v^2+w^2-v(4v+w)}{2(4v+w)} = -\frac{w(v-w)}{2(4v+w)}.$$

Since $0 < u < v$ in the first case, the first expression has a positive numerator and a positive denominator, so $S_+ > \frac{1}{2}v$. Since $0 < w < v$ in the second, the second expression is negative, so $S_- < \frac{1}{2}v$. Therefore

$$S_- < \tfrac{1}{2}v < S_+,$$

and the average speed is greater when $u > 0$, as required. Notice that this holds for any values of $|u|$ in the two cases, not just when they are equal: $\frac{1}{2}v$ separates every $u > 0$ journey from every $u < 0$ journey with the same v and a.

Checks

- Take $v = 2$, $a = 1$. With $u = 1$: $d_1 = \frac{1}{2}$, $d_2 = 2$, so $3d_1 = 1.5 < 2$, and the time is $\frac{7}{3}$, giving $S_+ = \frac{2.5}{7/3} \approx 1.071 > 1$. With $u = -1$: $d_1 = \frac{5}{6}$, so $2 < 3d_1 = 2.5 < 4$, and the time is 3, giving $S_- = \frac{17/6}{3} \approx 0.944 < 1$. A step-by-step numerical simulation of the motion reproduces these values.
- Limits: as $u \to 0$ or $|u| \to v$ both average speeds tend to $\frac{1}{2}v$, which is the average speed of the simple stop-start journey from rest. The inequalities also become equalities at the edges of the allowed range: $3d_1 \to d_2$ as $u \to 0$ in both parts, and $3d_1 \to 2d_2$ as $u \to -v$ in part (ii).

Common slips

- Using $d_1 = \frac{v^2-u^2}{6a}$ in part (ii). That is the displacement, and it would give $3d_1 < d_2$ again, contradicting what you are asked to show.
- Thinking the time also needs splitting when $u < 0$. It doesn't: $v = u + 3at_1$ is valid straight through the turning point.

Beyond STEP

The distinction between displacement and distance is the distinction between $\int v\,\mathrm{d}t$ and $\int |v|\,\mathrm{d}t$. The second is the arc length of the path, and it is what a car's odometer measures. In university mechanics, arc length becomes the natural parameter for describing a curved trajectory, leading to the tangential and normal components of acceleration. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops kinematics in this way.

WORKED SOLUTION

STEP I 2004, Question 11

Mechanics • Friction and Hinged Ladders

STEP I 2004, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q11

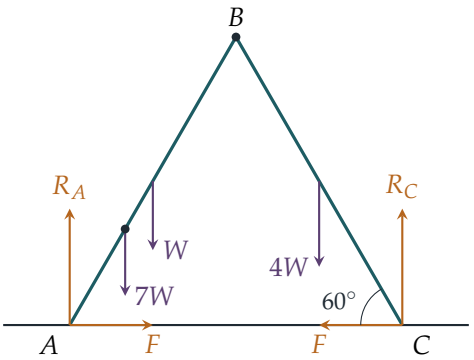
The key idea

Before anything slips, the whole frame is in equilibrium, so we can find both normal reactions from the whole system and the friction from one ladder on its own (taking moments about the hinge removes the unknown hinge force). The only horizontal external forces are the two frictional forces, so they are equal in size. Whichever foot has the *smaller* normal reaction needs the larger ratio F/R, and that foot is the one that goes first.

Solution

We model each ladder as a uniform rod of length $2a$, the hinge at B as smooth, and the decorator as a particle. Since $AB = BC$ and $\angle ABC = 60°$, triangle ABC is equilateral: $AC = 2a$ and each ladder makes $60°$ with the ground. Then B is a horizontal distance a from each foot and a height $a\sqrt{3}$ above the ground.

When the decorator is a third of the way up AB, she is a horizontal distance $\frac{1}{3}a$ from A. The centre of AB is $\frac{1}{2}a$ from A, and the centre of BC is $\frac{3}{2}a$ from A.



Normal reactions

Let the normal reactions at A and C be R_A and R_C. Resolving vertically for the whole system,

$$R_A + R_C = W + 4W + 7W = 12W.$$

Taking moments about A for the whole system (the hinge force is internal, so it does not appear):

$$R_C \cdot 2a = W \cdot \tfrac{1}{2}a + 7W \cdot \tfrac{1}{3}a + 4W \cdot \tfrac{3}{2}a = \tfrac{53}{6}Wa,$$

so

$$R_C = \tfrac{53}{12}W, \qquad R_A = 12W - \tfrac{53}{12}W = \tfrac{91}{12}W.$$

Friction

The feet tend to splay apart, so friction at each foot points inwards. Resolving horizontally for the whole system, the frictional forces are the only horizontal external forces, so they have the same magnitude; call it F.

To find F we need one ladder on its own. Take BC, which carries no decorator, and take moments about B so that the unknown hinge force drops out. About B, the reaction R_C has lever arm a, the weight $4W$ has lever arm $\frac{1}{2}a$, and the friction F (horizontal, at ground level) has lever arm $a\sqrt{3}$. The weight and F turn BC one way and R_C turns it the other, so

$$R_C \cdot a = 4W \cdot \tfrac{1}{2}a + F \cdot a\sqrt{3} \quad \Longrightarrow \quad F = \frac{R_C - 2W}{\sqrt{3}} = \frac{29W}{12\sqrt{3}}.$$

As a check, the same calculation for AB about B gives $R_A \cdot a = W \cdot \frac{1}{2}a + 7W \cdot \frac{2}{3}a + Fa\sqrt{3}$, so $F\sqrt{3} = \frac{91}{12}W - \frac{31}{6}W = \frac{29}{12}W$, which agrees.

Which ladder slips?

> **Ask yourself** The friction is the same at both feet, so how can we tell which one gives way?

A foot is on the point of slipping when the friction needed equals μ times the normal reaction there. Both feet need the same F, so the foot with the smaller normal reaction needs the larger ratio:

$$\frac{F}{R_A} = \frac{29}{91\sqrt{3}} \approx 0.184, \qquad \frac{F}{R_C} = \frac{29}{53\sqrt{3}} \approx 0.316.$$

So C reaches limiting friction first: it is the **heavier ladder** BC that slips, even though the decorator is on AB. Her weight presses A firmly into the ground, while C carries much less of the load.

We should also make sure nothing slipped earlier. If she is a fraction t of the way up,

the same working gives $R_C = (\frac{13}{4} + \frac{7}{2}t)W$ and $F\sqrt{3} = R_C - 2W = (\frac{5}{4} + \frac{7}{2}t)W$, so

$$\frac{F}{R_C} = \frac{1}{\sqrt{3}} \cdot \frac{\frac{5}{4} + \frac{7}{2}t}{\frac{13}{4} + \frac{7}{2}t} = \frac{1}{\sqrt{3}} \left(1 - \frac{2}{\frac{13}{4} + \frac{7}{2}t}\right),$$

which increases with t. Similarly F/R_A increases with t (its numerator rises and its denominator $R_A = 12W - R_C$ falls). So the required ratios grow steadily as she climbs, and the first slip happens exactly when F/R_C reaches μ, at $t = \frac{1}{3}$. Hence

$$\mu = \frac{29}{53\sqrt{3}} = \frac{29\sqrt{3}}{159} \approx 0.316.$$

Checks

- Moments for AB about B give the same F as moments for BC about B (shown above), so the reactions are consistent.
- With no decorator at all, R_C should exceed R_A because BC is heavier; indeed moments about A then give $R_C = \frac{13}{4}W$ and $R_A = \frac{7}{4}W$, adding to $5W$.
- $R_A = R_C$ when $t = \frac{11}{14}$; only beyond that point would AB be the one to slip, so at $t = \frac{1}{3}$ it must be BC.
- Numerically, recomputing every force from coordinates (and checking moments about B for both ladders) gives $\mu \approx 0.31591$.

Common slips

- Assuming the ladder carrying the decorator slips. Her weight increases the normal reaction at A, making that foot *more* secure.
- Taking moments about a foot for a single ladder, which brings in the unknown hinge force. Moments about B for one ladder avoid it.
- Forgetting that the friction has lever arm $a\sqrt{3}$ (the height of B) about the hinge.

Beyond STEP

This is the basic strategy of statics for jointed structures: treat the whole structure to find external reactions, then cut it at a joint to find what one member must supply. Questions of which contact fails first lead on to the analysis of frames and to the principle of virtual work, where equilibrium is found by considering small displacements consistent with the constraints. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

WORKED SOLUTION

STEP I 2004, Question 12

Probability and Statistics • Bayes' Theorem

STEP I 2004, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q12

The key idea

The sample from group 1 tells us, approximately, the conditional probability that a chip came from machine A *given* that it is in group 1. So we write that conditional probability in terms of λ, p and q using Bayes' theorem, set it equal to the observed proportion, and solve for λ. The faulty sorting changes only the probability that a chip from each machine lands in group 1; the method is identical.

Solution

Pick a chip at random from the factory's output. Let A be the event that it was made by machine A (so $\mathrm{P}(A) = \lambda$ and $\mathrm{P}(A') = 1 - \lambda$), let P be the event that it is perfect, and let G be the event that it is placed in group 1. We are told $\mathrm{P}(P \mid A) = p$ and $\mathrm{P}(P \mid A') = q$.

A chip drawn at random from group 1 is a chip drawn at random from the output, conditioned on G. Since the sample is large, the observed proportion $\frac{2}{5}$ is a good estimate of $\mathrm{P}(A \mid G)$, and our estimate of λ is the value that makes $\mathrm{P}(A \mid G)$ exactly $\frac{2}{5}$.

Perfect sorting

Here G is the same event as P. By Bayes' theorem,

$$\mathrm{P}(A \mid P) = \frac{\mathrm{P}(P \mid A)\,\mathrm{P}(A)}{\mathrm{P}(P \mid A)\,\mathrm{P}(A) + \mathrm{P}(P \mid A')\,\mathrm{P}(A')} = \frac{\lambda p}{\lambda p + (1-\lambda)q}.$$

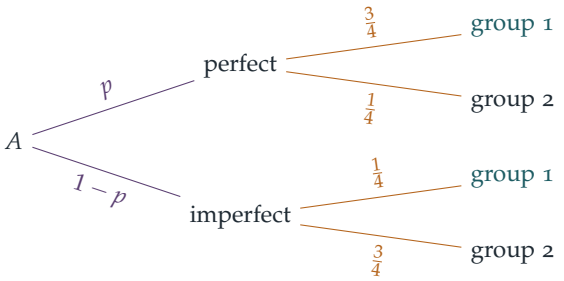
Setting this equal to $\frac{2}{5}$ and cross-multiplying (the denominator is positive):

$$\begin{aligned} 5\lambda p &= 2\lambda p + 2(1-\lambda)q \\ 3\lambda p + 2\lambda q &= 2q \\ \lambda &= \frac{2q}{3p+2q}, \end{aligned}$$

which is the required estimate, since $3p + 2q > 0$.

Faulty sorting

Now a chip reaches group 1 either by being perfect and correctly sorted, or by being imperfect and wrongly sorted. We assume the sorting errors happen independently of which machine made the chip, so that, given whether a chip is perfect, its sorting does not depend on its machine. The tree below shows the branches for a chip from machine A; machine B is the same with q in place of p.



Adding the two group-1 branches,

$$\mathrm{P}(G \mid A) = \tfrac{3}{4}p + \tfrac{1}{4}(1-p) = \frac{1+2p}{4}, \qquad \mathrm{P}(G \mid A') = \tfrac{3}{4}q + \tfrac{1}{4}(1-q) = \frac{1+2q}{4}.$$

So, by Bayes' theorem again (the factors of $\frac{1}{4}$ cancel),

$$\mathrm{P}(A \mid G) = \frac{\lambda(1+2p)}{\lambda(1+2p) + (1-\lambda)(1+2q)}.$$

Ask yourself Has the evidence changed? Should the $\frac{2}{5}$ still be used?

Yes: the sample was genuinely drawn from group 1, and $\frac{2}{5}$ of it genuinely came from machine A. What has changed is only our model of how chips got into group 1. So we set the new expression for $\mathrm{P}(A \mid G)$ equal to $\frac{2}{5}$:

$$\begin{aligned} 5\lambda(1+2p) &= 2\lambda(1+2p) + 2(1-\lambda)(1+2q) \\ \lambda\big[3(1+2p) + 2(1+2q)\big] &= 2(1+2q) \\ \lambda &= \frac{2(1+2q)}{5+6p+4q}. \end{aligned}$$

This is the new estimate of λ. Notice it is the first answer with p and q replaced by $\frac{1+2p}{4}$ and $\frac{1+2q}{4}$, the new probabilities of reaching group 1.

Checks

- If $p = q$ the machines are indistinguishable by quality, so group 1 should mirror production and λ should be $\frac{2}{5}$. Both formulae give this: $\frac{2p}{5p} = \frac{2}{5}$ and $\frac{2(1+2p)}{5+10p} = \frac{2}{5}$.

- Both estimates lie strictly between 0 and 1, since in each the numerator is positive and smaller than the denominator.
- A simulation of 4×10^5 chips with $\lambda = 0.3$, $p = 0.6$, $q = 0.8$ and faulty sorting gave 0.2667 of group 1 from machine A, against the formula's 0.2661; inverting the formula on the simulated proportion recovered $\lambda \approx 0.301$.

Common slips

- Equating $\frac{2}{5}$ to $\mathrm{P}(G \mid A)$ or to λp rather than to the conditional probability $\mathrm{P}(A \mid G)$.
- In the second part, forgetting the imperfect chips that are wrongly put into group 1, i.e. using $\frac{3}{4}p$ alone.

Beyond STEP

Choosing the parameter value that makes the observed data "fit exactly" is the *method of moments*; here it coincides with maximum likelihood estimation, which you will meet in a first statistics course. The faulty sorter is a simple model of measurement error (misclassification), a serious practical issue in medical testing, where false positives and false negatives distort the observed proportions in just this way. For more on Bayes' theorem and conditioning, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 2004, Question 13

Probability and Statistics • Maximum of Uniforms

STEP I 2004, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q13

The key idea

The maximum of several independent numbers is at most x exactly when *every* one of them is at most x. That turns the awkward "maximum" into a product of identical probabilities, giving the distribution function of X in one line. Everything else follows from it: differentiate for the density and the expectation, and set the distribution function equal to 0.05 to find the critical value of the test.

Solution

Part (i)

Let U_1, U_2, U_3 be the three numbers, independent and each uniform on $[0,1]$, so $\mathrm{P}(U_i \le x) = x$ for $0 \le x \le 1$. Since $X = \max(U_1, U_2, U_3)$,

$$X \le x \iff U_1 \le x,\ U_2 \le x \text{ and } U_3 \le x.$$

By independence,

$$F(x) = \mathrm{P}(X \le x) = x^3 \qquad (0 \le x \le 1).$$

In particular $\mathrm{P}(X \le 0.8) = 0.8^3 = 0.512$, as required.

Differentiating, X has density $f(x) = F'(x) = 3x^2$ on $[0,1]$, so

$$\mathrm{E}(X) = \int_0^1 x \cdot 3x^2 \,\mathrm{d}x = \left[\tfrac{3}{4}x^4\right]_0^1 = \frac{3}{4}.$$

This is sensible: the largest of three numbers should sit well above the mean $\frac{1}{2}$ of a single one.

Part (ii)

Now there are N independent numbers, each uniform on $[0, a]$, so $\mathrm{P}(U_i \le x) = x/a$ for $0 \le x \le a$, and the same argument gives

$$\mathrm{P}(X \le x) = \left(\frac{x}{a}\right)^N \qquad (0 \le x \le a),$$

with $\mathrm{P}(X \le x) = 1$ for $x \ge a$. Since X is continuous, $\mathrm{P}(X < x)$ is the same.

The critical value. The test rejects H_0 when $x < c$, and a 5% significance level means the probability of rejecting when H_0 is true ($a = 1$) is 0.05:

$$\mathrm{P}(X < c \mid a = 1) = c^N = 0.05, \qquad \text{so} \qquad c = 0.05^{1/N}.$$

Small values of x are evidence for $a < 1$, which is why the rejection region is $x < c$.

> **Ask yourself** How does "$x \le 0.8$ always leads to rejection" become an inequality in N?

Every $x \le 0.8$ is rejected exactly when every such x lies below c, that is, when $0.8 < c$. Since $t \mapsto t^N$ is increasing for $t > 0$, this is equivalent to

$$0.8^N < c^N = 0.05.$$

So we need the smallest integer N with $0.8^N < 0.05$. (Note that c grows towards 1 as N increases, so larger N makes the test more willing to reject.)

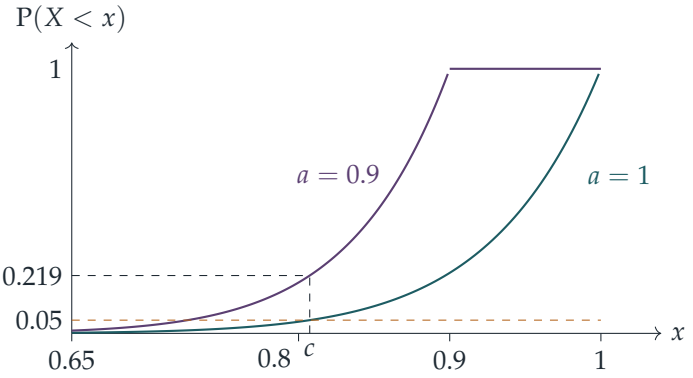
Using $2^{10} \approx 10^3$. Write $0.8 = 2^3/10$, so $0.8^N = 2^{3N}/10^N$.

- $N = 13$: $0.8^{13} = \dfrac{2^{39}}{10^{13}} = \dfrac{2^9 \cdot 2^{30}}{10^{13}} \approx \dfrac{512 \times 10^9}{10^{13}} = 0.0512 > 0.05.$
- $N = 14$: $0.8^{14} = \dfrac{2^{42}}{10^{14}} = \dfrac{2^2 \cdot 2^{40}}{10^{14}} \approx \dfrac{4 \times 10^{12}}{10^{14}} = 0.04 < 0.05.$

Since 0.8^N decreases as N increases, every $N \le 13$ also fails. So the smallest value is

$$N = 14.$$

(The approximation slightly underestimates powers of 2, since $2^{10} = 1024$; the true values $0.8^{13} \approx 0.0550$ and $0.8^{14} \approx 0.0440$ lead to the same conclusion.)



With $N = 14$, the critical value is $c = 0.05^{1/14} \approx 0.807$, just above 0.8.

If $a = 0.8$. All the numbers lie in $[0, 0.8]$, so certainly $X \leq 0.8 < c$. The null hypothesis is rejected with probability 1.

If $a = 0.9$. Now $c \approx 0.807 < 0.9$, so

$$\mathrm{P}(\text{reject}) = \mathrm{P}(X < c \mid a = 0.9) = \left(\frac{c}{0.9}\right)^{14} = \frac{c^{14}}{0.9^{14}} = \frac{0.05}{0.9^{14}}.$$

Here $0.9^2 = 0.81$, $0.9^4 = 0.6561$, $0.9^8 \approx 0.4305$, and so $0.9^{14} = 0.9^8 \cdot 0.9^4 \cdot 0.9^2 \approx 0.2288$. Therefore

$$\mathrm{P}(\text{reject} \mid a = 0.9) = \frac{0.05}{0.9^{14}} \approx 0.219.$$

So even with 14 observations, the test usually fails to detect that a is 0.9 rather than 1.

Checks

- A Monte Carlo simulation (2×10^5 trials) gave $\mathrm{P}(X \leq 0.8) \approx 0.512$ and $\mathrm{E}(X) \approx 0.750$ in part (i).
- In general, the maximum of n uniform numbers on $[0, 1]$ has mean $\frac{n}{n+1}$; for $n = 1$ this is $\frac{1}{2}$, as it should be, and for $n = 3$ it is $\frac{3}{4}$.
- With $N = 14$ and $c = 0.05^{1/14}$, simulation gave a rejection rate of about 0.0505 when $a = 1$ (the size of the test) and about 0.218 when $a = 0.9$, matching $0.05/0.9^{14} = 0.2186$.

Common slips

- Using $1 - 0.8^N$ in place of 0.8^N: the probability that the maximum is *small* is the product 0.8^N, not its complement.

- Taking the rejection probability at $a = 0.9$ to be $(0.8/0.9)^{14}$. The test rejects when $x < c$, and c is not 0.8.
- Stopping at $N = 13$ because 0.0512 "is about 0.05". We need strictly less than 0.05.

Beyond STEP

The maximum is an *order statistic*, and for the uniform distribution on $[0, a]$ it is a *sufficient statistic* for a: it contains all the information the sample holds about a. The rejection probability as a function of the true a is the *power function* of the test. Here it is $\min\{1,\ 0.05/a^N\}$ for $a \leq 1$, and comparing power functions is how statisticians choose between tests. Grimmett and Welsh, *Probability: An Introduction*, covers the distributions of maxima and minima; Blitzstein and Hwang, *Introduction to Probability*, treats order statistics in detail.

WORKED SOLUTION

STEP I 2004, Question 14

Probability and Statistics • Counting Arrangements

STEP I 2004, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q14

The key idea

Picture the whole chest being emptied one coin at a time into a single row. The pirates simply cut this row at the two lead coins: the first pirate gets the gold before the first lead coin, the second the gold between the lead coins, and the third the gold after the second lead coin. Every ordering of the coins is equally likely, so everything comes down to counting where the two lead coins can sit.

Solution

Imagine all $n+2$ coins drawn out in a random order, even though the third pirate actually just scoops up what is left. That doesn't change who gets which coins. The draws are random, so each of the $\binom{n+2}{2}$ choices of the two positions (out of $n+2$) for the lead coins is equally likely.

Let A, B and C be the numbers of gold coins taken by the first, second and third pirates. Then $A+B+C=n$, and the row of coins looks like this:



Here gold coins are amber and lead coins are plum. The lead positions determine (A,B,C), and every triple of non-negative integers with $A+B+C=n$ comes from exactly one choice of positions. So the $\binom{n+2}{2}=\frac{1}{2}(n+1)(n+2)$ triples (A,B,C) are all equally likely.

Part (i)

The first pirate gets no gold exactly when $A=0$, that is, when the very first coin drawn is lead. That has probability $\frac{2}{n+2}$, since each of the $n+2$ coins is equally likely to come

out first. Hence

$$\mathrm{P}(A \geq 1) = 1 - \frac{2}{n+2} = \frac{n}{n+2}.$$

(Counting gives the same answer: $A = 0$ leaves $B + C = n$, which has $n+1$ solutions, and $(n+1)/\frac{1}{2}(n+1)(n+2) = \frac{2}{n+2}$.)

Part (ii)

Ask yourself The second pirate starts drawing at a random time. Is she really no worse off than the first?

Yes, she is no worse off. We count the triples with $B = 0$: then $A + C = n$, which has $n+1$ solutions, exactly as in part (i). So

$$\mathrm{P}(B \geq 1) = 1 - \frac{n+1}{\frac{1}{2}(n+1)(n+2)} = \frac{n}{n+2}.$$

In fact, since all triples (A, B, C) are equally likely, the three shares are *exchangeable*: every pirate has the same chance $\frac{n}{n+2}$ of getting some gold, even the third.

Part (iii)

We need A, B and C all to be at least 1. Put $A = A' + 1$, $B = B' + 1$ and $C = C' + 1$. The triples we want then correspond to non-negative solutions of $A' + B' + C' = n - 3$. By stars and bars there are $\binom{n-1}{2}$ of these, which we can also see directly: we choose two of the $n-1$ gaps between adjacent gold coins for the lead coins. Hence, for $n \geq 1$,

$$\mathrm{P}(A \geq 1,\ B \geq 1,\ C \geq 1) = \frac{\binom{n-1}{2}}{\binom{n+2}{2}} = \frac{(n-1)(n-2)}{(n+1)(n+2)}.$$

This is 0 for $n = 1$ and $n = 2$, as it must be, since three pirates can't all have gold if there are fewer than three gold coins. (For $n = 0$ the probability is trivially 0; the formula doesn't apply there, because $\binom{n-1}{2}$ counts gaps that don't exist.)

Checks

- For $n = 1$ the arrangements of the coins are GLL, LGL and LLG, all equally likely. The first pirate gets the gold only in GLL, which has probability $\frac{1}{3} = \frac{1}{1+2}$, and the second only in LGL, also $\frac{1}{3}$.
- For $n = 3$ the only good arrangement is GLGLG, one of the $\binom{5}{2} = 10$ arrangements, and the formula gives $\frac{2 \cdot 1}{4 \cdot 5} = \frac{1}{10}$.
- Exact enumeration for $n = 1, \ldots, 14$ agrees with all three formulae. A simulation of 2×10^5 chests with $n = 5$ gave 0.715, 0.714 and 0.286, against $\frac{5}{7} \approx 0.714$, $\frac{5}{7}$ and $\frac{12}{42} = \frac{2}{7} \approx 0.286$.

Common slips

- Treating the second pirate's draws as starting afresh with a full chest. How many gold coins are left for her depends on how the first pirate did.
- Multiplying $\mathrm{P}(A \geq 1)$, $\mathrm{P}(B \geq 1)$ and $\mathrm{P}(C \geq 1)$ in part (iii). The shares are not independent, because they must add up to n.

Beyond STEP

The correspondence between positions of the lead coins and solutions of $A + B + C = n$ is the "stars and bars" bijection. It shows that the gaps between randomly placed markers are exchangeable. The continuous version, with points dropped uniformly on a stick, leads to the theory of uniform spacings and the Dirichlet distribution, and order statistics of this kind come up throughout first-year probability. Blitzstein and Hwang, *Introduction to Probability*, treats this counting and the symmetry arguments that go with it very well.

WORKED SOLUTION

STEP II 2004, Question 1

Pure • Equations with Surds

STEP II 2004, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q1

The key idea

Every equation involves $\sqrt{x}$ and $\sqrt{3x^2+1}$, so put $t=\sqrt{x}$ (with $t \geq 0$) to turn the first into a polynomial. Then isolate the remaining square root and square once. The result is a quartic in t, and it factorises easily because $t=0$ and $t=1$ are always roots. The real work is at the end: squaring can introduce false roots, so we keep only those for which the side we squared was not negative.

Solution

All three equations need $x \geq 0$ for $\sqrt{x}$ to be real, so we write $x=t^2$ with $t \geq 0$. Then $\sqrt{x}=t$ and $3x^2+1=3t^4+1$.

The squaring step works in only one direction. For real A and B,

$$\sqrt{A}=B \iff A=B^2 \text{ and } B \geq 0.$$

So after squaring we must check that the right-hand side is non-negative at each root.

Part (i)

Isolating the root gives

$$\sqrt{3t^4+1}=2t^2-t+1.$$

The right-hand side is always positive: its discriminant is $1-8<0$ and its leading coefficient is positive. So squaring loses nothing here, and every root of the squared equation is a genuine solution. Squaring,

$$\begin{aligned} 3t^4+1 &= (2t^2-t+1)^2 = 4t^4-4t^3+5t^2-2t+1, \\ 0 &= t^4-4t^3+5t^2-2t = t\,(t^3-4t^2+5t-2). \end{aligned}$$

The cubic vanishes at $t=1$ (since $1-4+5-2=0$), and dividing out the factor $t-1$ gives

$$t^3-4t^2+5t-2=(t-1)(t^2-3t+2)=(t-1)^2(t-2).$$

So $t\,(t-1)^2(t-2)=0$, giving $t=0,1,2$. All three are non-negative, so

$$x=0,\quad x=1,\quad x=4.$$

As a check, at $x=4$ we get $\sqrt{49}+2-8-1=0$.

Part (ii)

Now we have

$$\sqrt{3t^4+1}=-t^2+2t+1.$$

Squaring,

$$3t^4+1=(-t^2+2t+1)^2=t^4-4t^3+2t^2+4t+1,$$
$$0=2t^4+4t^3-2t^2-4t=2t\,(t^3+2t^2-t-2).$$

Grouping the terms of the cubic as $t^2(t+2)-(t+2)$ gives

$$2t\,(t+2)(t-1)(t+1)=0.$$

Since $t\geq 0$, only $t=0$ and $t=1$ remain. The right-hand side $-t^2+2t+1$ takes the values 1 and 2 there, both positive, so both roots are genuine:

$$x=0,\quad x=1.$$

Part (iii)

This time

$$\sqrt{3t^4+1}=t^2+2t-1.$$

Squaring,

$$3t^4+1=(t^2+2t-1)^2=t^4+4t^3+2t^2-4t+1,$$
$$0=2t^4-4t^3-2t^2+4t=2t\,(t^3-2t^2-t+2).$$

Grouping the cubic as $t^2(t-2)-(t-2)$ gives

$$2t\,(t-2)(t-1)(t+1)=0.$$

The non-negative roots are $t=0,1,2$.

Ask yourself Do all three of these values satisfy the original equation?

No. At $t=0$ the right-hand side t^2+2t-1 equals -1, but a square root can't be negative. So $t=0$ is a false root, introduced by the squaring. (Directly: at $x=0$ the left-hand side of (iii) is $1-0-0+1=2\neq 0$.) At $t=1$ and $t=2$ the right-hand side is 2 and 7, both positive, so these roots are genuine. Hence

$$x=1,\quad x=4.$$

There is a neat link between (ii) and (iii). Replacing t by $-t$ turns $-t^2+2t+1$ into $-(t^2+2t-1)$, and squaring removes the minus sign. So the squared equation in (iii) is the squared equation in (ii) with t replaced by $-t$, and its roots $0,\pm 1,2$ are the negatives of the roots $0,\pm 1,-2$ from (ii).

Checks

Substitute each answer into the original equation. For example, $x = 4$ in (iii) gives $\sqrt{49} - 4 - 4 + 1 = 0$. A fine numerical scan of each left-hand side for $0 \leq x \leq 50$ found no other zeros. In (i) the left-hand side touches zero at $x = 1$ without changing sign, which matches the double factor $(t-1)^2$.

Common slips

- Forgetting to check the sign of the side being squared, and so keeping $x = 0$ in (iii).
- Dividing by t (or by $\sqrt{x}$) and so losing the root $x = 0$ in (i) and (ii).
- Squaring with two square roots still on one side. Isolate $\sqrt{3x^2+1}$ first, so that a single squaring removes it.

Beyond STEP

The map $A \mapsto A^2$ is not injective on $\mathbb{R}$, so squaring both sides gives an implication, not an equivalence. Keeping track of which steps are reversible is the habit behind every careful solution of an equation, and in analysis it becomes the question of when a function has an inverse and on what domain. Bartle and Sherbert, *Introduction to Real Analysis*, sets this out carefully.

WORKED SOLUTION

STEP II 2004, Question 2

Pure • Quadratic Inequalities

STEP II 2004, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q2

The key idea

The modulus is the only awkward feature, and it disappears if we write $t = |x|$. Since $x^2 = |x|^2$, the inequality becomes an ordinary quadratic inequality in t, with the extra condition $t \geq 0$. Completing the square, or looking at the discriminant, then tells us everything. Each admissible interval of t gives *two* intervals of x, one on each side of the origin.

Solution

Throughout, put $t = |x| \geq 0$. Since $x^2 = t^2$, the inequality $(*)$ says exactly

$$q(t) = t^2 - \alpha t + 2 < 0, \qquad t = |x| \geq 0.$$

No solutions when $|\alpha| < 2\sqrt{2}$

Completing the square,

$$q(t) = \left(t - \frac{\alpha}{2}\right)^2 + 2 - \frac{\alpha^2}{4}.$$

If $|\alpha| < 2\sqrt{2}$ then $\alpha^2 < 8$, so $2 - \frac{\alpha^2}{4} > 0$. The squared term is never negative, so $q(t) > 0$ for *every* real t, and in particular for $t = |x|$. Hence $x^2 - \alpha|x| + 2 > 0$ for all x, and there is no value of x satisfying $(*)$.

The case $\alpha = 3$

Now $q(t) = t^2 - 3t + 2 = (t-1)(t-2)$, which is negative exactly when $1 < t < 2$. So $(*)$ holds if and only if $1 < |x| < 2$, and the solution set is

$$\{x : -2 < x < -1\} \cup \{x : 1 < x < 2\}.$$

The sum of lengths S

Let $\alpha > 2\sqrt{2}$. Then $\alpha^2 > 8$, so q has two distinct real roots

$$t_1 = \frac{\alpha - \sqrt{\alpha^2 - 8}}{2}, \qquad t_2 = \frac{\alpha + \sqrt{\alpha^2 - 8}}{2},$$

and, as q has positive leading coefficient, $q(t) < 0$ exactly when $t_1 < t < t_2$.

Ask yourself Do we have to worry that some of the interval $t_1 < t < t_2$ is at negative t?

No. The roots have product $t_1 t_2 = 2 > 0$ and sum $t_1 + t_2 = \alpha > 0$, so both are positive. Hence the whole interval (t_1, t_2) lies in $t > 0$, and every value of t in it is a genuine value of $|x|$.

So $(*)$ holds if and only if $t_1 < |x| < t_2$, that is,

$$-t_2 < x < -t_1 \qquad \text{or} \qquad t_1 < x < t_2.$$

These are two disjoint intervals (since $t_1 > 0$), each of length $t_2 - t_1 = \sqrt{\alpha^2 - 8}$. Therefore

$$S = 2\sqrt{\alpha^2 - 8}.$$

Since $\alpha^2 - 8 < \alpha^2$ and $\alpha > 0$, taking positive square roots gives $\sqrt{\alpha^2 - 8} < \alpha$, so

$$S < 2\alpha,$$

as required.

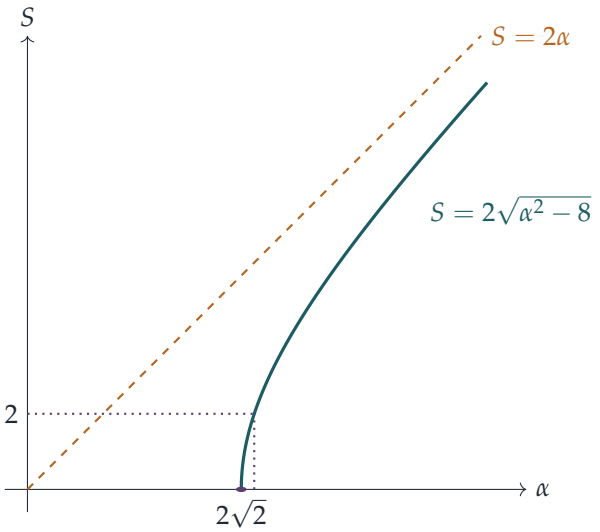
The graph of S against α

The function $S = 2\sqrt{\alpha^2 - 8}$ is defined for $\alpha > 2\sqrt{2}$ (for $\alpha \le 2\sqrt{2}$ there are no intervals at all: at $\alpha = 2\sqrt{2}$ we get $q(t) = (t - \sqrt{2})^2 \ge 0$, and whenever $\alpha \le 0$ we have $q(t) \ge 2$ for all $t \ge 0$). Its main features:

- $S \to 0$ as $\alpha \to 2\sqrt{2}^{+}$, and $\dfrac{\mathrm{d}S}{\mathrm{d}\alpha} = \dfrac{2\alpha}{\sqrt{\alpha^2 - 8}}$, which tends to infinity there: the curve starts at $(2\sqrt{2}, 0)$ with a vertical tangent.
- S is increasing, and $\dfrac{\mathrm{d}^2 S}{\mathrm{d}\alpha^2} = -\dfrac{16}{(\alpha^2 - 8)^{3/2}} < 0$, so the curve is concave.
- The line $S = 2\alpha$ is an asymptote, approached from below: since $(2\alpha - S)(2\alpha + S) = 4\alpha^2 - 4(\alpha^2 - 8) = 32$,

$$2\alpha - S = \frac{32}{2\alpha + S} \to 0 \quad \text{as } \alpha \to \infty.$$

In fact $\frac{\alpha^2}{8} - \frac{S^2}{32} = 1$, so the graph is one quarter of a hyperbola, with asymptote $S = 2\alpha$.



Checks

- At $\alpha = 3$ the formula gives $S = 2\sqrt{1} = 2$, matching the two intervals $(-2, -1)$ and $(1, 2)$ of length 1 each (the dotted point on the graph, at $\alpha = 3$).
- At $\alpha = 5$, $S = 2\sqrt{17} \approx 8.246$; measuring numerically the set of x in $[-20, 20]$ where $x^2 - 5|x| + 2 < 0$ gives the same value, comfortably below $2\alpha = 10$.
- For large α, $2\alpha - S \approx 8/\alpha$: at $\alpha = 100$ the gap is $0.0800\ldots$.

Common slips

- Forgetting that each interval of $|x|$ gives two intervals of x, and so getting $S = \sqrt{\alpha^2 - 8}$.
- Writing the $\alpha = 3$ answer as $1 < x < 2$ only, losing the negative half.
- Using the discriminant without noting that both roots in t are positive; for $\alpha < 0$ the same discriminant condition gives no solutions at all, because the roots are negative.

Beyond STEP

The function $x \mapsto x^2 - \alpha|x| + 2$ is even, so its solution set is symmetric, and the length S is the *measure* of the set where it is negative. Measuring the size of sets defined by inequalities is the starting point of measure theory and Lebesgue integration, met in second-year analysis. For a careful treatment of the limits, derivatives and inequalities used here, read Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2004, Question 3

Pure • Curve Sketching

STEP II 2004, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q3

The key idea

Everything comes from one curve. For C, factorise the derivative completely: the repeated factor $(x-2)$ survives differentiation, and what is left is a quadratic. Then the other two curves are built from C rather than started from scratch: in (i) we take $y = \pm\sqrt{f(x)}$, and in (ii) we feed x^2 into f. Keep track of where f is positive, where it has simple roots and where it has a repeated root; those details decide the shapes.

Solution

Write $f(x) = x(x+1)(x-2)^4$, so C is $y = f(x)$.

Stationary points of C

By the product rule, taking out the common factor $(x-2)^3$,

$$\begin{aligned} f'(x) &= (2x+1)(x-2)^4 + 4x(x+1)(x-2)^3 \\ &= (x-2)^3\left[(2x+1)(x-2) + 4x^2 + 4x\right] \\ &= (x-2)^3(6x^2 + x - 2) \\ &= (x-2)^3(3x+2)(2x-1). \end{aligned}$$

So the stationary points are at $x = -\frac{2}{3}$, $x = \frac{1}{2}$ and $x = 2$, with

$$f\left(-\tfrac{2}{3}\right) = \left(-\tfrac{2}{3}\right)\left(\tfrac{1}{3}\right)\left(-\tfrac{8}{3}\right)^4 = -\frac{8192}{729}, \qquad f\left(\tfrac{1}{2}\right) = \tfrac{1}{2}\cdot\tfrac{3}{2}\cdot\left(-\tfrac{3}{2}\right)^4 = \frac{243}{64}, \qquad f(2) = 0.$$

To classify them, look at the sign of each factor of f':

	$x < -\frac{2}{3}$	$-\frac{2}{3} < x < \frac{1}{2}$	$\frac{1}{2} < x < 2$	$x > 2$
$(x-2)^3$	$-$	$-$	$-$	$+$
$(3x+2)(2x-1)$	$+$	$-$	$+$	$+$
$f'(x)$	$-$	$+$	$-$	$+$

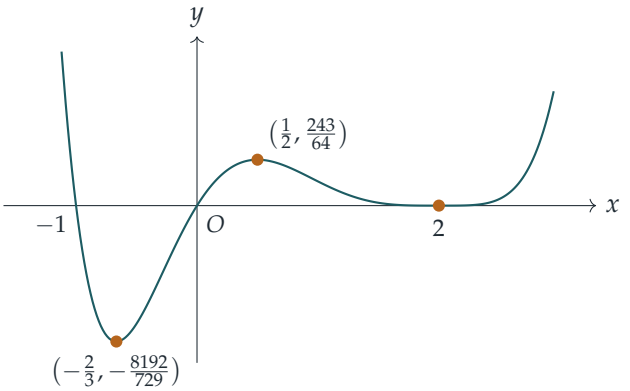
Hence

- $\left(-\frac{2}{3}, -\frac{8192}{729}\right)$ is a minimum (f' goes from $-$ to $+$);
- $\left(\frac{1}{2}, \frac{243}{64}\right)$ is a maximum (f' goes from $+$ to $-$);
- $(2, 0)$ is a minimum (f' goes from $-$ to $+$).

The last one is also clear without calculus: $f(x) \geq 0$ near $x = 2$ because $x(x+1) > 0$ there and $(x-2)^4 \geq 0$.

Sketching C

The curve crosses the x-axis at $x = -1$ and $x = 0$ (simple roots, with gradients $f'(-1) = -81$ and $f'(0) = 16$) and touches it at $x = 2$. Near $x = 2$, $f(x) \approx 2 \cdot 3\,(x-2)^4 = 6(x-2)^4$, so the curve is very flat there. Since f is a polynomial of degree 6 with positive leading coefficient, $y \to +\infty$ as $x \to \pm\infty$. The vertical scale in the sketch is compressed.



Part (i)

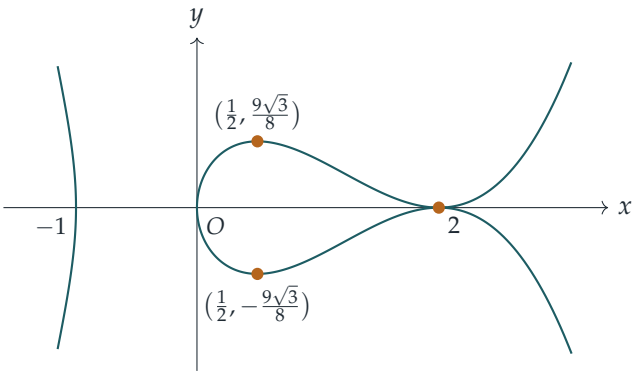
Now $y^2 = f(x)$, so $y = \pm\sqrt{f(x)}$. Three facts settle the shape.

- *Where the curve exists.* We need $f(x) \geq 0$, which from C means $x \leq -1$ or $x \geq 0$. There is no curve for $-1 < x < 0$.
- *Symmetry.* Replacing y by $-y$ leaves the equation unchanged, so the curve is symmetric in the x-axis. It is enough to sketch the upper half $y = \sqrt{f(x)} = (x-2)^2\sqrt{x(x+1)}$ and reflect it.
- *Stationary points.* For $y = \sqrt{f}$ with $f > 0$ we have $\mathrm{d}y/\mathrm{d}x = f'/(2\sqrt{f})$, so the upper half is stationary exactly where $f' = 0$ and $f > 0$. That leaves only $x = \frac{1}{2}$ (the point $x = -\frac{2}{3}$ is outside the domain), giving a maximum at $\left(\frac{1}{2}, \sqrt{243/64}\right) = \left(\frac{1}{2}, \frac{9\sqrt{3}}{8}\right)$, and a minimum at $\left(\frac{1}{2}, -\frac{9\sqrt{3}}{8}\right)$ on the lower half.

Ask yourself What happens at $(-1,0)$, $(0,0)$ and $(2,0)$, where $f = 0$?

At a *simple* root such as $x = 0$, $f(x) \approx 16x$, so near the origin $y^2 \approx 16x$: a sideways parabola, and the curve crosses the axis with a *vertical* tangent. The same happens at $x = -1$, where $y^2 \approx 81(-1 - x)$. At the *repeated* root $x = 2$, $y^2 \approx 6(x-2)^4$, so $y \approx \pm\sqrt{6}\,(x-2)^2$: two parabola-like arcs, one above and one below the axis, both touching the x-axis at $(2,0)$. So the upper and lower halves meet there with a common horizontal tangent; the upper half has a minimum and the lower half a maximum at $(2,0)$.

For large $|x|$, $y \approx \pm|x|^3$. So the curve is a closed loop from $(0,0)$ to $(2,0)$, pinched at $(2,0)$, continuing into two branches that open out to the right, together with a separate piece through $(-1,0)$ that opens out to the left.



Part (ii)

Here $y = x^2(x^2+1)(x^2-2)^4 = f(x^2)$. Two observations do most of the work.

- The curve is symmetric in the y-axis, since x appears only as x^2.
- As x runs over $[0,\infty)$, $u = x^2$ runs over $[0,\infty)$ increasing, so for $x \geq 0$ the curve is the part of C with $x \geq 0$, re-parametrised. In particular $y \geq 0$ everywhere, and the stretch of C with negative x (including its minimum) plays no part.

By the chain rule,

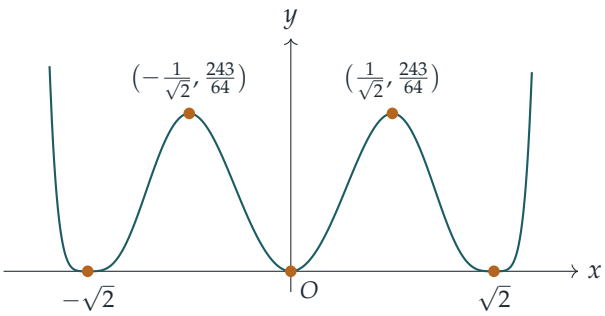
$$\frac{dy}{dx} = 2x\,f'(x^2) = 2x\,(x^2-2)^3(3x^2+2)(2x^2-1),$$

which vanishes when $x = 0$, $x^2 = \frac{1}{2}$ or $x^2 = 2$ (the factor $3x^2+2$ is never zero). On $u \geq 0$, f rises from 0 to its maximum at $u = \frac{1}{2}$, falls to 0 at $u = 2$ and then rises for ever, and $u = x^2$ preserves this order for $x \geq 0$. So the stationary points are

- a minimum at $(0,0)$, where $y \approx 16x^2$;
- maxima at $\left(\pm\frac{1}{\sqrt{2}}, \frac{243}{64}\right)$;

- minima at $(\pm\sqrt{2}, 0)$, where the curve touches the axis very flatly: $y \approx 384\,(x \mp \sqrt{2})^4$.

For large $|x|$, $y \approx x^{12}$, which rises very steeply.



Checks

A quick sanity check on the derivative: at $x = 0$, $f(x) = x(x+1)(x-2)^4 \approx 16x$, and indeed $f'(0) = (-8)(2)(-1) = 16$. A numerical check confirms the factorisation of f' against a finite-difference derivative at random points, gives $f(-\frac{2}{3}) \approx -11.237$ and $f(\frac{1}{2}) = 3.796875$, and finds the largest value of $\sqrt{f}$ on $[0, 2]$ to be $1.94856\ldots = \frac{9\sqrt{3}}{8}$, at $x = 0.5$. For (ii), $\mathrm{d}y/\mathrm{d}x$ changes sign at $x \approx \pm 0.707$ and ± 1.414 (and at 0), as predicted.

Common slips

- Expanding f before differentiating. Then f' is a quintic with no visible factors; keeping $(x-2)^3$ outside leaves only a quadratic to solve.
- In (i), drawing a cusp or a corner at $(0, 0)$ and $(-1, 0)$. At a simple root of f the tangent to $y^2 = f(x)$ is vertical.
- In (ii), copying the minimum of C at $x = -\frac{2}{3}$. Since $x^2 \geq 0$, only the part of C with $x \geq 0$ is used, and y is never negative.

Beyond STEP

The behaviour of $y^2 = f(x)$ near a root of f is the beginning of the study of singular points of algebraic curves: a simple root gives a smooth point with a vertical tangent, while a repeated root gives a place where two branches meet, as at $(2, 0)$. When f is a cubic, curves $y^2 = f(x)$ are *elliptic curves*, central in modern number theory. For a careful treatment of the calculus underlying curve sketching (monotonicity, the sign of the derivative and the chain rule), see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2004, Question 4

Pure • Optimisation in Geometry

STEP II 2004, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

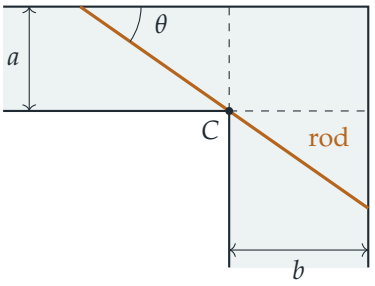
step.maths.org/questions/04-s2-q4

The key idea

Whatever happens, the rod (or table) has to swing from pointing along one corridor to pointing along the other, so at some moment it makes each angle θ in between with the walls. For a fixed angle, the most room there is lies along the line that just grazes the inner corner, and that length is easy to compute with right-angled triangles. So the object gets round only if its length is at most the *smallest* of these lengths over all angles, and calculus finds that smallest value.

Solution

Throughout, put the outer corner at the origin, with the outer walls along $y = 0$ (corridor of width a, running to the left) and $x = 0$ (corridor of width b, running downwards). The inner corner is then at $C = (-b, -a)$.



Part (i)

Suppose that at some moment the rod makes an acute angle θ with the walls of the a-corridor, sloping down from the outer wall $y = 0$ towards the outer wall $x = 0$ as in the figure.

The longest segment at angle θ. Take the line at angle θ through C. From C up to the wall

$y = 0$ is a vertical rise of a, so that piece has length $a\operatorname{cosec}\theta$; from C across to the wall $x = 0$ is a horizontal run of b, so that piece has length $b\sec\theta$. Hence the chord of this line between the two outer walls has length

$$f(\theta) = a\operatorname{cosec}\theta + b\sec\theta.$$

Any other line at angle θ that runs from one corridor into the other must pass on the outer side of C (otherwise it would cut through the inner walls). Such a line, together with the two outer walls, cuts off a smaller similar right-angled triangle, so its chord between the outer walls is shorter than $f(\theta)$. And if at this moment the rod lies wholly in the a-corridor, it fits in a strip of width a, so $L\sin\theta \le a$, giving $L \le a\operatorname{cosec}\theta < f(\theta)$; similarly $L \le b\sec\theta < f(\theta)$ if it lies wholly in the b-corridor. In every case,

$$L \le f(\theta) \quad \text{for each angle } \theta \text{ the rod passes through.}$$

As the rod turns from along the first corridor to along the second, θ varies continuously from near 0 to near $\frac{1}{2}\pi$, so it takes every value in between. Therefore $L \le f(\theta)$ for *every* $\theta \in (0, \frac{1}{2}\pi)$, that is, L is at most the minimum of f.

Minimising f. Differentiating,

$$f'(\theta) = -\frac{a\cos\theta}{\sin^2\theta} + \frac{b\sin\theta}{\cos^2\theta} = \frac{b\sin^3\theta - a\cos^3\theta}{\sin^2\theta\cos^2\theta}.$$

This is zero exactly when $\tan^3\theta = a/b$, which has a single solution α in $(0, \frac{1}{2}\pi)$, since tan is increasing there. The sign of f' is that of $b\tan^3\theta - a$, which is negative before α and positive after, so α gives the minimum. (Also $f \to \infty$ at both ends.) Taking $\theta = \alpha$ in $L \le f(\theta)$ gives

$$L \le a\operatorname{cosec}\alpha + b\sec\alpha, \qquad \tan^3\alpha = \frac{a}{b},$$

as required.

Part (ii)

Now the long sides of the table play the role of the rod. Its inner long side lies on a line at angle θ, and the outer long side lies on the parallel line a distance w further towards the outer corner. Both table corners on the outer side must lie inside the outer walls.

The room along the outer side. A line at angle θ whose distance from the outer corner O is d cuts off a right-angled triangle at O with hypotenuse c (the chord between the walls), legs $c\cos\theta$ and $c\sin\theta$, and altitude d from O. Twice its area is both $c\cos\theta \cdot c\sin\theta$ and $c\,d$, so

$$c = \frac{d}{\sin\theta\cos\theta} = 2d\operatorname{cosec}2\theta.$$

Chord length is proportional to the distance from O. Moving the line in by w (towards O) therefore shortens its chord by $2w\operatorname{cosec}2\theta$.

Ask yourself Why must the outer long side fit between the outer walls on the line that is w in from the line through C?

The inner long side cannot cross the inner corner, so it lies on the line through C or outside it. The outer long side is then on a line at least w further out, whose chord is at most $f(\theta) - 2w\operatorname{cosec} 2\theta$. (The two short sides run inwards from the outer corners, perpendicular to the long sides, and stay inside the corridors, so they impose nothing more.) If instead the whole table lies in the a-corridor, then its height across the strip is $l\sin\theta + w\cos\theta \le a$. The resulting bound $l \le (a - w\cos\theta)\operatorname{cosec}\theta$ is smaller than $g(\theta)$ below by $(b - w\sin\theta)\sec\theta > 0$, since $w < b$. The b-corridor case is the same. So, exactly as in part (i), success requires $l \le g(\theta)$ for all $\theta \in (0, \frac{1}{2}\pi)$, where

$$g(\theta) = a\operatorname{cosec}\theta + b\sec\theta - 2w\operatorname{cosec} 2\theta.$$

(The table must be narrower than both corridors, $w < a$ and $w < b$, or it cannot even travel along them.)

Minimising g. Since $\dfrac{\mathrm{d}}{\mathrm{d}\theta}\operatorname{cosec} 2\theta = -\dfrac{2\cos 2\theta}{\sin^2 2\theta}$,

$$g'(\theta) = -\frac{a\cos\theta}{\sin^2\theta} + \frac{b\sin\theta}{\cos^2\theta} + \frac{4w\cos 2\theta}{\sin^2 2\theta}.$$

Multiply by $\sin^2\theta\cos^2\theta = \frac{1}{4}\sin^2 2\theta > 0$. Then $g'(\theta) = 0$ exactly when

$$-a\cos^3\theta + b\sin^3\theta + w\cos 2\theta = 0, \qquad \text{i.e.} \qquad w\cos 2\theta = \cos^3\theta\,(a - b\tan^3\theta).$$

Write $\cos 2\theta = \cos^2\theta\,(1 - \tan^2\theta)$ and divide by $\cos^2\theta$. If $\theta = \frac{1}{4}\pi$ the equation would read $0 = \frac{1}{2\sqrt{2}}(a - b)$, which is impossible since $a \ne b$. So $\tan^2\theta \ne 1$ at a stationary point, and we may divide by $1 - \tan^2\theta$ to get

$$w = \left(\frac{a - b\tan^3\theta}{1 - \tan^2\theta}\right)\cos\theta.$$

As $\theta \to 0^+$, $g(\theta) \approx (a - w)/\theta \to +\infty$ because $w < a$; similarly $g \to +\infty$ as $\theta \to \frac{1}{2}\pi^-$ because $w < b$. So the continuous function g attains its minimum at an interior point, where $g' = 0$. Call this point β; it satisfies the displayed equation. Taking $\theta = \beta$ in $l \le g(\theta)$ gives

$$l \le a\operatorname{cosec}\beta + b\sec\beta - 2w\operatorname{cosec} 2\beta,$$

as required. (In fact $l \le \min g \le g(\beta)$ for *any* root β of the equation, so it doesn't matter which root is meant.)

Checks

- Putting $w = 0$ in part (ii) gives $\tan^3\beta = a/b$ and the bound of part (i), as it should.
- In part (i), $\sin\alpha = a^{1/3}/\sqrt{a^{2/3} + b^{2/3}}$ and $\cos\alpha = b^{1/3}/\sqrt{a^{2/3} + b^{2/3}}$, so the bound is $\left(a^{2/3} + b^{2/3}\right)^{3/2}$. For $a = 1$, $b = 8$ this is $5^{3/2} \approx 11.18$, and a numerical search for the minimum of f agrees.

- For a few hundred random $a \neq b$ and $w < \min(a, b)$, we found the minimum of g by brute force and confirmed that the minimising angle satisfies the equation for β. We also checked the chord lengths directly from coordinates.

Common slips

- Maximising f instead of minimising it. The rod has to survive the *worst* angle, so the bound is the minimum of the available length.
- Taking the table's *inner* side as the one limited by the outer walls. It is the outer side, on the line w nearer the outer corner, that is squeezed.
- Dividing by $1 - \tan^2 \beta$ without noticing that this needs $a \neq b$.

Beyond STEP

Part (i) is a classic "ladder round a corner" problem, and its answer $\left(a^{2/3} + b^{2/3}\right)^{3/2}$ is linked to the astroid: the lines through the inner corner sweep out an envelope. Part (ii) is a small case of the *moving sofa problem*: which shape of largest area can be carried round a unit-width right-angled corner? It was posed by Leo Moser in 1966 and has resisted a complete solution for most of its history. For more geometry-meets-calculus problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2004, Question 5

Pure • Integral Equations

STEP II 2004, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q5

The key idea

The unknown function sits inside an integral, which looks frightening until you expand $\sin(x+t)$ with the compound-angle formula. Then t comes outside the integral, and what is left inside are two fixed *numbers*. So f must be t plus a combination of $\sin t$ and $\cos t$, and the whole problem reduces to finding two constants from two linear equations. The two integrals in the first part are exactly the ones that equation needs.

Solution

The two integrals

Integrating by parts with $u = x$ and $\mathrm{d}v = \sin x\,\mathrm{d}x$:

$$\int_0^\pi x\sin x\,\mathrm{d}x = \Big[-x\cos x\Big]_0^\pi + \int_0^\pi \cos x\,\mathrm{d}x = (-\pi\cos\pi - 0) + \Big[\sin x\Big]_0^\pi = \pi.$$

Similarly, with $u = x$ and $\mathrm{d}v = \cos x\,\mathrm{d}x$:

$$\int_0^\pi x\cos x\,\mathrm{d}x = \Big[x\sin x\Big]_0^\pi - \int_0^\pi \sin x\,\mathrm{d}x = 0 - \Big[-\cos x\Big]_0^\pi = -(1+1) = -2.$$

We will also need three standard integrals over $[0,\pi]$:

$$\int_0^\pi \sin^2 x\,\mathrm{d}x = \int_0^\pi \cos^2 x\,\mathrm{d}x = \frac{\pi}{2}, \qquad \int_0^\pi \sin x\cos x\,\mathrm{d}x = 0.$$

The first two follow from $\sin^2 x = \frac{1}{2}(1-\cos 2x)$ and $\cos^2 x = \frac{1}{2}(1+\cos 2x)$, since $\cos 2x$ integrates to zero over a whole period; the third is $\frac{1}{2}\int_0^\pi \sin 2x\,\mathrm{d}x = 0$ for the same reason.

The form of f

Expand $\sin(x+t) = \sin x \cos t + \cos x \sin t$. The integration is over x, so $\sin t$ and $\cos t$ are constants and come out of the integral:

$$\begin{aligned} \mathrm{f}(t) &= t + \int_0^\pi \mathrm{f}(x)\big(\cos x \sin t + \sin x \cos t\big)\,\mathrm{d}x \\ &= t + \sin t \int_0^\pi \mathrm{f}(x)\cos x\,\mathrm{d}x + \cos t \int_0^\pi \mathrm{f}(x)\sin x\,\mathrm{d}x \\ &= t + A\sin t + B\cos t, \end{aligned}$$

where A and B are the two given integrals. Since they don't depend on t, they are genuine constants. This is the required form.

Finding A and B

? **Ask yourself** We don't know f, so how can we work out A and B?

We do know its *shape*: $\mathrm{f}(x) = x + A\sin x + B\cos x$. Putting this back into $(*)$ turns the integral into something we can evaluate in terms of A and B, and matching the two sides gives equations for them.

Substitute into the right-hand side of $(*)$, using the expansion of $\sin(x+t)$ from above:

$$t + \sin t \int_0^\pi (x + A\sin x + B\cos x)\cos x\,\mathrm{d}x + \cos t \int_0^\pi (x + A\sin x + B\cos x)\sin x\,\mathrm{d}x.$$

Using the integrals found earlier,

$$\begin{aligned} \int_0^\pi (x + A\sin x + B\cos x)\cos x\,\mathrm{d}x &= -2 + A\cdot 0 + B\cdot\frac{\pi}{2} = -2 + \frac{\pi}{2}B, \\ \int_0^\pi (x + A\sin x + B\cos x)\sin x\,\mathrm{d}x &= \pi + A\cdot\frac{\pi}{2} + B\cdot 0 = \pi + \frac{\pi}{2}A. \end{aligned}$$

So $(*)$ says that, for every t,

$$t + A\sin t + B\cos t = t + \left(-2 + \frac{\pi}{2}B\right)\sin t + \left(\pi + \frac{\pi}{2}A\right)\cos t.$$

Equating coefficients of $\sin t$ and $\cos t$ (justified because the identity holds for all t: putting $t = \frac{\pi}{2}$ and then $t = 0$ isolates each coefficient) gives

$$A = -2 + \frac{\pi}{2}B, \qquad B = \pi + \frac{\pi}{2}A.$$

Substituting the second into the first,

$$A = -2 + \frac{\pi^2}{2} + \frac{\pi^2}{4}A \quad\Longrightarrow\quad A\left(1 - \frac{\pi^2}{4}\right) = \frac{\pi^2}{2} - 2, \quad \text{i.e.} \quad A\cdot\frac{4-\pi^2}{4} = -\frac{4-\pi^2}{2}.$$

Since $\pi^2 \neq 4$, we may divide by $4 - \pi^2$, giving $A = -2$. Then $B = \pi + \frac{\pi}{2}(-2) = 0$. Therefore

$$A = -2, \qquad B = 0, \qquad \text{so} \qquad \mathrm{f}(t) = t - 2\sin t.$$

Checks

Check the answer against the definitions of A and B directly. With $\mathrm{f}(x) = x - 2\sin x$,

$$\int_0^{\pi} \mathrm{f}(x)\cos x\,\mathrm{d}x = -2 - 2\cdot 0 = -2 = A, \qquad \int_0^{\pi} \mathrm{f}(x)\sin x\,\mathrm{d}x = \pi - 2\cdot\frac{\pi}{2} = 0 = B.$$

Numerically, Simpson's rule gives $\int_0^{\pi} x\sin x\,\mathrm{d}x \approx 3.14159$ and $\int_0^{\pi} x\cos x\,\mathrm{d}x \approx -2.00000$. We also evaluated the right-hand side of $(*)$ with $\mathrm{f}(t) = t - 2\sin t$ at fifty random values of t, and it matched $\mathrm{f}(t)$ to within 10^{-9} every time.

Common slips

- Forgetting the minus sign in $\int_0^{\pi} x\cos x\,\mathrm{d}x$: the boundary term vanishes and $-\int_0^{\pi}\sin x\,\mathrm{d}x = -2$.
- Mixing up which integral multiplies $\sin t$. The $\cos x\sin t$ part of $\sin(x+t)$ gives $A\sin t$ with $A = \int \mathrm{f}(x)\cos x\,\mathrm{d}x$, not $\int \mathrm{f}(x)\sin x\,\mathrm{d}x$.
- Dividing by $1 - \frac{\pi^2}{4}$ without noting that it is non-zero.

Beyond STEP

Equation $(*)$ is a *Fredholm integral equation of the second kind*. Its kernel $\sin(x+t) = \sin x\cos t + \cos x\sin t$ is a finite sum of products of a function of x and a function of t (a "degenerate" or "separable" kernel), and that is exactly why the problem collapses to a 2×2 linear system. The condition $\pi^2 \neq 4$ is the statement that this system has a non-zero determinant; if it were zero there could be no solution or infinitely many, which is the integral-equation version of a singular matrix. For the linear algebra behind this, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP II 2004, Question 6

Pure • Vector Decomposition

STEP II 2004, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q6

The key idea

Splitting a vector into a part along $\mathbf{a}$ and a part perpendicular to $\mathbf{a}$ is just projection: since $\mathbf{a}$ is a unit vector, the component of any vector $\mathbf{v}$ along it is $(\mathbf{v}\cdot\mathbf{a})\,\mathbf{a}$. The second part is the same idea one dimension up. The vectors $\mathbf{a}$ and $\mathbf{q}$ are perpendicular directions in Π, and $\mathbf{R}$ supplies the third, perpendicular direction. So we can either project onto each direction in turn or write down unknown multiples and impose the perpendicularity conditions with dot products. We do the second and use the first as a check.

Solution

The vectors $\mathbf{p}$ and $\mathbf{q}$

Since $\mathbf{p}$ is parallel to $\mathbf{a}$, write $\mathbf{p}=\lambda\mathbf{a}$. Then

$$\mathbf{q}=\mathbf{a}+\mathbf{b}-\lambda\mathbf{a}=(1-\lambda)\mathbf{a}+\mathbf{b},$$

which lies in Π automatically, because $\mathbf{a}$ and $\mathbf{b}$ do. For $\mathbf{q}$ to be perpendicular to $\mathbf{a}$ we need

$$0=\mathbf{q}\cdot\mathbf{a}=(1-\lambda)\,|\mathbf{a}|^2+\mathbf{a}\cdot\mathbf{b}=(1-\lambda)+3,$$

using $|\mathbf{a}|=1$ and $\mathbf{a}\cdot\mathbf{b}=3$. So $\lambda=4$ and

$$\boxed{\mathbf{p}=4\mathbf{a},\qquad \mathbf{q}=\mathbf{b}-3\mathbf{a}.}$$

Indeed $\mathbf{q}\cdot\mathbf{a}=3-3=0$, and $\mathbf{p}+\mathbf{q}=\mathbf{a}+\mathbf{b}$. (Equivalently, $\mathbf{p}$ is the projection $\big((\mathbf{a}+\mathbf{b})\cdot\mathbf{a}\big)\mathbf{a}=4\mathbf{a}$.)

The vectors $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R}$

First, a few facts we will need. With $|\mathbf{b}|=5$,

$$|\mathbf{q}|^2=|\mathbf{b}|^2-6\,\mathbf{a}\cdot\mathbf{b}+9|\mathbf{a}|^2=25-18+9=16,$$

so $|\mathbf{q}| = 4$. In particular $\mathbf{q} \neq \mathbf{0}$, so $\mathbf{b}$ is not parallel to $\mathbf{a}$, and $\mathbf{a}$ and $\mathbf{b}$ span the plane Π.

Write $\mathbf{P} = \alpha\mathbf{a}$ and $\mathbf{Q} = \beta\mathbf{q} = \beta(\mathbf{b} - 3\mathbf{a})$. Then $\mathbf{R}$ is forced to be

$$\mathbf{R} = \mathbf{a} + \mathbf{b} + \mathbf{c} - \alpha\mathbf{a} - \beta(\mathbf{b} - 3\mathbf{a}) = \mathbf{c} + u\,\mathbf{a} + v\,\mathbf{b}, \qquad \text{where } u = 1 - \alpha + 3\beta,\ v = 1 - \beta.$$

Ask yourself How do we turn "$\mathbf{R}$ is perpendicular to Π" into equations?

A vector is perpendicular to a plane exactly when it is perpendicular to two non-parallel vectors in that plane. We have just seen that $\mathbf{a}$ and $\mathbf{b}$ are such a pair. So the condition is $\mathbf{R} \cdot \mathbf{a} = 0$ and $\mathbf{R} \cdot \mathbf{b} = 0$. Using $\mathbf{a} \cdot \mathbf{c} = -2$, $\mathbf{b} \cdot \mathbf{c} = 2$, $\mathbf{a} \cdot \mathbf{b} = 3$, $|\mathbf{a}|^2 = 1$ and $|\mathbf{b}|^2 = 25$:

$$\mathbf{R} \cdot \mathbf{a} = -2 + u + 3v = 0,$$
$$\mathbf{R} \cdot \mathbf{b} = 2 + 3u + 25v = 0.$$

The first gives $u = 2 - 3v$. Substituting into the second gives $8 + 16v = 0$, so

$$v = -\tfrac{1}{2}, \qquad u = \tfrac{7}{2}.$$

Then $\beta = 1 - v = \tfrac{3}{2}$ and $\alpha = 1 + 3\beta - u = 1 + \tfrac{9}{2} - \tfrac{7}{2} = 2$. Therefore

$$\boxed{\mathbf{P} = 2\mathbf{a}, \qquad \mathbf{Q} = \tfrac{3}{2}(\mathbf{b} - 3\mathbf{a}), \qquad \mathbf{R} = \mathbf{c} + \tfrac{7}{2}\mathbf{a} - \tfrac{1}{2}\mathbf{b}.}$$

Adding these, the coefficient of $\mathbf{a}$ is $2 - \tfrac{9}{2} + \tfrac{7}{2} = 1$, the coefficient of $\mathbf{b}$ is $\tfrac{3}{2} - \tfrac{1}{2} = 1$ and the coefficient of $\mathbf{c}$ is 1, so $\mathbf{P} + \mathbf{Q} + \mathbf{R} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ as required.

Finally, $\mathbf{R} \neq \mathbf{0}$: if it were zero, $\mathbf{c} = \tfrac{1}{2}\mathbf{b} - \tfrac{7}{2}\mathbf{a}$ would lie in Π, contrary to the hypothesis that $\mathbf{c}$ is not parallel to Π. That hypothesis is what makes $\mathbf{R}$ a genuine normal vector.

A second view: projecting onto perpendicular axes

The unit vectors $\mathbf{a}$, $\hat{\mathbf{q}} = \tfrac{1}{4}(\mathbf{b} - 3\mathbf{a})$ and a unit normal $\mathbf{n}$ to Π form an orthonormal set. So $\mathbf{P}$ and $\mathbf{Q}$ are the projections of $\mathbf{v} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ onto the first two:

$$\mathbf{v} \cdot \mathbf{a} = 1 + 3 - 2 = 2, \qquad \text{so } \mathbf{P} = 2\mathbf{a};$$
$$\mathbf{v} \cdot \hat{\mathbf{q}} = \tfrac{1}{4}\left(\mathbf{a} \cdot \mathbf{q} + \mathbf{b} \cdot \mathbf{q} + \mathbf{c} \cdot \mathbf{q}\right) = \tfrac{1}{4}(0 + 16 + 8) = 6, \qquad \text{so } \mathbf{Q} = 6\hat{\mathbf{q}} = \tfrac{3}{2}(\mathbf{b} - 3\mathbf{a}),$$

using $\mathbf{b} \cdot \mathbf{q} = 25 - 9 = 16$ and $\mathbf{c} \cdot \mathbf{q} = 2 + 6 = 8$. What is left over, $\mathbf{R}$, is the component of $\mathbf{c}$ normal to Π. This agrees with the answer above.

Checks

- Directly: $\mathbf{R} \cdot \mathbf{a} = -2 + \tfrac{7}{2} - \tfrac{3}{2} = 0$ and $\mathbf{R} \cdot \mathbf{b} = 2 + \tfrac{21}{2} - \tfrac{25}{2} = 0$.
- A concrete case: take $\mathbf{a} = \mathbf{i}$, $\mathbf{b} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{c} = -2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, which satisfy all the given conditions. Then $\mathbf{q} = 4\mathbf{j}$, $\mathbf{P} = 2\mathbf{i}$, $\mathbf{Q} = 6\mathbf{j}$ and $\mathbf{R} = \mathbf{k}$, and these add up to $\mathbf{a} + \mathbf{b} + \mathbf{c} = 2\mathbf{i} + 6\mathbf{j} + \mathbf{k}$.
- We also tested the formulae on 2000 randomly oriented configurations satisfying the conditions. Every time, $\mathbf{q} \cdot \mathbf{a} = 0$, $|\mathbf{q}| = 4$, $\mathbf{R}$ was normal to Π and the three vectors summed to $\mathbf{a} + \mathbf{b} + \mathbf{c}$, all to 10^{-9}.

Common slips

- Taking $\mathbf{p} = (\mathbf{a}\cdot\mathbf{b})\mathbf{a} = 3\mathbf{a}$, which is the projection of $\mathbf{b}$ alone. We need the projection of $\mathbf{a}+\mathbf{b}$, which is $4\mathbf{a}$.
- Imposing only $\mathbf{R}\cdot\mathbf{a} = 0$. Perpendicularity to a plane needs two independent conditions.
- Assuming $\mathbf{P} = \mathbf{p}$ and $\mathbf{Q} = \mathbf{q}$. Adding $\mathbf{c}$ changes the in-plane components too, because $\mathbf{c}$ itself has a component in Π.

Beyond STEP

This is the Gram–Schmidt process in miniature. From $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ we built mutually perpendicular directions $\mathbf{a}$, $\mathbf{b}-3\mathbf{a}$ and $\mathbf{c}+\frac{7}{2}\mathbf{a}-\frac{1}{2}\mathbf{b}$, and then decomposed a vector along them. The same procedure works in any number of dimensions. It underlies orthogonal projection, least-squares fitting and the QR factorisation of a matrix. Strang's *Introduction to Linear Algebra* develops all of these.

WORKED SOLUTION

STEP II 2004, Question 7

Pure • Interval Bisection

STEP II 2004, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q7

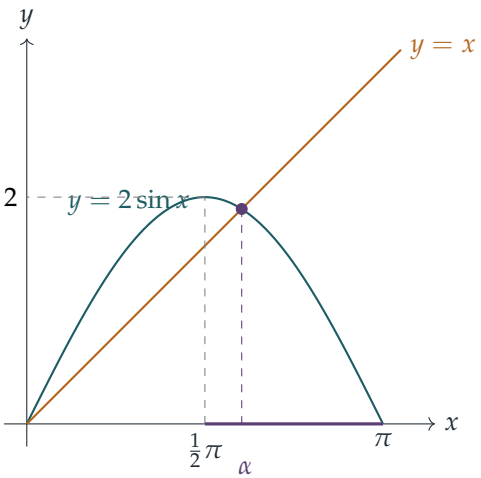
The key idea

A root of $2\sin x - x = 0$ is a place where the curve $y = 2\sin x$ meets the line $y = x$, so the graphical part is about one sine curve and one straight line. The bisection steps then come down to one question each time: is $2\sin c$ bigger or smaller than c at the midpoint? The half-angle formula turns the sines of $\frac{5}{8}\pi$ and $\frac{9}{16}\pi$ into nested square roots, and comparing *squares* of two positive numbers keeps the arithmetic to the given approximations.

Solution

Exactly one root in I_0

We have $f(x) = 0 \iff 2\sin x = x$, so the roots are the x-coordinates of the points where $y = 2\sin x$ meets $y = x$.



On $[\frac{1}{2}\pi, \pi]$:

- at $x = \frac{1}{2}\pi$ the curve is at height 2 and the line at height $\frac{1}{2}\pi < 2$ (since $\pi < 4$), so the curve is above the line;
- at $x = \pi$ the curve is at height 0 and the line at height $\pi > 0$, so the curve is below the line;
- the curve $y = 2\sin x$ is decreasing on this interval while the line $y = x$ is increasing.

Both graphs are continuous, so the curve must cross the line somewhere in the interval; and since one is falling and the other rising, once the curve is below the line it can never come back above it. So they cross exactly once, and $f(x) = 0$ has exactly one root α in I_0. In terms of f: $f(\frac{1}{2}\pi) = 2 - \frac{1}{2}\pi > 0$, $f(\pi) = -\pi < 0$, and f is strictly decreasing there.

In every bisection step below, we keep the half on which f changes sign. Since $f(\frac{1}{2}\pi) > 0$, the sign of $f(c_n)$ decides everything: if $f(c_n) < 0$ the root is to the left of c_n, and if $f(c_n) > 0$ it is to the right.

Finding I_1

Here $c_0 = \frac{3}{4}\pi$ and

$$f(\tfrac{3}{4}\pi) = 2 \cdot \frac{1}{\sqrt{2}} - \tfrac{3}{4}\pi \approx 1.4 - \tfrac{3}{4}\sqrt{10}.$$

Since $\sqrt{10} > 3$, we have $\frac{3}{4}\sqrt{10} > 2.25 > 1.4$, so $f(c_0) < 0$. With $f(a_0) > 0$ this gives $f(a_0)f(c_0) < 0$, so

$$I_1 = [\tfrac{1}{2}\pi, \tfrac{3}{4}\pi].$$

Finding I_2

Now $c_1 = \frac{1}{2}(\frac{1}{2}\pi + \frac{3}{4}\pi) = \frac{5}{8}\pi$, and we need the sign of $2\sin\frac{5}{8}\pi - \frac{5}{8}\pi$.

> **?** **Ask yourself** How can we find $\sin\frac{5}{8}\pi$ without a calculator?

Use $\sin\frac{5}{8}\pi = \sin(\pi - \frac{5}{8}\pi) = \sin\frac{3}{8}\pi = \cos\frac{1}{8}\pi$, and then the double-angle formula $\cos 2\theta = 2\cos^2\theta - 1$ with $\theta = \frac{1}{8}\pi$:

$$4\cos^2\tfrac{1}{8}\pi = 2\big(1 + \cos\tfrac{1}{4}\pi\big) = 2 + \sqrt{2}.$$

So $(2\sin\frac{5}{8}\pi)^2 = 2 + \sqrt{2}$. Both $2\sin\frac{5}{8}\pi$ and $\frac{5}{8}\pi$ are positive, so we may compare their squares:

$$(2\sin\tfrac{5}{8}\pi)^2 = 2 + \sqrt{2} \approx 3.4, \qquad (\tfrac{5}{8}\pi)^2 \approx \tfrac{25}{64} \cdot 10 = \tfrac{250}{64} \approx 3.9.$$

Hence $2\sin\frac{5}{8}\pi < \frac{5}{8}\pi$, that is, $f(c_1) < 0$. Again $f(a_1) = f(\frac{1}{2}\pi) > 0$, so $f(a_1)f(c_1) < 0$ and

$$I_2 = [\tfrac{1}{2}\pi, \tfrac{5}{8}\pi],$$

as required.

Finding I_3

Now $c_2 = \frac{1}{2}(\frac{1}{2}\pi + \frac{5}{8}\pi) = \frac{9}{16}\pi$. As before, $\sin\frac{9}{16}\pi = \sin\frac{7}{16}\pi = \cos\frac{1}{16}\pi$, and the double-angle formula with $\theta = \frac{1}{16}\pi$ gives

$$(2\sin\tfrac{9}{16}\pi)^2 = 4\cos^2\tfrac{1}{16}\pi = 2\big(1+\cos\tfrac{1}{8}\pi\big) = 2 + 2\cos\tfrac{1}{8}\pi = 2+\sqrt{2+\sqrt{2}},$$

using $2\cos\frac{1}{8}\pi = \sqrt{2+\sqrt{2}}$ from the previous step. With $\sqrt{2} \approx 1.4$, we have $\sqrt{2+\sqrt{2}} \approx \sqrt{3.4} > 1.8$ (because $1.8^2 = 3.24$), so

$$(2\sin\tfrac{9}{16}\pi)^2 > 3.8, \qquad (\tfrac{9}{16}\pi)^2 \approx \tfrac{81}{256}\cdot 10 = \tfrac{810}{256} < 3.2.$$

So $2\sin\frac{9}{16}\pi > \frac{9}{16}\pi$, that is, $f(c_2) > 0$. This time $f(a_2)f(c_2) > 0$, while $f(c_2)f(b_2) = f(\frac{9}{16}\pi)\,f(\frac{5}{8}\pi) < 0$. Therefore

$$I_3 = [\tfrac{9}{16}\pi, \tfrac{5}{8}\pi].$$

The approximations are safe here: at each midpoint the two quantities being compared differ by about 0.5 or more, far more than the error introduced by $\frac{1}{\sqrt{2}} \approx 0.7$ or $\pi \approx \sqrt{10}$.

Checks

- Running the bisection numerically gives the same intervals, and the root is $\alpha \approx 1.8955$, which lies between $\frac{9}{16}\pi \approx 1.767$ and $\frac{5}{8}\pi \approx 1.963$.
- Exact values agree with the approximate ones: $f(\frac{3}{4}\pi) \approx -0.942$, $f(\frac{5}{8}\pi) \approx -0.116$ and $f(\frac{9}{16}\pi) \approx 0.194$.
- Each interval has half the length of the one before: $\frac{1}{2}\pi$, $\frac{1}{4}\pi$, $\frac{1}{8}\pi$, $\frac{1}{16}\pi$.

Common slips

- Showing a sign change but not explaining why there is only *one* root. A sign change alone gives at least one; you need the decreasing-versus-increasing argument for uniqueness.
- Comparing $2\sin c$ and c by squaring without first noting that both are positive.

Beyond STEP

The sign-change argument is the intermediate value theorem, one of the first big theorems in a university analysis course; its proof needs the completeness of the real numbers, and bisection is essentially that proof run as an algorithm. Bisection always converges, but slowly (one binary digit per step). Methods such as Newton–Raphson converge much faster, but can fail, so numerical analysis is largely about trading safety against speed. For the theory, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2004, Question 8

Pure • Comparing Differential Equations

STEP II 2004, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q8

The key idea

While $0 \leq x < 1$ the right-hand side is positive, so x increases with t and we may turn the equation over: t becomes a function of x, namely the time taken to climb from 0 to height x. A bigger gradient at every height means every height is reached sooner, so the given inequalities between the right-hand sides immediately order the three graphs, one above the next. Whether the climb to $x = 1$ takes finite or infinite time is then a question about whether an improper integral converges.

Solution

Throughout, write $x_n(t)$ for the solution with exponent n. At $t = 0$ we have $x = 0$ and gradient $(1-0)^{1/n} = 1$ for every n, so all the curves leave the origin along the line $x = t$. While $0 \leq x < 1$ the gradient is strictly positive, so x is increasing and

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{(1-x^n)^{1/n}}, \qquad \text{so} \qquad t_n(x) = \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^n)^{1/n}}$$

is the time at which the solution reaches height x, for $0 \leq x < 1$.

Part (i)

With $n = 1$ the equation is $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 1 - x$. Separating (fine, since $1 - x > 0$ along the solution),

$$t = \int_0^x \frac{\mathrm{d}\xi}{1-\xi} = -\ln(1-x), \qquad \text{so} \qquad x = 1 - \mathrm{e}^{-t}.$$

The graph starts at the origin with gradient 1, increases throughout and is concave (its gradient e^{-t} decreases), and approaches the asymptote $x = 1$ as $t \to \infty$ without ever reaching it: $t_1(x) = -\ln(1-x) \to \infty$ as $x \to 1$. The lowest curve in the figure below shows it.

Part (ii)

The inequality. For $0 < x < 1$ both $1 - x$ and $(1 - x^2)^{1/2}$ are positive, so the inequality is equivalent to the one obtained by squaring:

$$(1-x)^2 < 1 - x^2 \iff 1 - 2x + x^2 < 1 - x^2 \iff 2x^2 < 2x \iff x < 1,$$

where the last step divides by $2x > 0$. Since $x < 1$ is true, so is $1 - x < (1 - x^2)^{1/2}$.

The sketch. At every height x in $(0, 1)$ the $n = 2$ curve is steeper than the $n = 1$ curve. Hence, for $0 < x < 1$,

$$t_2(x) = \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^2)^{1/2}} < \int_0^x \frac{\mathrm{d}\xi}{1-\xi} = t_1(x),$$

because the integrand on the left is strictly smaller on $(0, x]$. So the $n = 2$ solution reaches every height strictly sooner, and since both solutions are increasing, at every time $t > 0$ (while $x_2 < 1$) the $n = 2$ curve lies strictly above the $n = 1$ curve. Both leave the origin with gradient 1, so near the origin they are very close, and they separate as t grows.

> **Ask yourself** The two curves have the same gradient at the origin. Why can't the $n = 1$ curve sneak above later?

To get above, it would have to catch up with the $n = 2$ curve at some height x, that is, reach x at the same time. But $t_2(x) < t_1(x)$ for every $x \in (0, 1)$: the slower climber is always strictly late. (Equivalently: wherever the curves are at the same height, the $n = 2$ one is steeper, so it can never be overtaken.)

Solving. Put $x = \sin y$ with $0 \le y < \frac{1}{2}\pi$, and $y = 0$ at $t = 0$. Then $\cos y > 0$, so $(1 - x^2)^{1/2} = (\cos^2 y)^{1/2} = \cos y$, and the equation becomes

$$\cos y \frac{\mathrm{d}y}{\mathrm{d}t} = \cos y \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 1 \quad \Longrightarrow \quad y = t,$$

using $y = 0$ at $t = 0$. So the solution is

$$x = \sin t \qquad \text{for } 0 \le t < \tfrac{1}{2}\pi.$$

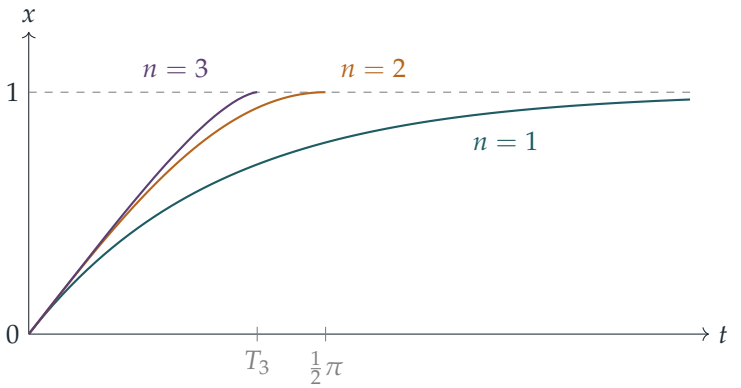
This agrees with $t_2(x) = \arcsin x$. Unlike the case $n = 1$, the solution reaches $x = 1$ in finite time, at $t = \frac{1}{2}\pi$, and arrives there with gradient $\cos \frac{1}{2}\pi = 0$, touching the line $x = 1$.

Part (iii)

Using the given inequality $(1 - x^2)^{1/2} < (1 - x^3)^{1/3}$ on $(0, 1)$ exactly as before, the $n = 3$ curve is steeper than the $n = 2$ curve at every height, so

$$t_3(x) = \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^3)^{1/3}} < \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^2)^{1/2}} = \arcsin x \qquad (0 < x < 1).$$

So the $n = 3$ curve leaves the origin with gradient 1 like the others and lies strictly above $x = \sin t$ for $t > 0$. Its gradient $(1 - x^3)^{1/3}$ falls to 0 as $x \to 1$, so it also flattens out as it meets the line $x = 1$.



Reaching $x = 1$ before $t = \frac{1}{2}\pi$. The sketch shows the $n = 3$ curve above $\sin t$, which itself reaches 1 at $\frac{1}{2}\pi$; so the $n = 3$ curve must get to 1 no later than that. To see that it is strictly earlier, let $x \to 1$ in the comparison of integrals. The time to reach $x = 1$ is

$$T_3 = \int_0^1 \frac{\mathrm{d}\xi}{(1-\xi^3)^{1/3}}.$$

This improper integral converges: since $1 - \xi^3 = (1-\xi)(1+\xi+\xi^2) \geq 1 - \xi$ on $[0,1]$, the integrand is at most $(1-\xi)^{-1/3}$, whose integral over $[0,1]$ is $\frac{3}{2}$. And since the integrand is strictly less than $(1-\xi^2)^{-1/2}$ on $(0,1)$,

$$T_3 < \int_0^1 \frac{\mathrm{d}\xi}{(1-\xi^2)^{1/2}} = \Big[\arcsin\xi\Big]_0^1 = \tfrac{1}{2}\pi.$$

So the $n = 3$ solution reaches $x = 1$ at a time $T_3 < \frac{1}{2}\pi$, as required. (Once there it stays at $x = 1$, since the gradient is 0 and cannot become negative.)

Checks

- Substituting back: $\frac{\mathrm{d}}{\mathrm{d}t}(1 - \mathrm{e}^{-t}) = \mathrm{e}^{-t} = 1 - x$, and $\frac{\mathrm{d}}{\mathrm{d}t}\sin t = \cos t = (1 - \sin^2 t)^{1/2}$ for $0 \leq t \leq \frac{1}{2}\pi$. A numerical (Runge–Kutta) solution of each equation agrees with these formulae.
- Numerically, $T_3 \approx 1.2092$, comfortably less than $\frac{1}{2}\pi \approx 1.5708$. In fact $T_3 = \dfrac{2\pi}{3\sqrt{3}}$ exactly (see below), which matches to seven decimal places.
- At height $x = 0.9$ the three times are $-\ln 0.1 \approx 2.30$, $\arcsin 0.9 \approx 1.12$ and $t_3(0.9) \approx 0.98$, in the predicted order.

Common slips

- Arguing "the gradient is bigger, so the curve is higher" without noting that the comparison is of gradients *at the same height*, not at the same time. Turning to t as a function of x makes the argument honest.

- Writing $(1-\sin^2 y)^{1/2} = \cos y$ without saying why $\cos y \geq 0$: it needs $0 \leq y \leq \frac{1}{2}\pi$.
- Continuing $x = \sin t$ past $t = \frac{1}{2}\pi$: there it would decrease, but the equation forces $\mathrm{d}x/\mathrm{d}t \geq 0$.

Beyond STEP

The arrival at $x = 1$ in finite time, with the solution then sitting at $x = 1$ for ever, happens because $(1-x^n)^{1/n}$ is not Lipschitz at $x = 1$; this is exactly where the uniqueness theorem for differential equations breaks down (for $n = 2$, both $x = 1$ and $x = \sin t$ pass through $(\frac{1}{2}\pi, 1)$). The arrival time is a Beta function: substituting $u = \xi^n$ gives $T_n = \frac{1}{n}B(\frac{1}{n}, 1-\frac{1}{n}) = \dfrac{\pi/n}{\sin(\pi/n)}$, which is $\frac{1}{2}\pi$ for $n = 2$, $\frac{2\pi}{3\sqrt{3}}$ for $n = 3$, and decreases towards 1 as $n \to \infty$. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness of solutions and comparison of solutions.

WORKED SOLUTION

STEP II 2004, Question 9

Mechanics • Centre of Gravity

STEP II 2004, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q9

The key idea

A hemisphere resting on its curved surface is touching the plane at a point of a sphere, and the normal to a sphere always passes through its centre. So the reaction acts along the vertical through O, the centre of the base, and in equilibrium the centre of gravity of whatever is resting there must lie on that same vertical line. Each part of the question is then just "find the centre of gravity, and tilt until it is directly below O". Coordinates fixed in the body make this painless.

Solution

The model and the equilibrium condition

Treat the hemisphere as a rigid body touching a horizontal plane at a single point C, and the particle as a point mass fixed to the rim. The only forces are the weights and the contact force at C. Since C is on the spherical surface, the normal at C passes through O, and since the plane is horizontal this normal is vertical. So C is directly below O, and O stays at height r however the body rolls.

In equilibrium the contact force must balance the total weight, so it is vertical (there is no friction) and acts along the same line as the weight. Hence:

in equilibrium, the centre of gravity of the system lies on the vertical line through O.

It lies below O, because the body sits on the curved side of the base.

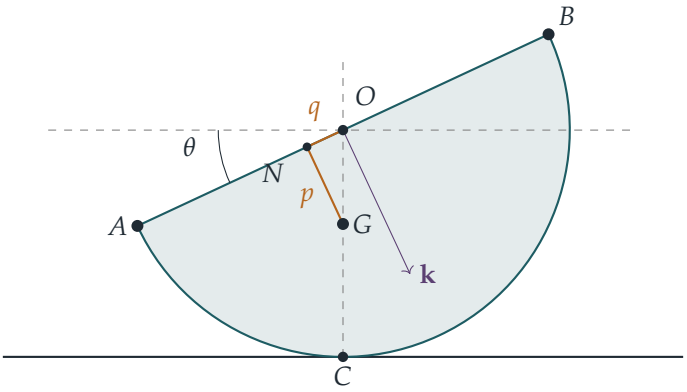
Body coordinates and the initial position

Fix two perpendicular unit vectors in the body: $\mathbf{k}$ along the axis of symmetry, from O towards the curved surface, and $\mathbf{e}$ in the plane of the base. Since G is a distance p from the base and $\sqrt{p^2+q^2}$ from O, its foot N on the base is a distance q from O. Choosing

e to point from O towards N, we get

$$\overrightarrow{OG} = p\,\mathbf{k} + q\,\mathbf{e}.$$

The base is perpendicular to **k**, so the angle the base makes with the horizontal equals the angle between **k** and the vertical. In the initial position $\overrightarrow{OG}$ is vertically downwards, so this angle θ satisfies $\tan\theta = q/p$.



Which rim point is lowest? A point on the rim is $r(\cos\phi\,\mathbf{e} + \sin\phi\,\mathbf{f})$ from O, where **f** is the third unit vector, perpendicular to both. Its depth below O is its component along the downward unit vector $(p\mathbf{k} + q\mathbf{e})/\sqrt{p^2+q^2}$, namely

$$\frac{rq\cos\phi}{\sqrt{p^2+q^2}},$$

which is greatest when $\phi = 0$ (taking $q > 0$; if $q = 0$ the base is horizontal and there is no single lowest point). So

$$\overrightarrow{OA} = r\,\mathbf{e}, \qquad \overrightarrow{OB} = -r\,\mathbf{e}.$$

In words: A is on the same side of the axis as G, in the vertical plane through O and G, and the whole problem stays in that plane.

The particle at A

The centre of gravity G_A of hemisphere and particle is given by

$$(M+m)\,\overrightarrow{OG_A} = M(p\,\mathbf{k} + q\,\mathbf{e}) + mr\,\mathbf{e} = Mp\,\mathbf{k} + (Mq + mr)\,\mathbf{e}.$$

In the new equilibrium $\overrightarrow{OG_A}$ is vertical, so, exactly as for θ, the angle between **k** and the vertical (which is the angle between the base and the horizontal) is the angle between $\overrightarrow{OG_A}$ and **k**:

$$\tan\alpha = \frac{Mq + mr}{Mp}, \qquad \text{so} \qquad \alpha = \tan^{-1}\left(\frac{Mq + mr}{Mp}\right).$$

As we would expect, the extra weight at A pulls A further down, and $\alpha > \theta$.

Ask yourself Why is the base angle the angle between $\overrightarrow{OG_A}$ and the axis?

The base is perpendicular to the axis $\mathbf{k}$, and the horizontal is perpendicular to the vertical. Rotating both lines of a pair through a right angle doesn't change the angle between them, so the angle between base and horizontal equals the angle between $\mathbf{k}$ and the vertical. In equilibrium the vertical through O is the line OG_A, whose components along $\mathbf{k}$ and $\mathbf{e}$ are Mp and $Mq + mr$ (after dividing by $M + m$). The tangent of the angle it makes with $\mathbf{k}$ is the ratio of these.

The particle at B

Now the particle is at $-r\,\mathbf{e}$, so the new centre of gravity G_B satisfies

$$(M+m)\,\overrightarrow{OG_B} = Mp\,\mathbf{k} + (Mq - mr)\,\mathbf{e}, \qquad \text{giving} \qquad \tan\beta = \frac{Mq - mr}{Mp}.$$

Because m is small we have $mr < Mq$, so $\overrightarrow{OG_B}$ is still on the same side of the axis as before and the base tilts the same way, through the smaller angle β.

The angle between the two positions

By the compound-angle formula,

$$\begin{aligned}\tan(\alpha-\beta) &= \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta} = \frac{\dfrac{2mr}{Mp}}{1 + \dfrac{(Mq+mr)(Mq-mr)}{M^2p^2}} \\ &= \frac{2mMrp}{M^2p^2 + M^2q^2 - m^2r^2} = \frac{2mMrp}{M^2\,(p^2+q^2) - m^2r^2},\end{aligned}$$

multiplying top and bottom by M^2p^2 in the third step. This is the required result.

Checks

- If $q = 0$ (for instance a uniform hemisphere) the two positions are mirror images: $\tan\alpha = mr/(Mp)$ and $\beta = -\alpha$. The formula gives $\tan 2\alpha = 2t/(1-t^2)$ with $t = mr/(Mp)$, the double-angle formula, as it should.
- If $m \to 0$, then α and β both tend to θ and the right-hand side tends to 0.
- With $M = 1$, $m = 0.1$, $r = 1$, $p = \frac{3}{8}$, $q = \frac{1}{8}$ we get $\tan\alpha = 0.6$ and $\tan\beta = \frac{1}{15}$, so $\alpha \approx 30.96°$ and $\beta \approx 3.81°$. Both sides of the result equal $\frac{20}{39} \approx 0.5128$. We also found the equilibrium angles numerically by minimising the height of the centre of gravity for 2000 random sets of values, and they agreed with the formulae for $\tan\alpha$ and $\tan\beta$ to within 10^{-13}.

Common slips

- Putting G vertically below the *contact point of the original position* rather than below O. The contact point moves as the body rolls; O is the point that is always directly above it.
- Taking A on the wrong side of the axis. The lowest rim point is on the same side as G, which is why the particle at A *increases* the tilt.
- Treating β as positive when $mr > Mq$. Then the base tilts the other way and the formula holds only with β counted as negative; this is what the condition that m is small rules out.

Beyond STEP

The principle used here is that a rigid body in equilibrium under gravity and a single contact force has its centre of mass on the line of that force. Equivalently, the centre of mass is at a stationary height, and checking that height as the body rolls also decides whether the equilibrium is stable. That is the start of the energy method for stability that first-year mechanics develops for rocking and rolling bodies. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers centres of mass and rigid-body equilibrium in this spirit.

WORKED SOLUTION

STEP II 2004, Question 10

Mechanics • Friction on Inclined Plane

STEP II 2004, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q10

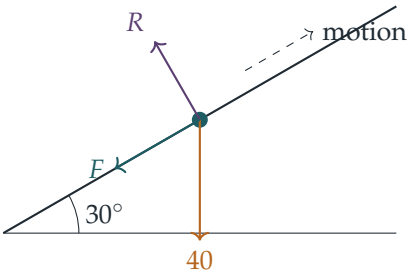
The key idea

Friction always opposes the motion, so a particle sliding up the plane has a different acceleration from one sliding down. Work out the two accelerations first: the numbers are chosen to make them very tidy. Then P goes up and stops, and comes back down to meet Q, which is still on its way up. Writing both positions as "distance below the highest point" makes the collision equation a difference of two squares.

(The printed paper has a misprint in part (i): the "b" in the collision time should be T.)

Solution

We model each particle as a particle on a rough plane, and take positions x measured from A up ℓ, with the positive direction up the slope.



Perpendicular to the plane there is no acceleration, so the normal reaction is

$$R = 4 \cdot 10 \cos 30^\circ = 20\sqrt{3} \text{ N},$$

and since each particle is sliding, friction is limiting:

$$F = \mu R = \frac{1}{5\sqrt{3}} \cdot 20\sqrt{3} = 4 \text{ N}.$$

The weight component down the slope is $40\sin 30^\circ = 20\,\text{N}$.

Moving up: friction acts down the slope, so $4a = -(20+4)$ and the deceleration is $6\,\text{m}\,\text{s}^{-2}$.

Moving down: friction acts up the slope, so the acceleration down the slope is $(20-4)/4 = 4\,\text{m}\,\text{s}^{-2}$.

Since $20 > 4$, a particle that comes to rest does not stay at rest: the limiting friction cannot hold it, so it slides back down.

Part (i)

Motion of P. Going up, $v = 6 - 6t$, so P stops at $t = 1$, after travelling $\frac{1}{2}\cdot 6\cdot 1 = 3\,\text{m}$. For $0 \le t \le 1$, $x_P = 6t - 3t^2 = 3 - 3(1-t)^2$. Afterwards P slides down from rest at $x = 3$ with acceleration 4, so

$$x_P = 3 - 2(t-1)^2 \qquad (t \ge 1),$$

which holds until P returns to A, when $2(t-1)^2 = 3$, that is $t = 1 + \sqrt{3/2}$.

Motion of Q. Exactly the same motion shifted by T: with $u = t - T$,

$$x_Q = 6u - 3u^2 = 3 - 3(1-u)^2 \qquad (0 \le u \le 1).$$

Ask yourself Could they collide while both are still going up?

No. While both are going up, Q follows exactly the path P followed T seconds earlier, and x is increasing along that path, so P is strictly ahead. They can only meet once P has turned round, that is for $t \ge 1$. Also Q cannot reach its own highest point ($x = 3$, at $u = 1$) first, because P has already left $x = 3$ by then and is below it. So at the collision P is descending and Q ascending.

The collision. Write $\tau = t - 1 \ge 0$ for the time since P turned. Then $1 - u = 1 - (t - T) = T - \tau$, and $x_P = x_Q$ gives

$$2\tau^2 = 3(T-\tau)^2.$$

Both τ and $T - \tau = 1 - u$ are non-negative (the second because Q is still going up), so we may take positive square roots:

$$\sqrt{2}\,\tau = \sqrt{3}\,(T-\tau)$$

$$\tau = \frac{\sqrt{3}\,T}{\sqrt{3}+\sqrt{2}} = \sqrt{3}(\sqrt{3}-\sqrt{2})\,T = (3-\sqrt{6})\,T,$$

rationalising with $(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2}) = 1$. Hence the collision is at

$$t = (3-\sqrt{6})\,T + 1$$

seconds after P is projected, as required.

It remains to see where the condition on T is used. This time must come before P gets back to A, so we need $\tau < \sqrt{3/2}$:

$$(3-\sqrt{6})\,T < \sqrt{\tfrac{3}{2}} \iff T < \frac{\sqrt{3/2}\,(3+\sqrt{6})}{3} = \sqrt{\tfrac{3}{2}} + \tfrac{\sqrt{9}}{3} = 1 + \sqrt{\tfrac{3}{2}},$$

since $\sqrt{3/2}\cdot\sqrt{6} = 3$. This is exactly the given condition. (The same inequality also guarantees $t \geq T$, so that Q has been launched: $(3-\sqrt{6})T + 1 \geq T \iff T \leq 1/(\sqrt{6}-2) = 1+\sqrt{3/2}$.)

Part (ii)

With $T = 1+\sqrt{2/3}$,

$$\tau = (3-\sqrt{6})\left(1+\sqrt{\tfrac{2}{3}}\right) = 3 - \sqrt{6} + 3\sqrt{\tfrac{2}{3}} - \sqrt{6}\sqrt{\tfrac{2}{3}} = 3 - \sqrt{6} + \sqrt{6} - 2 = 1,$$

so the collision is at $t = 2$. This value of T is less than $1+\sqrt{3/2}$, so part (i) applies.

The energy lost to friction is the friction force times the total distance slid, because friction has constant magnitude 4 N and always opposes the motion.

- P slides 3 m up and then $2\tau^2 = 2$ m down: 5 m in all.
- Q has not turned, and is at the same point as P, namely $x = 3-2 = 1$: it has slid 1 m.

So the energy lost is

$$4 \times (5+1) = 24 \text{ J}.$$

Checks

An energy balance confirms this. Initially the kinetic energy is $2 \times \frac{1}{2}\cdot 4 \cdot 6^2 = 144$ J. At $t = 2$, P has speed $4 \times 1 = 4$ (KE 32 J); Q has $u = 1-\sqrt{2/3}$ and speed $6(1-u) = 6\sqrt{2/3}$, so $v^2 = 24$ (KE 48 J). Both are 1 m up the slope, $\frac{1}{2}$ m higher, a gain of $2 \times 4 \cdot 10 \cdot \frac{1}{2} = 40$ J of potential energy. And $144 - 32 - 48 - 40 = 24$ J. A numerical step-by-step simulation of both particles for random $T < 1+\sqrt{3/2}$ gave collision times matching $(3-\sqrt{6})T+1$ (for example ≈ 2.000 when $T \approx 1.816$), with 24.000 J lost to friction.

Common slips

- Using the same acceleration for P on the way down as on the way up: friction reverses direction, so it is 4, not 6.
- Forgetting the friction on Q: Q has been sliding too, and its 1 m contributes 4 J.
- Taking the wrong square root in $2\tau^2 = 3(T-\tau)^2$: the sign is fixed by $\tau \geq 0$ and $T - \tau \geq 0$.

Beyond STEP

Friction is the simplest example of a non-conservative force: the work it does depends on the path, not just the endpoints, which is why we needed the total distance slid rather than the displacement. This is why mechanical energy is conserved for some systems and not others, an idea made precise with line integrals and potentials in a first-year mechanics course. Kleppner and Kolenkow's *An Introduction to Mechanics* treats work, conservative forces and friction carefully.

WORKED SOLUTION

STEP II 2004, Question 11

Mechanics • Power and Resistance

STEP II 2004, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

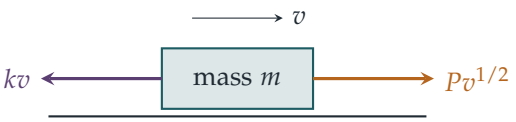
step.maths.org/questions/04-s2-q11

The key idea

Power is force times speed, so an engine working at full power $Pv^{3/2}$ pushes with a force $Pv^{1/2}$. Newton's second law then gives a differential equation with a $\sqrt{v}$ in it, and the substitution $u = \sqrt{v}$ turns it into a linear first-order equation that we can solve at once. After the engines are switched off, each speed decays exponentially, and the trick in part (iii) is to compare the two decay rates.

Solution

We model each train as a particle moving along a straight horizontal track. While the engine is on at full power, the forces along the track are the driving force and the resistance kv.



If an engine develops power H at speed v, its driving force is H/v. So at full power the driving force on A is $Pv^{3/2}/v = Pv^{1/2}$, and on B it is $2Pv^{1/2}$.

Part (i)

For A, Newton's second law along the track gives

$$m\frac{\mathrm{d}v_A}{\mathrm{d}t} = Pv_A^{1/2} - kv_A.$$

Put $u = v_A^{1/2}$, so that $v_A = u^2$ and $\dfrac{\mathrm{d}v_A}{\mathrm{d}t} = 2u\dfrac{\mathrm{d}u}{\mathrm{d}t}$. Then

$$2mu\frac{\mathrm{d}u}{\mathrm{d}t} = Pu - ku^2, \qquad \text{so} \qquad 2m\frac{\mathrm{d}u}{\mathrm{d}t} = P - ku,$$

dividing by u, which is allowed because the train is always moving ($u > 0$). This is linear (and separable):

$$\int \frac{2m\,\mathrm{d}u}{P-ku} = \int \mathrm{d}t \quad \Longrightarrow \quad -\frac{2m}{k}\ln(P-ku) = t + \text{const},$$

so $P - ku = Ce^{-kt/2m}$ for some constant C. (Here $P - ku > 0$ at the start, when u is tiny, and it stays positive because the solution never reaches the equilibrium $u = P/k$.)

Ask yourself What does "crawling along at very low speeds" tell us?

It gives the initial condition. The speeds at $t = 0$ are positive (so dividing by u was fine) but negligibly small, so we take $u = 0$ at $t = 0$. Then $C = P$, and

$$u = \frac{P}{k}\left(1 - e^{-kt/2m}\right), \qquad \text{so} \qquad v_A = \frac{P^2\left(1 - e^{-kt/2m}\right)^2}{k^2},$$

as required. (If the train were exactly at rest, $v \equiv 0$ would also satisfy the original equation, since there would be no driving force at all; this is why the question has the trains already moving slowly.)

For B the equation is $2m\dfrac{\mathrm{d}v_B}{\mathrm{d}t} = 2Pv_B^{1/2} - kv_B$. The same substitution gives $4m\dfrac{\mathrm{d}u}{\mathrm{d}t} = 2P - ku$, which is the equation for A with P replaced by $2P$ and m by $2m$. So

$$v_B = \frac{4P^2\left(1 - e^{-kt/4m}\right)^2}{k^2}.$$

Part (ii)

Both speeds are positive, so $9v_A = 4v_B$ is equivalent to $3\sqrt{v_A} = 2\sqrt{v_B}$. Write $x = e^{-kt/4m}$, so that $e^{-kt/2m} = x^2$, and $0 < x < 1$ for $t > 0$. The condition becomes

$$\frac{3P}{k}\left(1 - x^2\right) = \frac{4P}{k}(1 - x) \quad \Longrightarrow \quad 3(1-x)(1+x) = 4(1-x).$$

Since $x \neq 1$ we may divide by $1 - x$, giving $3 + 3x = 4$, so $x = \frac{1}{3}$ (that is, $t = \frac{4m}{k}\ln 3$). Then $1 - x^2 = \frac{8}{9}$ and $1 - x = \frac{2}{3}$, so

$$v_A = \frac{64P^2}{81k^2}, \qquad v_B = \frac{16P^2}{9k^2}.$$

As a quick check, $9v_A = \frac{64P^2}{9k^2} = 4v_B$.

Part (iii)

Measure time τ from the moment the engines are switched off, and let $V_A = \frac{64P^2}{81k^2}$ and $V_B = \frac{16P^2}{9k^2}$ be the speeds at that moment. Now only resistance acts:

$$m\frac{\mathrm{d}v_A}{\mathrm{d}\tau} = -kv_A, \qquad 2m\frac{\mathrm{d}v_B}{\mathrm{d}\tau} = -kv_B,$$

so

$$v_A = V_A\,\mathrm{e}^{-k\tau/m}, \qquad v_B = V_B\,\mathrm{e}^{-k\tau/2m}.$$

The heavier train B loses speed at half the exponential rate, so squaring v_B matches the exponentials:

$$v_B^2 = V_B^2\,\mathrm{e}^{-k\tau/m} = \frac{V_B^2}{V_A}\,v_A.$$

Finally

$$\frac{V_B^2}{V_A} = \frac{256P^4}{81k^4}\cdot\frac{81k^2}{64P^2} = \frac{4P^2}{k^2},$$

so $v_B^2 = \dfrac{4P^2}{k^2}v_A$, that is, $k^2v_B^2 = 4P^2v_A$ for all later times, as required.

Checks

- As $t\to\infty$, $v_A \to P^2/k^2$, which is where the driving force $Pv^{1/2}$ equals the resistance kv. Similarly $v_B \to 4P^2/k^2$, where $2Pv^{1/2} = kv$.
- For small t, $v_A \approx P^2t^2/(4m^2)$ and $v_B \approx P^2t^2/(4m^2)$: the trains start off together, as they should, since each has driving force per unit mass $Pv^{1/2}/m$.
- With $m = 1.7$, $P = 2.3$, $k = 0.9$ we stepped both equations of motion numerically (Runge–Kutta) from a tiny initial speed and matched the formulae of part (i) to about 10^{-5}; the moment $9v_A = 4v_B$ gave $\mathrm{e}^{-kt/4m} = 0.33333$, $v_A = 5.1602$ and $v_B = 11.6104$, and stepping on with the engines off kept $k^2v_B^2$ equal to $4P^2v_A$ (for example both 37.8739 two seconds later).

Common slips

- Using $Pv^{3/2}$ as the force. It is a power; the force is the power divided by the speed.
- Forgetting that the condition in part (ii) is a quadratic in $x = \mathrm{e}^{-kt/4m}$ with the root $x = 1$ (that is, $t = 0$, when both speeds are zero) as well as $x = \frac{1}{3}$.

Beyond STEP

The substitution $u = \sqrt{v}$ is a special case of the method for Bernoulli equations, $y' + p(t)y = q(t)y^n$, where $u = y^{1-n}$ always gives a linear equation. The remark about the train exactly at rest is a first glimpse of uniqueness theory: because $\sqrt{v}$ has infinite slope at $v = 0$, the equation has more than one solution from $v = 0$, which the Picard–Lindelöf theorem explains. Boyce and DiPrima, *Elementary Differential Equations*, covers both; for the mechanics of power and resistance, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2004, Question 12

Probability and Statistics • Continuous Distributions

STEP II 2004, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q12

The key idea

Everything flows from one antiderivative: xe^{-ax^2} is, up to a constant, the derivative of e^{-ax^2}. That gives the normalising constant and a closed form for the cumulative distribution function, so the median comes from solving one exponential equation. The mode is where the curve peaks, provided that peak lies inside $[0,1]$; otherwise the density is still climbing at $x=1$. The conditional probability then needs only values of the cumulative distribution function at m and h.

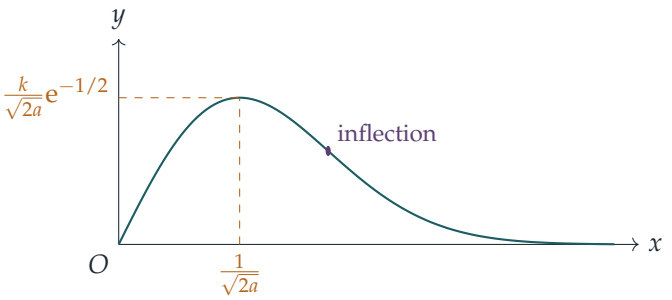
Solution

The sketch

Let $g(x) = kxe^{-ax^2}$ for $x \geq 0$. Then $g(0) = 0$, $g(x) > 0$ for $x > 0$, and by the product rule

$$g'(x) = ke^{-ax^2}(1 - 2ax^2),$$

which is positive for $0 \leq x < 1/\sqrt{2a}$ and negative afterwards. So g rises from the origin (with gradient k there) to a single maximum at $x = 1/\sqrt{2a}$, where $y = \dfrac{k}{\sqrt{2a}}e^{-1/2}$, and then decreases. As $x \to \infty$ the exponential beats the linear factor, so $y \to 0$: the x-axis is an asymptote. (Differentiating again shows a point of inflection at $x = \sqrt{3/(2a)}$, where the curve switches from concave to convex.)



The constant k

Since $\dfrac{\mathrm{d}}{\mathrm{d}x}e^{-ax^2} = -2axe^{-ax^2}$, for $0 \le t \le 1$,

$$\int_0^t kxe^{-ax^2}\,\mathrm{d}x = \left[-\frac{k}{2a}e^{-ax^2}\right]_0^t = \frac{k}{2a}\left(1-e^{-at^2}\right).$$

The total probability is 1, so taking $t = 1$ gives $\dfrac{k}{2a}(1-e^{-a}) = 1$, that is,

$$k = \frac{2a}{1-e^{-a}},$$

as required. (The denominator is positive because $a > 0$.) Substituting this back, the cumulative distribution function on $[0,1]$ is

$$F(t) = P(X \le t) = \frac{1-e^{-at^2}}{1-e^{-a}}.$$

The mode

The density is the curve we sketched, cut off at $x = 1$. From the sketch, on $[0,1]$ it is largest at the turning point if that lies in the interval, and otherwise at the right-hand end.

- If $a > \frac{1}{2}$, then $2a > 1$, so $1/\sqrt{2a} < 1$: the peak is inside the interval and $m = \dfrac{1}{\sqrt{2a}}$.
- If $a < \frac{1}{2}$, then $1/\sqrt{2a} > 1$, so $f'(x) > 0$ throughout $[0,1)$: the density is increasing on the whole interval and $m = 1$.

(When $a = \frac{1}{2}$ both formulae give $m = 1$.)

The median

The median h satisfies $F(h) = \frac{1}{2}$:

$$1-e^{-ah^2} = \tfrac{1}{2}\left(1-e^{-a}\right) \quad\Longleftrightarrow\quad e^{-ah^2} = \frac{1+e^{-a}}{2}.$$

Taking logarithms and using $h > 0$,

$$h = \sqrt{\frac{1}{a}\ln\left(\frac{2}{1+\mathrm{e}^{-a}}\right)}.$$

This is real since $1+\mathrm{e}^{-a} < 2$, so the logarithm is positive.

When is $h > m$?

If $a \le \frac{1}{2}$ then $m = 1$, and $h < 1$ (half the probability lies below h, and none above 1), so $h < m$. So $h > m$ can happen only when $a > \frac{1}{2}$, and then $m = 1/\sqrt{2a}$. Comparing squares (both are positive),

$$\begin{aligned} h > m &\iff \frac{1}{a}\ln\left(\frac{2}{1+\mathrm{e}^{-a}}\right) > \frac{1}{2a} \iff \ln\left(\frac{2}{1+\mathrm{e}^{-a}}\right) > \frac{1}{2} \\ &\iff \frac{2}{1+\mathrm{e}^{-a}} > \mathrm{e}^{1/2} \iff 1+\mathrm{e}^{-a} < 2\mathrm{e}^{-1/2} \\ &\iff \mathrm{e}^{-a} < 2\mathrm{e}^{-1/2} - 1 \iff a > -\ln\bigl(2\mathrm{e}^{-1/2}-1\bigr). \end{aligned}$$

We multiplied by $a > 0$ in the first step, used that exp is increasing in the third and that ln is increasing in the last. The last step also needs $2\mathrm{e}^{-1/2} - 1 > 0$, which holds because $\mathrm{e}^{1/2} < 2$ (as $\mathrm{e} < 4$).

> **Ask yourself** The chain above assumed $a > \frac{1}{2}$. Doesn't the result need that as an extra condition?

No, and that's why the next part asks for it. We show below that $c = -\ln\bigl(2\mathrm{e}^{-1/2}-1\bigr)$ exceeds $\frac{1}{2}$. So if $a > c$ then automatically $a > \frac{1}{2}$, the formula $m = 1/\sqrt{2a}$ applies, and the chain shows $h > m$. Together with the case $a \le \frac{1}{2}$, this shows that $h > m$ exactly when $a > c$.

The constant exceeds $\frac{1}{2}$

Since ln is increasing,

$$-\ln\bigl(2\mathrm{e}^{-1/2}-1\bigr) > \tfrac{1}{2} \iff 2\mathrm{e}^{-1/2} - 1 < \mathrm{e}^{-1/2} \iff \mathrm{e}^{-1/2} < 1,$$

and the last inequality is true. (Numerically, $c \approx 1.546$.)

The conditional probability

Suppose $a > c$. Then $m < h$, so the event $\{X > m\} \cap \{X < h\}$ is the event $\{m < X < h\}$. Since $P(X < h) = \frac{1}{2}$,

$$P(X > m \mid X < h) = \frac{P(m < X < h)}{P(X < h)} = \frac{F(h) - F(m)}{\frac{1}{2}} = 1 - 2F(m).$$

With $m = 1/\sqrt{2a}$ we have $am^2 = \frac{1}{2}$, so $F(m) = \dfrac{1 - \mathrm{e}^{-1/2}}{1 - \mathrm{e}^{-a}}$. Hence

$$P(X > m \mid X < h) = \frac{(1 - \mathrm{e}^{-a}) - 2(1 - \mathrm{e}^{-1/2})}{1 - \mathrm{e}^{-a}} = \frac{2\mathrm{e}^{-1/2} - \mathrm{e}^{-a} - 1}{1 - \mathrm{e}^{-a}},$$

as required.

Checks

- At $a = c$ the numerator $2\mathrm{e}^{-1/2} - \mathrm{e}^{-a} - 1$ is exactly 0, matching $m = h$ there; it is positive for $a > c$, as a probability should be.
- As $a \to \infty$ the answer tends to $2\mathrm{e}^{-1/2} - 1 \approx 0.213$, which is less than 1.
- With $a = 2.5$: $k \approx 5.447$, $m \approx 0.447$, $h \approx 0.496$, and numerical integration of the density gives $P(X > m \mid X < h) \approx 0.1427$, agreeing with the formula. For $a = 1$ we get $m \approx 0.707 > h \approx 0.616$, consistent with $1 < c$.

Common slips

- Giving $m = 1/\sqrt{2a}$ in every case: when $a < \frac{1}{2}$ the turning point lies beyond $x = 1$, where the density is zero.
- Forgetting that the k inside F must be the normalised value, so that $F(1) = 1$.
- Writing $P(X > m \mid X < h) = P(X > m)/P(X < h)$; the numerator must be the probability of *both* events.

Beyond STEP

This density is a truncated Rayleigh distribution: without the cut-off at $x = 1$, $kx\mathrm{e}^{-ax^2}$ with $k = 2a$ is the distribution of the distance from the origin of a point whose two coordinates are independent normal variables. Comparing mean, median and mode, and how truncation shifts them, is a recurring theme in a first statistics course. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read more about continuous random variables and their density and distribution functions.

WORKED SOLUTION

STEP II 2004, Question 13

Probability and Statistics • Optimal Stopping

STEP II 2004, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q13

The key idea

Think one draw at a time. If she has safely collected n white balls, one more draw either adds £1 or wipes out the £n she is holding, so its expected effect is easy to write down. The total expected reward for a target of N balls goes up exactly when the expected gain of the extra draw is positive, so the best target is the last one at which that gain is still non-negative.

Solution

Throughout we assume $1 \le r < b$, so there is at least one red and at least one white ball. (With $r = 0$ she simply takes every ball.)

The expected gain on the next draw

Suppose her first n draws were all white. The bag now holds $b - n$ balls: r red and $b - n - r$ white, each equally likely to come out next (we need $n \le b - r - 1$ for there to be a white ball left; in any case $b - n \ge r \ge 1$).

- With probability $\dfrac{b-n-r}{b-n}$ the ball is white, and her reward goes from £n to £$(n+1)$: a gain of $+1$.
- With probability $\dfrac{r}{b-n}$ the ball is red, and she loses the £n she would otherwise have had: a gain of $-n$.

So her expected gain from the next draw is

$$G_n = \frac{(b-n-r)\cdot 1 + r\cdot(-n)}{b-n} = \frac{b-r-n(r+1)}{b-n}.$$

The denominator is positive, so $G_n = 0$ exactly when $b - r = n(r+1)$, that is when

$$n = \frac{b-r}{r+1},$$

as required. More usefully, $G_n > 0$ when $n < \dfrac{b-r}{r+1}$ and $G_n < 0$ when $n > \dfrac{b-r}{r+1}$.

The best number of balls

Let P_N be the probability that the first N draws are all white, and let E_N be her expected total reward if she aims to remove N balls. She gets £N if those N balls are all white and nothing otherwise, so

$$E_N = NP_N.$$

Ask yourself How does the expected gain on one draw relate to the total expected reward E_N?

Compare aiming for N with aiming for $N+1$. The two plans differ only if the first N balls are all white (otherwise she has already lost under both), which happens with probability P_N; and in that case the extra draw changes her reward by an amount with expected value G_N. Hence

$$E_{N+1} - E_N = P_N\, G_N = P_N\, \frac{b-r-N(r+1)}{b-N}.$$

(One can also check this directly from $P_{N+1} = P_N \cdot \frac{b-r-N}{b-N}$.)

For $N \le b-r$ we have $P_N > 0$, so $E_{N+1} - E_N$ has the sign of $b-r-N(r+1)$. Writing $k = N+1$, this is the sign of

$$(b+1) - k(r+1).$$

Put $m = \dfrac{b+1}{r+1}$. Then $E_k > E_{k-1}$ when $k < m$, $E_k = E_{k-1}$ when $k = m$, and $E_k < E_{k-1}$ when $k > m$. So the sequence $E_0, E_1, E_2, \ldots$ increases strictly up to $E_{\lfloor m \rfloor}$ (with a tie $E_{m-1} = E_m$ if m is an integer) and then decreases strictly, at least while $k - 1 \le b - r$. Beyond that $P_N = 0$ and $E_N = 0$, which is no better.

We should check that $\lfloor m \rfloor$ is a sensible target, i.e. $\lfloor m \rfloor \le b-r$. If $b = r+1$ then $m = \frac{r+2}{r+1}$, so $\lfloor m \rfloor = 1 = b-r$. If $b \ge r+2$ then

$$(b-r)(r+1) - (b+1) = r(b-r-1) - 1 \ge r - 1 \ge 0,$$

so $m \le b-r$. Either way the maximum of E_N is attained at

$$n = \left\lfloor \frac{b+1}{r+1} \right\rfloor,$$

the integer part of $\frac{b+1}{r+1}$, as required.

This target is also the best she can do with any stopping rule: the only thing she ever learns before deciding whether to carry on is that every ball so far was white (if not, the game is over), so every rule amounts to choosing a number N in advance.

The case $r = 1$

With one red ball, the first N draws avoid it exactly when the red ball is in one of the last $b - N$ positions, all positions being equally likely. So

$$P_N = \frac{b-N}{b}, \qquad E_N = \frac{N(b-N)}{b}.$$

The best target is $n = \left\lfloor \frac{b+1}{2} \right\rfloor$.

b even. Then $\frac{b+1}{2} = \frac{b}{2} + \frac{1}{2}$, so $n = \frac{b}{2}$ and

$$E_n = \frac{\frac{b}{2} \cdot \frac{b}{2}}{b} = \frac{b}{4}.$$

Her expected total reward is £$\frac{1}{4}b$, as required.

b odd. Then $n = \frac{b+1}{2}$, so $b - n = \frac{b-1}{2}$ and

$$E_n = \frac{\frac{b+1}{2} \cdot \frac{b-1}{2}}{b} = \frac{b^2-1}{4b}.$$

Her expected total reward is £$\frac{b^2-1}{4b}$. (Here $m = \frac{b+1}{2}$ is an integer, so aiming for $\frac{b-1}{2}$ balls gives the same value, as the symmetric formula $N(b-N)/b$ shows.)

Checks

- For $r = 1$, $E_N = N(b-N)/b$ is a downward parabola in N with vertex at $N = b/2$, which matches both answers: exactly $b/4$ at the vertex when b is even, and slightly less, $\frac{b}{4} - \frac{1}{4b}$, when b is odd.
- Take $b = 7$, $r = 2$: the formula gives $n = \lfloor 8/3 \rfloor = 2$, and direct computation gives $E_0, \ldots, E_5 = 0,\ \frac{5}{7},\ \frac{20}{21},\ \frac{6}{7},\ \frac{4}{7},\ \frac{5}{21}$, peaking at $N = 2$. We confirmed this by enumerating all 7! orderings of the balls, and checked the formula for n against a full optimal-stopping calculation for every $b \le 24$.
- For $b = 9$, $r = 1$ the answer is $\frac{80}{36} = \frac{20}{9} \approx 2.22$, just under $\frac{9}{4}$.

Common slips

- Forgetting that a red ball costs her the £n already collected, and writing the expected gain as just $\frac{b-n-r}{b-n}$.
- Confusing "n balls already drawn" with "target of n balls": the gain G_n is zero at $n = \frac{b-r}{r+1}$, which is why the best *target* is one more, $\frac{b-r}{r+1} + 1 = \frac{b+1}{r+1}$.
- Missing the tie when $\frac{b+1}{r+1}$ is an integer: then both n and $n-1$ are optimal.

Beyond STEP

This is a small *optimal stopping* problem, and the argument is the "one-step look-ahead" rule: stop as soon as one more step has negative expected gain. That rule is optimal whenever the problem is *monotone*, meaning that once continuing looks bad it stays bad, as here, where G_n decreases in n. The same ideas lead to the secretary problem and to martingale methods in first-year and second-year probability. Grimmett and Welsh, *Probability: An Introduction*, is a good place to build the conditional-expectation tools behind it.

WORKED SOLUTION

STEP II 2004, Question 14

Probability and Statistics • Inclusion-Exclusion Principle

STEP II 2004, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q14

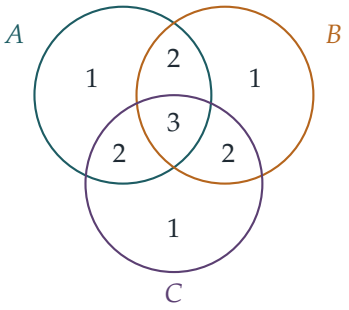
The key idea

Inclusion–exclusion is a counting correction: adding $\mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C)$ counts overlaps too often, and the pairwise and triple terms put the count right. For the puddings it is much easier to work with the *complement*: let E_i be the event that pudding i is empty. The probability that *some* pudding is empty is a union, which is exactly what inclusion–exclusion handles, and each intersection of the E_i has a very simple probability.

Solution

The inclusion–exclusion formula

Split $A \cup B \cup C$ into the seven disjoint regions of the Venn diagram. The probability of any event made of whole regions is the sum of the probabilities of its regions, so it is enough to check that each region is counted exactly once on the right-hand side.



The numbers in the diagram show how many of A, B, C each region lies in. Take a region lying in exactly k of the three events, where $k = 1$, 2 or 3.

- It appears in k of the terms $\mathrm{P}(A), \mathrm{P}(B), \mathrm{P}(C)$.

- It appears in $\binom{k}{2}$ of the pairwise intersections, because it lies in $A \cap B$, say, exactly when it lies in both A and B.
- It appears in the triple intersection only if $k = 3$, that is, $\binom{k}{3}$ times.

So the net number of times it is counted is

$$k - \binom{k}{2} + \binom{k}{3} = \begin{cases} 1 - 0 + 0 = 1 & (k = 1), \\ 2 - 1 + 0 = 1 & (k = 2), \\ 3 - 3 + 1 = 1 & (k = 3). \end{cases}$$

Every region of $A \cup B \cup C$ is counted exactly once, and regions outside the union are not counted at all. So the right-hand side equals $\mathrm{P}(A \cup B \cup C)$.

(Equivalently, apply the two-event rule $\mathrm{P}(X \cup Y) = \mathrm{P}(X) + \mathrm{P}(Y) - \mathrm{P}(X \cap Y)$ to $X = A \cup B$ and $Y = C$, then to $(A \cap C) \cup (B \cap C)$.)

Each pudding gets at least one sixpence

Model. Stirring thoroughly and dividing into equal portions means that each sixpence lands in pudding 1, 2 or 3 with probability $\frac{1}{3}$ each, independently of the other sixpences. Take $r \geq 1$.

Let E_i be the event that pudding i contains no sixpence.

- E_1 happens when every one of the r sixpences avoids pudding 1. Each does so with probability $\frac{2}{3}$, so by independence $\mathrm{P}(E_1) = \left(\frac{2}{3}\right)^r$, and likewise for E_2, E_3.
- $E_1 \cap E_2$ happens when every sixpence goes into pudding 3, so $\mathrm{P}(E_1 \cap E_2) = \left(\frac{1}{3}\right)^r$, and likewise for the other pairs.
- $E_1 \cap E_2 \cap E_3$ would need all three puddings empty, which is impossible since $r \geq 1$. So its probability is 0.

By inclusion–exclusion, the probability that at least one pudding is empty is

$$p_r = \mathrm{P}(E_1 \cup E_2 \cup E_3) = 3\left(\tfrac{2}{3}\right)^r - 3\left(\tfrac{1}{3}\right)^r.$$

The probability that each pudding contains at least one sixpence is the complement:

$$1 - 3\left(\tfrac{2}{3}\right)^r + 3\left(\tfrac{1}{3}\right)^r = \frac{3^r - 3 \cdot 2^r + 3}{3^r}.$$

At least six sixpences

We need the values of r for which $p_r < \frac{1}{3}$. Direct calculation gives

$$p_5 = \frac{3 \cdot 32 - 3}{243} = \frac{93}{243} = \frac{31}{81} > \frac{27}{81} = \frac{1}{3}, \qquad p_6 = \frac{3 \cdot 64 - 3}{729} = \frac{189}{729} = \frac{7}{27} < \frac{9}{27} = \frac{1}{3}.$$

To be sure that no smaller r works, we show that p_r decreases. For $r \geq 1$,

$$p_{r+1} - p_r = 3\left(\tfrac{2}{3}\right)^r\left(\tfrac{2}{3} - 1\right) - 3\left(\tfrac{1}{3}\right)^r\left(\tfrac{1}{3} - 1\right) = -\left(\tfrac{2}{3}\right)^r + 2\left(\tfrac{1}{3}\right)^r = \left(\tfrac{1}{3}\right)^r(2 - 2^r),$$

which is 0 for $r = 1$ and negative for $r \geq 2$. So $p_1 = p_2 \geq p_3 \geq p_4 \geq p_5 = \frac{31}{81}$, and every $r \leq 5$ has $p_r > \frac{1}{3}$. (With $r = 0$ every pudding is certainly empty.) Hence the chance that some pudding has no sixpence is below $\frac{1}{3}$ only if $r \geq 6$; and since p_r keeps decreasing, every $r \geq 6$ does give $p_r \leq p_6 < \frac{1}{3}$. So the cook must use 6 or more sixpences, as required.

Two sixpences in each pudding

Now $r = 6$. Let T be the event that each pudding contains exactly two sixpences, and S the event that each pudding contains at least one. From the previous part,

$$\mathrm{P}(S) = 1 - p_6 = 1 - \frac{7}{27} = \frac{20}{27}.$$

There are $3^6 = 729$ equally likely ways to send the six (distinguishable) sixpences to the puddings. The number giving two to each pudding is the number of ways to split six objects into three labelled pairs, $\dfrac{6!}{2!\,2!\,2!} = 90$. So $\mathrm{P}(T) = \frac{90}{729} = \frac{10}{81}$.

Ask yourself Why is $\mathrm{P}(T \cap S)$ just $\mathrm{P}(T)$?

If every pudding has two sixpences then certainly every pudding has at least one, so $T \subseteq S$ and $T \cap S = T$. Therefore

$$\mathrm{P}(T \mid S) = \frac{\mathrm{P}(T)}{\mathrm{P}(S)} = \frac{10/81}{20/27} = \frac{10}{81} \cdot \frac{27}{20} = \frac{1}{6}.$$

Checks

For $r = 1$ and $r = 2$ the formula gives a probability 0 that every pudding has a sixpence, which is right: you can't fill three puddings with fewer than three coins. For $r = 3$ it gives $\frac{27-24+3}{27} = \frac{2}{9}$, which matches the direct count $3!/3^3 = 6/27$. An exact enumeration of all 3^r assignments for $r \leq 9$ agrees with the formula, confirms $p_5 = \frac{31}{81} \approx 0.383$ and $p_6 = \frac{7}{27} \approx 0.259$, and gives the final answer $\frac{1}{6} \approx 0.167$ (a simulation of 200 000 puddings gave 0.166).

Common slips

- Treating the sixpences as indistinguishable and counting the 28 ways of writing $6 = a + b + c$ as equally likely. They aren't: $(2, 2, 2)$ can happen in 90 ways, while $(6, 0, 0)$ can happen in only one.
- Checking only $r = 6$ and $r = 5$. "Must use 6 or more" also needs every smaller r to fail, which is why the monotonicity step matters.
- Forgetting that the triple intersection is empty, or giving it probability $(0)^r$ without comment.

Beyond STEP

The general inclusion–exclusion principle for n events, with alternating sums of all k-fold intersections, is proved by exactly the counting argument above, using $\sum_{j\geq 1}(-1)^{j-1}\binom{k}{j} = 1$. Applied to n puddings it counts *surjections*, and connects to Stirling numbers of the second kind and to the classic coupon-collector problem. Grimmett and Welsh, *Probability: An Introduction*, develops inclusion–exclusion and these occupancy problems.

WORKED SOLUTION

STEP III 2004, Question 1

Pure • Hyperbolic Integrals

STEP III 2004, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q1

The key idea

Both integrands are a hyperbolic function times something built from the *other* hyperbolic function, and $\sinh x\,\mathrm{d}x$ and $\cosh x\,\mathrm{d}x$ are exactly the differentials of $\cosh x$ and $\sinh x$. So the substitutions $c = \cosh x$ and $s = \sinh x$ turn them into rational integrals: one needing partial fractions, the other an arctangent. The crucial observation for the "hence" is that the two denominators are the same function, since $2\cosh^2 x - 1 = 1 + 2\sinh^2 x$.

Solution

The first integral

Put $c = \cosh x$, so $\mathrm{d}c = \sinh x\,\mathrm{d}x$; the limits $x = 0$ and $x = a$ become $c = 1$ and $c = \cosh a$. Then

$$\int_0^a \frac{\sinh x}{2\cosh^2 x - 1}\,\mathrm{d}x = \int_1^{\cosh a} \frac{\mathrm{d}c}{2c^2 - 1}.$$

The denominator factorises as $(\sqrt{2}\,c - 1)(\sqrt{2}\,c + 1)$, and

$$\frac{1}{(\sqrt{2}\,c - 1)(\sqrt{2}\,c + 1)} = \frac{1}{2}\left(\frac{1}{\sqrt{2}\,c - 1} - \frac{1}{\sqrt{2}\,c + 1}\right),$$

as you can confirm by recombining: the numerator is $\frac{1}{2}\big((\sqrt{2}c + 1) - (\sqrt{2}c - 1)\big) = 1$. Each term integrates to a logarithm with an extra factor $1/\sqrt{2}$ from the chain rule. On $c \geq 1$ both $\sqrt{2}c \pm 1$ are positive, so no modulus signs are needed:

$$\begin{aligned}\int_1^{\cosh a} \frac{\mathrm{d}c}{2c^2 - 1} &= \frac{1}{2\sqrt{2}}\Big[\ln(\sqrt{2}\,c - 1) - \ln(\sqrt{2}\,c + 1)\Big]_1^{\cosh a} \\ &= \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}\cosh a - 1}{\sqrt{2}\cosh a + 1}\right) - \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2} - 1}{\sqrt{2} + 1}\right).\end{aligned}$$

Since $-\ln\frac{\sqrt{2}-1}{\sqrt{2}+1} = \ln\frac{\sqrt{2}+1}{\sqrt{2}-1}$, this is exactly the required result.

The second integral

Now put $s = \sinh x$, so $\mathrm{d}s = \cosh x\,\mathrm{d}x$, with limits $s = 0$ and $s = \sinh a$:

$$\int_0^a \frac{\cosh x}{1+2\sinh^2 x}\,\mathrm{d}x = \int_0^{\sinh a} \frac{\mathrm{d}s}{1+(\sqrt{2}\,s)^2} = \frac{1}{\sqrt{2}}\Big[\arctan(\sqrt{2}\,s)\Big]_0^{\sinh a}$$
$$= \frac{1}{\sqrt{2}}\arctan\big(\sqrt{2}\sinh a\big).$$

The infinite integral

Ask yourself The two integrands have different-looking denominators. How can we subtract them?

Using $\cosh^2 x = 1+\sinh^2 x$,

$$2\cosh^2 x - 1 = 2 + 2\sinh^2 x - 1 = 1 + 2\sinh^2 x,$$

so the two denominators are identical, and

$$\int_0^a \frac{\cosh x - \sinh x}{1+2\sinh^2 x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\arctan(\sqrt{2}\sinh a) - \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}\cosh a - 1}{\sqrt{2}\cosh a + 1}\right)$$
$$- \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right).$$

Now let $a \to \infty$. Since $\sinh a \to \infty$, $\arctan(\sqrt{2}\sinh a) \to \frac{\pi}{2}$. Since $\cosh a \to \infty$,

$$\frac{\sqrt{2}\cosh a - 1}{\sqrt{2}\cosh a + 1} = \frac{1 - 1/(\sqrt{2}\cosh a)}{1 + 1/(\sqrt{2}\cosh a)} \to 1,$$

so its logarithm tends to 0, by continuity of ln at 1. All three terms have finite limits, so the infinite integral converges and

$$\int_0^\infty \frac{\cosh x - \sinh x}{1+2\sinh^2 x}\,\mathrm{d}x = \frac{\pi}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right),$$

as required.

The final integral

Ask yourself What do $\cosh x - \sinh x$ and $1+2\sinh^2 x$ look like in terms of e^x?

With $u = e^x$ we have $\cosh x - \sinh x = e^{-x} = 1/u$, and, using the double-angle formula,

$$1 + 2\sinh^2 x = \cosh 2x = \frac{u^2 + u^{-2}}{2} = \frac{u^4+1}{2u^2}.$$

Also $\mathrm{d}u = e^x\,\mathrm{d}x$, so $\mathrm{d}x = \mathrm{d}u/u$, and the limits $x = 0$, $x \to \infty$ become $u = 1$, $u \to \infty$. Hence

$$\int_0^\infty \frac{\cosh x - \sinh x}{1 + 2\sinh^2 x}\,\mathrm{d}x = \int_1^\infty \frac{1}{u}\cdot\frac{2u^2}{1+u^4}\cdot\frac{\mathrm{d}u}{u} = 2\int_1^\infty \frac{\mathrm{d}u}{1+u^4}.$$

Halving the previous result,

$$\int_1^\infty \frac{\mathrm{d}u}{1+u^4} = \frac{\pi}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right).$$

This tidies up: $\dfrac{\sqrt{2}+1}{\sqrt{2}-1} = \dfrac{(\sqrt{2}+1)^2}{(\sqrt{2}-1)(\sqrt{2}+1)} = (\sqrt{2}+1)^2$, so the logarithm is $2\ln(1+\sqrt{2})$ and

$$\int_1^\infty \frac{\mathrm{d}u}{1+u^4} = \frac{\pi}{4\sqrt{2}} - \frac{\ln(1+\sqrt{2})}{2\sqrt{2}} \approx 0.2437.$$

Checks

- At $a = 0$ the first formula gives $\frac{1}{2\sqrt{2}}\left(\ln\frac{\sqrt{2}-1}{\sqrt{2}+1} + \ln\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) = 0$, as it should for an integral over an empty interval.
- The final answer is positive and less than $\int_1^\infty u^{-4}\,\mathrm{d}u = \frac{1}{3}$, which is sensible since $1 + u^4 > u^4$.
- Simpson's rule on $\int_1^\infty \frac{\mathrm{d}u}{1+u^4}$ (after $u = 1/t$ it becomes $\int_0^1 \frac{t^2}{1+t^4}\,\mathrm{d}t$) gives 0.243748, agreeing with the formula to twelve decimal places. The infinite integral in x checks numerically as 0.487495, twice this.

Common slips

- Forgetting the factor $\frac{1}{\sqrt{2}}$ from integrating $\frac{1}{\sqrt{2}c \pm 1}$ or $\frac{1}{1+2s^2}$.
- Not changing the limits when substituting, or leaving the answer in terms of c or s.
- In the last part, forgetting that $\mathrm{d}x = \mathrm{d}u/u$, which gives $2u/(1+u^4)$ and a divergent integral.

Beyond STEP

The final result is part of a classic: $\int_0^\infty \frac{\mathrm{d}u}{1+u^4} = \frac{\pi}{2\sqrt{2}}$, and the substitution $u \mapsto 1/u$ shows the pieces over $[0,1]$ and $[1,\infty)$ are $\int_0^1 \frac{1}{1+t^4}\,\mathrm{d}t$ and $\int_0^1 \frac{t^2}{1+t^4}\,\mathrm{d}t$, whose sum and difference are elementary. In a first-year course you would also meet the slicker route through contour integration and residues, which evaluates $\int_0^\infty \frac{\mathrm{d}u}{1+u^n} = \frac{\pi/n}{\sin(\pi/n)}$ in one go. Needham's *Visual Complex Analysis* is a beautiful place to learn that idea.

WORKED SOLUTION

STEP III 2004, Question 2

Pure • Asymptotes and Curve Sketching

STEP III 2004, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q2

The key idea

Write $f(x)$ as the line $y = x - 4$ plus a correction term. The correction is a perfect square on top, so it vanishes *twice* at one point, and that double zero is exactly what tangency means. For the roots of $f(x) = 0$, clear the denominator: you get a difference of two squares, which factorises at once.

Solution

Throughout, write

$$f(x) = (x-4) - \frac{16(2x+1)^2}{x^2(x-4)}.$$

Part (i)

The denominator vanishes at $x = 0$ and $x = 4$. The numerator $16(2x+1)^2$ is non-zero at both, so the vertical asymptotes are

$$x = 0 \qquad \text{and} \qquad x = 4.$$

For large $|x|$ the fraction has a quadratic on top and a cubic underneath:

$$\frac{16(2x+1)^2}{x^2(x-4)} = \frac{64x^2+64x+16}{x^3-4x^2} \approx \frac{64}{x} \to 0 \quad \text{as } x \to \pm\infty,$$

so $f(x) - (x-4) \to 0$, and the oblique asymptote is

$$y = x - 4.$$

Ask yourself How can an asymptote also be a tangent?

An asymptote only describes what happens far away, so nothing stops the line from touching the curve at some finite point as well. To find where the line and curve meet, set $\mathrm{f}(x) = x - 4$:

$$\mathrm{f}(x) - (x-4) = -\frac{16(2x+1)^2}{x^2(x-4)} = 0 \iff x = -\tfrac{1}{2}.$$

Because the factor $(2x+1)$ is squared, $x = -\frac{1}{2}$ is a *double* root of $\mathrm{f}(x) - (x-4) = 0$. This means the line touches the curve there. To check directly, let $D(x) = \mathrm{f}(x) - (x-4) = (2x+1)^2\, h(x)$ with $h(x) = -16/\big(x^2(x-4)\big)$. The function h is differentiable near $x = -\frac{1}{2}$, so

$$D'(x) = 4(2x+1)h(x) + (2x+1)^2 h'(x),$$

which is 0 at $x = -\frac{1}{2}$. Hence $\mathrm{f}'(-\frac{1}{2}) = 1$, the gradient of the asymptote. Also $\mathrm{f}(-\frac{1}{2}) = -\frac{9}{2}$, which lies on $y = x - 4$. So the oblique asymptote is the tangent to the curve at $\big(-\frac{1}{2}, -\frac{9}{2}\big)$.

Part (ii)

Put everything over the common denominator $x^2(x-4)$:

$$\mathrm{f}(x) = \frac{x^2(x-4)^2 - 16(2x+1)^2}{x^2(x-4)} = \frac{\big[x(x-4)\big]^2 - \big[4(2x+1)\big]^2}{x^2(x-4)}.$$

The numerator is a difference of two squares:

$$\begin{aligned}\big[x^2 - 4x - 4(2x+1)\big]\big[x^2 - 4x + 4(2x+1)\big] &= (x^2 - 12x - 4)(x^2 + 4x + 4)\\ &= (x^2 - 12x - 4)(x+2)^2.\end{aligned}$$

So

$$\mathrm{f}(x) = \frac{(x+2)^2(x^2 - 12x - 4)}{x^2(x-4)}.$$

The factor $(x+2)^2$ shows that $x = -2$ is a double root of $\mathrm{f}(x) = 0$, and $x = -2$ is not a zero of the denominator, as required. The other two roots come from $x^2 - 12x - 4 = 0$:

$$x = 6 \pm 2\sqrt{10} \approx 12.32,\ -0.32.$$

Part (iii)

We collect the features we need for the sketch.

Near $x = 0$. As $x \to 0$, $\mathrm{f}(x) \approx -\dfrac{16}{x^2 \cdot (-4)} = \dfrac{4}{x^2}$, so $\mathrm{f}(x) \to +\infty$ from *both* sides.

Near $x = 4$. As $x \to 4$, $\mathrm{f}(x) \approx -\dfrac{16 \cdot 81}{16(x-4)} = -\dfrac{81}{x-4}$. So $\mathrm{f}(x) \to +\infty$ as $x \to 4^-$ and $\mathrm{f}(x) \to -\infty$ as $x \to 4^+$.

Which side of the oblique asymptote? $\mathrm{f}(x) - (x-4) = -16(2x+1)^2/\big(x^2(x-4)\big)$. This is positive for $x < 4$ (apart from $x = 0$ and $x = -\frac{1}{2}$) and negative for $x > 4$. So the curve

lies above $y = x - 4$ on the left of $x = 4$, touching it only at $x = -\frac{1}{2}$, and below it on the right.

Sign of f. From the factorised form, the factors $(x+2)^2$ and x^2 are never negative. So the sign of f is the sign of $(x^2 - 12x - 4)/(x-4)$:

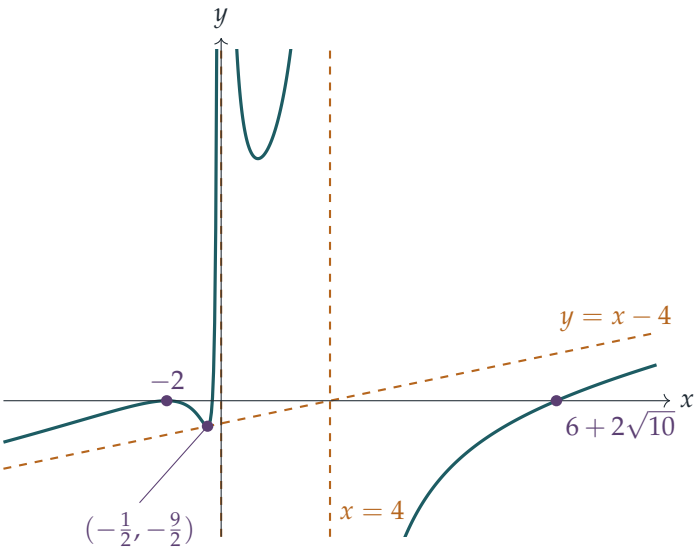
- $\mathrm{f} < 0$ for $x < 6 - 2\sqrt{10}$ (except $\mathrm{f}(-2) = 0$) and for $4 < x < 6 + 2\sqrt{10}$;
- $\mathrm{f} > 0$ for $6 - 2\sqrt{10} < x < 4$ ($x \neq 0$) and for $x > 6 + 2\sqrt{10}$.

The curve stays negative on both sides of $x = -2$ and touches the x-axis there, so $(-2, 0)$ is a local maximum.

Turning points. These aren't needed for a STEP sketch, but they make it more accurate. Differentiating gives

$$\mathrm{f}'(x) = 1 + \frac{16(2x+1)(2x^2+3x-8)}{x^3(x-4)^2}.$$

On $(-2,\ 6 - 2\sqrt{10})$ the curve starts and ends at height 0 and is negative in between, so it has a minimum there. Numerically this minimum is at about $(-0.51,\ -4.50)$, just left of the point of tangency. On $(0, 4)$ the curve tends to $+\infty$ at both ends, so it has a minimum there too, at about $(1.35,\ 42.7)$. Clearing denominators in $\mathrm{f}'(x) = 0$ gives $(x+2)(x^4 - 10x^3 + 100x^2 - 72x - 64) = 0$. The quartic has exactly these two real roots, so there are no other turning points. In particular, f increases throughout $x > 4$.



The sketch is drawn to scale, which makes the dip between $x = -2$ and $x \approx -0.32$ look shallow. The lowest point there is only about 4.5 below the axis, and the curve touches the asymptote at $(-\frac{1}{2}, -\frac{9}{2})$.

Checks

- $f(-\frac{1}{2}) = -\frac{9}{2} - 0 = -\frac{9}{2}$, which is on $y = x - 4$. A numerical derivative at $x = -\frac{1}{2}$ gives 1.000.
- Expand $(x+2)^2(x^2 - 12x - 4)$ and compare with $x^2(x-4)^2 - 16(2x+1)^2$: both are $x^4 - 8x^3 - 48x^2 - 64x - 16$.
- $f(1000) - (1000 - 4) \approx -0.064 \approx -64/1000$, as the $64/x$ estimate predicts.

Common slips

- Forgetting that near $x = 0$ the curve goes to $+\infty$ on *both* sides, because of the x^2. The two vertical asymptotes behave differently.
- Proving tangency by showing only that the line meets the curve. You also need the matching gradient, or equivalently the repeated root.
- Expanding the quartic numerator in part (ii) and then trying to spot its roots, instead of seeing the difference of two squares.

Beyond STEP

A line meets a curve with a "double root" exactly when it is tangent. That is the idea of *multiplicity of intersection*, which is basic to algebraic geometry and leads to results such as Bézout's theorem on how many times two curves meet. Rational functions are approximated at infinity by their polynomial part, and the size of the remainder ($64/x$ here) tells you how fast the curve approaches the asymptote. This is a first taste of asymptotic expansions. Siklos, *Advanced Problems in Mathematics*, has more curve-sketching problems in this style.

WORKED SOLUTION

STEP III 2004, Question 3

Pure • Convexity and Series Bounds

STEP III 2004, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q3

The key idea

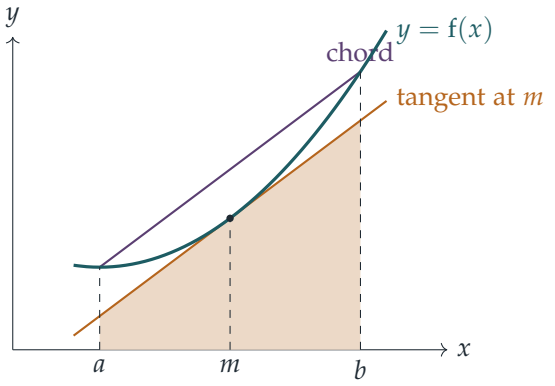
If $f'' > 0$ the curve bends upwards. So it lies above every tangent line and below every chord. The tangent at the midpoint and the chord from end to end each cut off a trapezium whose area is easy to write down, and these two areas trap the integral. With $f(x) = 1/x^2$ on $[n-1, n]$ the integral telescopes. Adding the inequalities over n then gives bounds on $\sum 1/n^2$.

Solution

The two bounds on the integral

Write $m = \frac{1}{2}(a+b)$. Since $f'' > 0$ on $[a,b]$, the gradient f' is strictly increasing, so the curve is convex. This gives two facts.

- *The curve lies above its tangent at m.* Let $t(x) = f(m) + f'(m)(x-m)$ and $g = f - t$. Then $g(m) = g'(m) = 0$ and $g'' = f'' > 0$, so g' is increasing. Hence $g' < 0$ to the left of m and $g' > 0$ to the right, so g has a strict minimum 0 at m, and $f(x) > t(x)$ for $x \neq m$.
- *The curve lies below its chord.* Let $c(x)$ be the straight line through $(a, f(a))$ and $(b, f(b))$, and $h = f - c$. Then $h(a) = h(b) = 0$ and $h'' > 0$. If h had a value ≥ 0 inside (a,b), it would have a maximum there with $h'' \leq 0$, which is impossible. So $f(x) < c(x)$ for $a < x < b$.



Now compare areas over $[a,b]$. The area under the tangent is a trapezium of width $b-a$. Its mean height is the height of t at the midpoint, $t(m)=\mathrm{f}(m)$, so the area is $(b-a)\,\mathrm{f}(m)$ (shaded). The area under the chord is a trapezium with parallel sides $\mathrm{f}(a)$ and $\mathrm{f}(b)$, so the area is $(b-a)\frac{1}{2}\big(\mathrm{f}(a)+\mathrm{f}(b)\big)$. The curve lies strictly between the two lines except at isolated points. So

$$(b-a)\,\mathrm{f}\Big(\frac{a+b}{2}\Big) < \int_a^b \mathrm{f}(x)\,\mathrm{d}x < (b-a)\,\frac{\mathrm{f}(a)+\mathrm{f}(b)}{2}.$$

(The sketch shows a positive f. The argument is really about $\int(\mathrm{f}-t)>0$ and $\int(c-\mathrm{f})>0$, so the sign of f doesn't matter.)

Choosing a, b and f

The middle term $\frac{1}{n-1}-\frac{1}{n}$ looks like $\big[-\frac{1}{x}\big]_{n-1}^{n}$, so take $\mathrm{f}(x)=1/x^2$, $a=n-1$, $b=n$. Then $b-a=1$, and $\mathrm{f}''(x)=6/x^4>0$ on $[n-1,n]$ because $n-1\geq 1>0$. Also

$$\int_{n-1}^{n}\frac{\mathrm{d}x}{x^2}=\frac{1}{n-1}-\frac{1}{n}, \qquad \mathrm{f}\Big(\frac{2n-1}{2}\Big)=\frac{4}{(2n-1)^2},$$

so the first part gives

$$\frac{4}{(2n-1)^2} < \frac{1}{n-1}-\frac{1}{n} < \frac{1}{2}\Big(\frac{1}{n^2}+\frac{1}{(n-1)^2}\Big) \qquad (n\geq 2),$$

as required.

Deducing the two bounds on 1

Write $S=\sum_{n=1}^{\infty}\frac{1}{n^2}$. This series converges, because for $n\geq 2$ we have $\frac{1}{n^2}<\frac{1}{n(n-1)}=\frac{1}{n-1}-\frac{1}{n}$, and those terms telescope. Add the inequalities for $n=2,3,\ldots,N$. The middle terms telescope to $1-\frac{1}{N}$, which tends to 1.

Ask yourself When we let $N \to \infty$, strict inequalities usually become $\leq$. How do we keep them strict?

Look at the gaps. For each $n \geq 2$ the gap $d_n = \left(\frac{1}{n-1} - \frac{1}{n}\right) - \frac{4}{(2n-1)^2}$ is positive. So the infinite sum of the gaps is at least its first term, $d_2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} > 0$. Every series here converges, so we may sum term by term, and the difference of the sums is strictly positive. The same argument works for the right-hand inequality.

On the left, $n = 2, 3, 4, \ldots$ gives $2n - 1 = 3, 5, 7, \ldots$, so

$$4\left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots\right) < 1.$$

On the right,

$$\sum_{n=2}^{\infty} \frac{1}{2}\left(\frac{1}{n^2} + \frac{1}{(n-1)^2}\right) = \frac{1}{2}(S-1) + \frac{1}{2}S = \frac{1}{2} + (S-1),$$

which gives

$$1 < \frac{1}{2} + \left(\frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots\right).$$

Comparing with the odd squares

Group the left-hand series in consecutive pairs, an odd term followed by an even term. For $k \geq 2$,

$$\frac{1}{2}\left(\frac{1}{(2k-1)^2} + \frac{1}{(2k)^2}\right) < \frac{1}{2}\left(\frac{1}{(2k-1)^2} + \frac{1}{(2k-1)^2}\right) = \frac{1}{(2k-1)^2},$$

since $2k > 2k - 1 > 0$. With $k = 2, 3, 4, \ldots$ the pairs on the left are $(3^2, 4^2)$, $(5^2, 6^2), \ldots$, which together use every term $\frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, \ldots$ exactly once. (Grouping is allowed because all the terms are positive and the series converges.) The right-hand sides are $\frac{1}{3^2}, \frac{1}{5^2}, \ldots$. Summing over k, with strictness kept as before, gives

$$\frac{1}{2}\left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \cdots\right) < \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots.$$

The bounds on S

Lower bound. From the right-hand inequality, $1 < \frac{1}{2} + (S-1)$, so $S > \frac{3}{2}$.

Upper bound. Let $T = \frac{1}{3^2} + \frac{1}{4^2} + \cdots = S - 1 - \frac{1}{4}$. Combining the last comparison with the left-hand inequality gives

$$\frac{1}{2}T < \frac{1}{3^2} + \frac{1}{5^2} + \cdots < \frac{1}{4},$$

so $T < \frac{1}{2}$. Hence $S = 1 + \frac{1}{4} + T < \frac{7}{4}$. Altogether

$$\frac{3}{2} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \frac{7}{4}.$$

Checks

- The true value is $S = \pi^2/6 \approx 1.6449$, which lies inside $(1.5, 1.75)$.
- Numerically, $4\sum_{k\geq 2}(2k-1)^{-2} = 4(\pi^2/8 - 1) \approx 0.935 < 1$, and $\frac{1}{2} + (S-1) \approx 1.145 > 1$.
- For the comparison, $\frac{1}{2}T \approx 0.197 < 0.234 \approx \sum_{k\geq 2}(2k-1)^{-2}$.
- Try $n = 2$: $\frac{4}{9} \approx 0.444 < \frac{1}{2} < \frac{5}{8} = 0.625$. A check over $2 \leq n \leq 2000$ with exact fractions confirms all three inequalities.

Common slips

- Forgetting the missing $\frac{1}{4}$: the bracket starts at $\frac{1}{3^2}$, so $T = S - \frac{5}{4}$, not $S - 1$. Using $S - 1$ leads to the false claim $S < \frac{3}{2}$.
- Getting the inequalities the wrong way round. For a curve that bends upwards, the tangent (midpoint) estimate is too small and the chord (trapezium) estimate is too big.
- Passing to infinite sums and quietly keeping "$<$" without saying why it survives the limit.

Beyond STEP

The two bounds are the midpoint rule and the trapezium rule. Convexity tells you which way each one errs, and Taylor's theorem shows that both errors shrink like $(b-a)^3$. The exact value $\sum 1/n^2 = \pi^2/6$ is Euler's solution of the Basel problem. A first analysis course proves it with Fourier series. It also makes the "strict inequalities in the limit" step rigorous. Abbott's *Understanding Analysis* covers both convergence of series and convex functions.

WORKED SOLUTION

STEP III 2004, Question 4

Pure • Tangent Circles and Series

STEP III 2004, Question 4

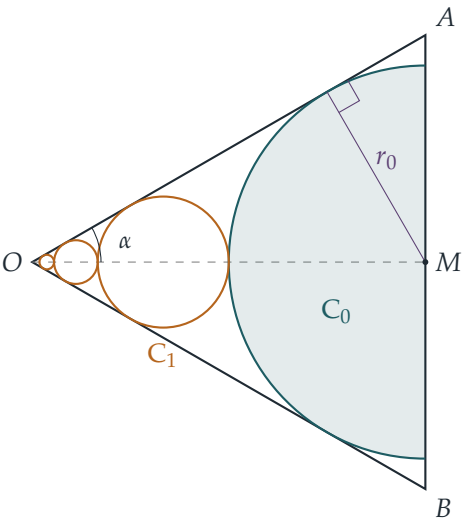
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q4

The key idea

Every circle that touches both OA and OB has its centre on the bisector of angle AOB, and its distance from O is fixed by its radius: a right-angled triangle with angle α at O gives distance $= r/\sin\alpha$. Writing "the two circles touch" as an equation in these distances links consecutive radii, and the radii form a geometric progression. The areas then form a geometric series, and the last part is a one-variable maximisation (or a well-chosen value of α).

Solution



The ratio of radii

Let M be the midpoint of AB. Since $OA = OB$, the line OM bisects angle AOB and is perpendicular to AB; each of the angles AOM and BOM is α.

Take any circle of radius r touching both lines OA and OB. Its centre is equidistant from the two lines, so it lies on the bisector OM. If the centre is P and the circle touches OA at T, then $PT \perp OA$, $PT = r$ and angle $POT = \alpha$, so in the right-angled triangle OTP

$$OP = \frac{r}{\sin\alpha}.$$

This applies to C_0 (centre M) and to every C_n. Write $d_n = r_n / \sin\alpha$ for the distance from O to the centre of C_n.

The circle C_{n+1} sits in the corner between C_n and O (it must lie inside the angle, on the O side of C_n, since it touches the sides OA and OB, not their extensions). Two circles touching externally have centres a distance $r_n + r_{n+1}$ apart, and here both centres lie on OM, so

$$d_n - d_{n+1} = r_n + r_{n+1}.$$

Substituting $d_n = r_n / \sin\alpha$ and multiplying by $\sin\alpha$:

$$\begin{aligned} r_n - r_{n+1} &= (r_n + r_{n+1})\sin\alpha \\ r_n(1 - \sin\alpha) &= r_{n+1}(1 + \sin\alpha), \end{aligned}$$

and therefore

$$\frac{r_{n+1}}{r_n} = \frac{1 - \sin\alpha}{1 + \sin\alpha},$$

as required. This works for $n = 0$ too: C_1 touches the curved part of the semicircle, which is the half of C_0 facing O.

The total area

Write $k = \dfrac{1 - \sin\alpha}{1 + \sin\alpha}$. Since $0 < \sin\alpha < 1$, we have $0 < k < 1$, and $r_n = k^n r_0$. The semicircle has area $\frac{1}{2}\pi r_0^2$ and C_n has area $\pi k^{2n} r_0^2$. The series $\sum_{n\geq 1} k^{2n}$ is geometric with ratio $k^2 < 1$, so it converges to $k^2/(1 - k^2)$. Thus

$$S = \pi r_0^2 \left(\frac{1}{2} + \frac{k^2}{1 - k^2} \right).$$

Now

$$1 - k^2 = \frac{(1 + \sin\alpha)^2 - (1 - \sin\alpha)^2}{(1 + \sin\alpha)^2} = \frac{4\sin\alpha}{(1 + \sin\alpha)^2}, \qquad \text{so} \qquad \frac{k^2}{1 - k^2} = \frac{(1 - \sin\alpha)^2}{4\sin\alpha}.$$

Hence

$$S = \pi r_0^2 \, \frac{2\sin\alpha + 1 - 2\sin\alpha + \sin^2\alpha}{4\sin\alpha} = \frac{1 + \sin^2\alpha}{4\sin\alpha} \, \pi r_0^2,$$

which is the result we wanted.

Comparing with the triangle

From the first part, $OM = d_0 = r_0/\sin\alpha$. In the right-angled triangle OMA, $AM = OM\tan\alpha$, so the area of triangle OAB is

$$T = OM \cdot AM = OM^2 \tan\alpha = \frac{r_0^2}{\sin^2\alpha} \cdot \frac{\sin\alpha}{\cos\alpha} = \frac{r_0^2}{\sin\alpha\cos\alpha}.$$

(The semicircle does fit on the base: $r_0 = OM\sin\alpha < OM\tan\alpha = AM$.) Dividing,

$$\frac{S}{T} = \frac{\pi}{4}(1+\sin^2\alpha)\cos\alpha.$$

Ask yourself Which value of α should we try?

As $\alpha \to 0$ the ratio tends to $\pi/4 \approx 0.785$, just short of 0.8, so very thin triangles don't work. A natural first try is a right angle at O, that is $\alpha = \pi/4$. Then $\sin^2\alpha = \frac{1}{2}$ and $\cos\alpha = 1/\sqrt{2}$, so

$$\frac{S}{T} = \frac{\pi}{4} \cdot \frac{3}{2} \cdot \frac{1}{\sqrt{2}} = \frac{3\pi}{8\sqrt{2}}.$$

This exceeds $\frac{4}{5}$ exactly when $15\pi > 32\sqrt{2}$. Both sides are positive, so we may square: the condition becomes $225\pi^2 > 2048$, that is $\pi^2 > 9.102\ldots$. Since $\pi > 3.1$, we have $\pi^2 > 9.61$, so the inequality holds. Hence for $\alpha = \pi/4$ (and, by continuity, for all α close to $\pi/4$) the circles cover more than four fifths of the triangle.

For the record, the best possible α comes from maximising $f(c) = (2-c^2)c$ with $c = \cos\alpha$: $f'(c) = 2 - 3c^2 = 0$ gives $\cos\alpha = \sqrt{2/3}$, i.e. $\sin\alpha = 1/\sqrt{3}$, and the maximum ratio is $\pi\sqrt{6}/9 \approx 0.855$.

Checks

- As $\alpha \to \pi/2$ the triangle flattens and $k \to 0$: the circles C_n vanish and $S \to \frac{1}{2}\pi r_0^2$, matching $\frac{1+1}{4}\pi r_0^2$.
- With $\alpha = \pi/6$, $k = \frac{1}{3}$ and $S = \pi r_0^2\left(\frac{1}{2} + \frac{1/9}{8/9}\right) = \frac{5}{8}\pi r_0^2$; the formula gives $\frac{1+1/4}{2}\pi r_0^2 = \frac{5}{8}\pi r_0^2$.
- Numerically, building 3000 circles by direct tangency for random α reproduces both formulae to about 10^{-15}, and the ratio at $\alpha = \pi/4$ is 0.8330.

Common slips

- Counting C_0 as a full circle: it is a semicircle, which is why the $\frac{1}{2}$ appears.
- Using $d_n + d_{n+1} = r_n + r_{n+1}$ or $d_n - d_{n+1} = r_n - r_{n+1}$: the circles touch externally and both centres are on the same side of O.
- Trying a small α for the last part: the ratio tends to $\pi/4 < \frac{4}{5}$ there.

Beyond STEP

The chain of circles is generated by a single dilation (enlargement) centred at O with scale factor k: it maps the corner region to itself and each circle to the next. Self-similar configurations like this lead to fractals and to the Apollonian circle packings, where the curvatures of mutually tangent circles obey Descartes' circle theorem. Inversion, a transformation that turns circles into lines, makes many such tangency problems easy; Needham's *Visual Complex Analysis* treats it beautifully.

WORKED SOLUTION

STEP III 2004, Question 5

Pure • Trigonometric Equations

STEP III 2004, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q5

The key idea

Two cosines are equal exactly when their angles agree up to sign and a multiple of 2π. Passing to tangents throws away the information that distinguishes θ from $\theta + \pi$, so the implication only goes one way, and that is the point of the counterexample. In parts (i) and (ii) we put $a\cos x + b\sin x$ into the form $R\cos(x - \alpha)$, use the first result to get $\tan x$ in terms of $\tan\omega$, and then use the sizes of the angles to choose the correct quadrant.

Solution

The opening result

If $\cos(x - \alpha) = \cos\beta$, then $x - \alpha = \pm\beta + 2n\pi$ for some integer n, since cosine is even and has period 2π. So

$$x = \alpha + \beta + 2n\pi \quad \text{or} \quad x = \alpha - \beta + 2n\pi.$$

Since tan has period π, and so certainly period 2π, this gives $\tan x = \tan(\alpha + \beta)$ or $\tan x = \tan(\alpha - \beta)$ (whenever these tangents are defined).

The converse fails because tan has period π but cos does not. Take $\alpha = \beta = 0$ and $x = \pi$. Then $\tan x = 0 = \tan(\alpha + \beta)$, but

$$\cos(x - \alpha) = \cos\pi = -1 \neq 1 = \cos\beta.$$

In general, $x = \alpha + \beta + \pi$ gives $\cos(x - \alpha) = -\cos\beta$.

Two facts about ω will be useful. Since $\tan\omega = \frac{4}{3}$, we have $\cos\omega = \frac{3}{5}$ and $\sin\omega = \frac{4}{5}$. Since $\frac{4}{3} > 1$, we also have $\frac{\pi}{4} < \omega < \frac{\pi}{2}$.

Part (i)

Here $\sqrt{1^2+7^2} = 5\sqrt{2}$, so we divide by $5\sqrt{2}$:

$$\tfrac{1}{5\sqrt{2}}\cos x - \tfrac{7}{5\sqrt{2}}\sin x = \tfrac{1}{\sqrt{2}}.$$

Let φ be the acute angle with $\cos\varphi = \frac{1}{5\sqrt{2}}$ and $\sin\varphi = \frac{7}{5\sqrt{2}}$, so that $\tan\varphi = 7$. The equation becomes

$$\cos(x+\varphi) = \cos\tfrac{\pi}{4},$$

which is the opening result with $\alpha = -\varphi$ and $\beta = \frac{\pi}{4}$. Using $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$,

$$\tan(\alpha+\beta) = \tan\left(\tfrac{\pi}{4} - \varphi\right) = \frac{1-7}{1+7} = -\frac{3}{4},$$
$$\tan(\alpha-\beta) = -\tan\left(\varphi + \tfrac{\pi}{4}\right) = -\frac{7+1}{1-7} = \frac{4}{3}.$$

Ask yourself Each of these tangent values belongs to two angles in $[0, 2\pi]$, but the equation has only two solutions there. Which two are they?

As the counterexample showed, the tangent alone can't decide this, so we go back to $x = \alpha \pm \beta + 2n\pi$ and work out where $\alpha \pm \beta$ actually lies. Since $\tan\varphi = 7 > 1$, we have $\frac{\pi}{4} < \varphi < \frac{\pi}{2}$.

- $\alpha + \beta = \frac{\pi}{4} - \varphi$ lies in $(-\frac{\pi}{4}, 0)$ and has tangent $-\frac{3}{4}$. Now $\tan(\frac{\pi}{2} - \omega) = \cot\omega = \frac{3}{4}$, and $\frac{\pi}{2} - \omega \in (0, \frac{\pi}{4})$. Hence $\alpha + \beta = \omega - \frac{\pi}{2}$, and adding 2π gives $x = \frac{3\pi}{2} + \omega$.
- $\alpha - \beta = -\varphi - \frac{\pi}{4}$ lies in $(-\frac{3\pi}{4}, -\frac{\pi}{2})$ and has tangent $\frac{4}{3}$. The angle $\omega - \pi$ also lies in this interval and has tangent $\frac{4}{3}$. Tangent is one-to-one on an interval of length less than π that avoids its asymptotes, so $\alpha - \beta = \omega - \pi$. Adding 2π gives $x = \pi + \omega$.

Both values lie in $[0, 2\pi]$ because $0 < \omega < \frac{\pi}{2}$, and no other choice of n gives a value in this range. Therefore

$$x = \pi + \omega \quad \text{or} \quad x = \tfrac{3\pi}{2} + \omega.$$

To confirm, at $x = \pi + \omega$ we get $\cos x - 7\sin x = -\frac{3}{5} + \frac{28}{5} = 5$, and at $x = \frac{3\pi}{2} + \omega$ we have $\cos x = \sin\omega$ and $\sin x = -\cos\omega$, which give $\frac{4}{5} + \frac{21}{5} = 5$.

Part (ii)

Here $\sqrt{2^2+11^2} = \sqrt{125} = 5\sqrt{5}$, so

$$\tfrac{2}{5\sqrt{5}}\cos x + \tfrac{11}{5\sqrt{5}}\sin x = \tfrac{2}{\sqrt{5}}.$$

Let α be acute with $\cos\alpha = \frac{2}{5\sqrt{5}}$ and $\sin\alpha = \frac{11}{5\sqrt{5}}$, so $\tan\alpha = \frac{11}{2}$. Let β be acute with $\cos\beta = \frac{2}{\sqrt{5}}$, so $\sin\beta = \frac{1}{\sqrt{5}}$ and $\tan\beta = \frac{1}{2}$. The equation is $\cos(x-\alpha) = \cos\beta$, and

$$\tan(\alpha-\beta) = \frac{\frac{11}{2} - \frac{1}{2}}{1 + \frac{11}{4}} = \frac{5}{15/4} = \frac{4}{3},$$
$$\tan(\alpha+\beta) = \frac{\frac{11}{2} + \frac{1}{2}}{1 - \frac{11}{4}} = \frac{6}{-7/4} = -\frac{24}{7}.$$

The second value is $\tan 2\omega$, because

$$\tan 2\omega = \frac{2\cdot\frac{4}{3}}{1 - \frac{16}{9}} = \frac{8/3}{-7/9} = -\frac{24}{7}.$$

Again we place the angles. Both α and β are acute, so

- $\alpha - \beta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ with tangent $\frac{4}{3}$, so $\alpha - \beta = \omega$;
- $\alpha + \beta \in (0, \pi)$ with negative tangent, so it lies in $(\frac{\pi}{2}, \pi)$. Since $\frac{\pi}{4} < \omega < \frac{\pi}{2}$, the angle 2ω also lies in $(\frac{\pi}{2}, \pi)$ and has the same tangent, so $\alpha + \beta = 2\omega$.

Both ω and 2ω are in $[0, 2\pi]$, and adding $\pm 2\pi$ takes them outside it. Hence the solutions are

$$x = \omega \quad \text{and} \quad x = 2\omega,$$

and one solution is twice the other, as required. As a direct check, $x = \omega$ gives $\frac{6}{5} + \frac{44}{5} = 10$. For $x = 2\omega$ we have $\cos 2\omega = \frac{9-16}{25} = -\frac{7}{25}$ and $\sin 2\omega = \frac{24}{25}$, which give $-\frac{14}{25} + \frac{264}{25} = 10$.

Checks

- Each equation should have exactly two roots in $[0, 2\pi]$, because $5 < 5\sqrt{2}$ and $10 < 5\sqrt{5}$, so the right-hand side lies strictly between $-R$ and R.
- Numerically, $\omega \approx 0.9273$. A sign-change search finds roots 4.0689 and 5.6397 for (i), which are $\pi + \omega$ and $\frac{3\pi}{2} + \omega$. For (ii) it finds 0.9273 and 1.8546, which are ω and 2ω.

Common slips

- Solving $\tan x = \frac{4}{3}$ or $\tan x = -\frac{3}{4}$ and listing all four angles in $[0, 2\pi]$. Two of them are spurious, for exactly the reason given by the counterexample.
- Taking φ in (i) with the wrong sign, for example writing $\cos(x - \varphi)$ instead of $\cos(x + \varphi)$ for $\cos x - 7\sin x$.

Beyond STEP

The difficulty here is that tan is not injective, so "$\tan x = \tan y$" only tells you x modulo π. In complex analysis the same issue appears in the definition of the argument and of branches of $\log z$, where angles are known only up to multiples of 2π. In that setting, $a\cos x + b\sin x = \mathrm{Re}\left((a - \mathrm{i}b)\mathrm{e}^{\mathrm{i}x}\right)$, and R and α are just the modulus and argument of $a + \mathrm{i}b$. Needham's *Visual Complex Analysis* develops this geometric view beautifully.

WORKED SOLUTION

STEP III 2004, Question 6

Pure • Recurrence Relations

STEP III 2004, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q6

The key idea

The expression F_n is a quantity attached to each *consecutive pair* of terms, and the opening identity tells us exactly when it stays the same from one pair to the next. For the recurrence in (i) the second factor vanishes, so F_n is an invariant and we only need its value at the start. Part (ii) runs the argument backwards: if F_n is constant, the product must vanish, so at every step one of the two factors is zero. That gives a choice at each step, and the period-4 sequence comes from choosing well.

Solution

The identity

Writing out both terms, the w_{n-1}^2 terms cancel:

$$\begin{aligned} F_n - F_{n-1} &= \left(w_n^2 + w_{n-1}^2 - 4w_n w_{n-1}\right) - \left(w_{n-1}^2 + w_{n-2}^2 - 4w_{n-1}w_{n-2}\right) \\ &= \left(w_n^2 - w_{n-2}^2\right) - 4w_{n-1}\left(w_n - w_{n-2}\right) \\ &= (w_n - w_{n-2})(w_n + w_{n-2}) - 4w_{n-1}(w_n - w_{n-2}) \\ &= (w_n - w_{n-2})(w_n + w_{n-2} - 4w_{n-1}), \end{aligned}$$

which holds for $n \geq 2$ (we need w_{n-2} to exist), as required.

One more observation will be useful throughout: F_n is *symmetric* in w_n and w_{n-1}. Swapping them leaves $w_n^2 + w_{n-1}^2 - 4w_n w_{n-1}$ unchanged.

Part (i)

Take $w_n = u_n$, and let F_n be the corresponding quantity. For $n \geq 2$ the recurrence says $u_n + u_{n-2} - 4u_{n-1} = 0$, so the second factor in the identity is zero and

$$F_n = F_{n-1} \quad (n \geq 2).$$

Hence $F_n = F_1$ for every $n \geq 1$, and

$$F_1 = u_1^2 + u_0^2 - 4u_1u_0 = 4 + 1 - 8 = -3.$$

So $u_n^2 + u_{n-1}^2 - 4u_nu_{n-1} = -3$, that is, $u_n^2 + u_{n-1}^2 = 4u_nu_{n-1} - 3$ for all $n \geq 1$, as required. (The sequence runs $1, 2, 7, 26, 97, \ldots$.)

Part (ii)(a)

Putting $n = 1$ and $v_0 = 1$ into $(*)$:

$$v_1^2 + 1 = 4v_1 - 3 \iff v_1^2 - 4v_1 + 4 = 0 \iff (v_1 - 2)^2 = 0,$$

so $v_1 = 2$. There is no choice here, because it is a repeated root.

Now take $w_n = v_n$. Condition $(*)$ says precisely that $F_n = -3$ for every $n \geq 1$. So for $n \geq 2$ both F_n and F_{n-1} equal -3, and the identity gives

$$0 = F_n - F_{n-1} = (v_n - v_{n-2})(v_n + v_{n-2} - 4v_{n-1}).$$

A product of two real numbers is zero only if one of them is zero, so either $v_n = v_{n-2}$ or $v_n = 4v_{n-1} - v_{n-2}$, as required.

Ask yourself Is this all? Do both options really give a valid next term?

Yes. For fixed v_{n-1}, condition $(*)$ is the quadratic $x^2 - 4v_{n-1}x + (v_{n-1}^2 + 3) = 0$ in $x = v_n$. We know $x = v_{n-2}$ is a root, because F is symmetric and $F_{n-1} = -3$. The roots add up to $4v_{n-1}$, so the other root is $4v_{n-1} - v_{n-2}$. So the two options in (a) are exactly the two roots. We may pick either at each step, and the result always satisfies $(*)$. (When the roots coincide, as at $n = 1$, there is only one option.)

Part (ii)(b)

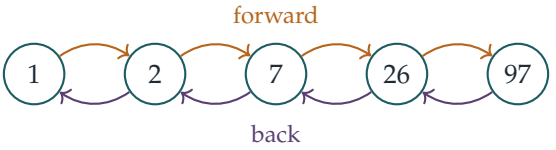
Here $v_0 = 1$, as required, and every consecutive pair (v_{n-1}, v_n) is either $(1, 2)$ or $(2, 1)$. Since F is symmetric, both give

$$v_n^2 + v_{n-1}^2 - 4v_nv_{n-1} = 1 + 4 - 8 = -3,$$

so $(*)$ holds for every $n \geq 1$. In the language of (a), this sequence always takes the option $v_n = v_{n-2}$.

Part (ii)(c)

We have $v_0 = 1$ and $v_1 = 2$. By (a), each later term is either the term two places back ("step back") or $4\times$(previous) $-$ (the one before) ("step forward"). From $(1, 2)$, stepping forward gives 7, then 26, 97, and so on, which is the sequence u_n. Stepping back undoes the last step forward.



So the sequence is a walk along this chain, starting at 1 and moving to a neighbour each time. Since 1 has only one neighbour, the walk must go to 2 straight after visiting 1.

To get period 4 with $v_0 = 1$, we need $v_4 = 1$. The period-2 sequence of (b) just shuttles between 1 and 2. To do something new, we go one step further:

$$v_0 = 1, \quad v_1 = 2, \quad v_2 = 4 \cdot 2 - 1 = 7, \quad v_3 = v_1 = 2, \quad v_4 = v_2 \text{ or } 4 \cdot 2 - 7 = 1.$$

Choosing $v_4 = 1$, and then repeating, gives

$$v_n = \begin{cases} 1 & n \equiv 0 \pmod 4, \\ 2 & n \text{ odd}, \\ 7 & n \equiv 2 \pmod 4, \end{cases} \qquad \text{i.e.} \quad 1, 2, 7, 2, 1, 2, 7, 2, \ldots$$

To check $(*)$ directly: every consecutive pair is $(1,2)$, $(2,7)$, $(7,2)$ or $(2,1)$. By symmetry we only need

$$1 + 4 - 8 = -3 \quad \text{and} \quad 4 + 49 - 56 = -3,$$

which both hold. So $(*)$ holds for all $n \geq 1$, and the sequence has period 4 (and not period 2, since $v_2 \neq v_0$).

In fact this is the only answer of exact period 4. After $1, 2$, the walk needs $v_4 = 1$, so it can only reach 7 at v_2 and must come straight back.

Checks

- Compute u_n up to $n = 40$ and confirm that $u_n^2 + u_{n-1}^2 - 4u_n u_{n-1} = -3$ every time. For example, $(26, 97)$: $676 + 9409 - 10088 = -3$.
- Test the opening identity on thousands of random integer triples.
- List every sequence allowed by (a) up to v_8, starting $1, 2$. The only ones with $v_{k+4} = v_k$ throughout are $1, 2, 1, 2, \ldots$ and $1, 2, 7, 2, \ldots$.

Common slips

- Concluding $F_n - F_{n-1} = 0$ in (ii) without saying why. It is because $(*)$ makes *every* F_n equal to -3.
- Forgetting that v_1 is forced. The quadratic has the repeated root 2, so no case split is needed.
- In (c), checking only one period's worth of pairs and forgetting the "wrap-around" pair $(v_3, v_4) = (2, 1)$.

Beyond STEP

The quantity F_n is an *invariant* of the recurrence: a conserved quantity, like energy in mechanics. The equation $x^2 + y^2 - 4xy = -3$ is a Markov-type (or Pell-type) Diophantine equation. Swapping one root of the quadratic for the other, $x \mapsto 4y - x$, is the classic "Vieta jumping" move, which generates all its integer solutions from the smallest one. Davenport's *The Higher Arithmetic* covers Pell's equation and its solutions by recurrences.

WORKED SOLUTION

STEP III 2004, Question 7

Pure • Reduction Formulae

STEP III 2004, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q7

The key idea

The integrand of I_{n+1} is the integrand of I_n multiplied by $\frac{t}{t+1}$, so a bound on that one factor gives a bound linking consecutive integrals. Integration by parts gives an exact link too, and putting the inequality and the equality side by side squeezes out a bound on I_n alone. The exact link telescopes down to I_1, which is a logarithm we know, and that turns everything into a series for $\ln 2$ with an error term we can control.

Solution

The inequality $I_{n+1} < \frac{1}{2} I_n$

On $0 \leq t \leq 1$,

$$\frac{t}{t+1} = 1 - \frac{1}{t+1},$$

which increases with t, so its greatest value is $\frac{1}{2}$, taken only at $t = 1$. Now write the integrand of I_{n+1} as

$$\frac{t^n}{(t+1)^{n+1}} = \frac{t}{t+1} \cdot \frac{t^{n-1}}{(t+1)^n}.$$

Hence

$$\frac{1}{2} I_n - I_{n+1} = \int_0^1 \left(\frac{1}{2} - \frac{t}{t+1}\right) \frac{t^{n-1}}{(t+1)^n} \, \mathrm{d}t.$$

The integrand here is continuous, never negative, and strictly positive for $0 < t < 1$ (both factors are positive there). So the integral is strictly positive, and $I_{n+1} < \frac{1}{2} I_n$.

? **Ask yourself** Why is the inequality strict, when $\frac{t}{t+1}$ actually reaches $\frac{1}{2}$?

It reaches $\frac{1}{2}$ only at the single point $t = 1$. A continuous non-negative function that is positive on a whole interval has a positive integral, so one point of equality makes no difference.

The recurrence

Integrate I_{n+1} by parts with $u = t^n$ and $\dfrac{\mathrm{d}v}{\mathrm{d}t} = (t+1)^{-(n+1)}$, so that $v = -\dfrac{1}{n(t+1)^n}$ (valid as $n \geq 1$):

$$\begin{aligned} I_{n+1} &= \Big[-\frac{t^n}{n(t+1)^n}\Big]_0^1 + \int_0^1 \frac{nt^{n-1}}{n(t+1)^n}\,\mathrm{d}t \\ &= -\frac{1}{n\,2^n} + I_n, \end{aligned}$$

since the bracket is $-\frac{1}{n2^n}$ at $t = 1$ and 0 at $t = 0$. This is the required result.

The bound on I_n

Substitute the recurrence into the inequality:

$$I_n - \frac{1}{n\,2^n} = I_{n+1} < \frac{1}{2}I_n \quad\Longrightarrow\quad \frac{1}{2}I_n < \frac{1}{n\,2^n} \quad\Longrightarrow\quad I_n < \frac{1}{n\,2^{n-1}}.$$

The series for $\ln 2$

First,

$$I_1 = \int_0^1 \frac{1}{t+1}\,\mathrm{d}t = \big[\ln(t+1)\big]_0^1 = \ln 2.$$

The recurrence says $I_r - I_{r+1} = \dfrac{1}{r\,2^r}$ for each $r \geq 1$. Adding these for $r = 1, \ldots, n$, the left-hand side telescopes:

$$I_1 - I_{n+1} = \sum_{r=1}^{n} \frac{1}{r\,2^r}, \qquad \text{so} \qquad \ln 2 = \sum_{r=1}^{n} \frac{1}{r\,2^r} + I_{n+1}.$$

The bounds $\frac{2}{3} < \ln 2 < \frac{17}{24}$

We have two facts about the remainder: $I_{n+1} > 0$, because its integrand is positive on $(0, 1]$, and $I_{n+1} < \dfrac{1}{(n+1)2^n}$, by the bound above with $n+1$ in place of n. So for every n,

$$\sum_{r=1}^{n} \frac{1}{r\,2^r} < \ln 2 < \sum_{r=1}^{n} \frac{1}{r\,2^r} + \frac{1}{(n+1)2^n}.$$

The partial sums are

$$\tfrac{1}{2}, \qquad \tfrac{1}{2} + \tfrac{1}{8} = \tfrac{5}{8}, \qquad \tfrac{5}{8} + \tfrac{1}{24} = \tfrac{16}{24} = \tfrac{2}{3}.$$

? **Ask yourself** Which value of n should we use for each bound?

Different ones. For the lower bound take $n = 3$: $\ln 2 > \frac{2}{3}$. For the upper bound take $n = 2$, where the error bound is $\frac{1}{3\cdot 4} = \frac{1}{12}$:

$$\ln 2 < \tfrac{5}{8} + \tfrac{1}{12} = \tfrac{15}{24} + \tfrac{2}{24} = \tfrac{17}{24}.$$

Together, $\frac{2}{3} < \ln 2 < \frac{17}{24}$, as required.

Checks

Numerically $\ln 2 \approx 0.6931$, which does lie between $\frac{2}{3} \approx 0.6667$ and $\frac{17}{24} \approx 0.7083$. Simpson's rule gives $I_3 \approx 0.0681$, comfortably below the bound $\frac{1}{12} \approx 0.0833$, and indeed $\frac{5}{8} + 0.0681 = 0.6931$. We also checked the recurrence and both inequalities numerically for n up to 24.

Common slips

- Using the same n for both bounds: $n = 3$ gives the upper bound $\frac{2}{3} + \frac{1}{32} = \frac{67}{96}$, which is true and sharper, but it is not the $\frac{17}{24}$ asked for.
- Getting the sign of the boundary term wrong in the integration by parts; v is negative.
- Forgetting to say why $I_{n+1} > 0$, which the strict lower bound needs.

Beyond STEP

This is the series $\ln 2 = \sum_{r \geq 1} \frac{1}{r 2^r}$, which is $-\ln(1 - x) = \sum x^r / r$ at $x = \frac{1}{2}$, with an explicit bound on the remainder after n terms. Estimating the remainder by an integral is exactly how Taylor's theorem with integral remainder works, a staple of first-year analysis. See Abbott, *Understanding Analysis*, for the theory of power series and their remainders.

WORKED SOLUTION

STEP III 2004, Question 8

Pure • Differential Equations

STEP III 2004, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q8

The key idea

The equation is not linear in y, because of the $1/y$ term. But multiplying through by y turns $y\,\frac{\mathrm{d}y}{\mathrm{d}x}$ into half the derivative of y^2, and the $1/y$ term becomes a plain function of x. So in terms of $u = y^2$ the equation is linear (an equation of Bernoulli type), and an integrating factor finishes it off. The three curves are then conics, and identifying them makes the sketch easy.

Solution

The substitution

Let $u = y^2$. By the chain rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2y\left(f(x)\,y + \frac{g(x)}{y}\right) = 2f(x)\,y^2 + 2g(x).$$

Replacing y^2 by u,

$$\frac{\mathrm{d}u}{\mathrm{d}x} - 2f(x)\,u = 2g(x),$$

which is linear in u: u and $\frac{\mathrm{d}u}{\mathrm{d}x}$ appear only to the first power, with coefficients depending on x alone. This is what we had to show.

Solving the equation

Here $f(x) = \frac{1}{x}$ and $g(x) = -1$, so the linear equation is

$$\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{2}{x}\,u = -2.$$

The integrating factor is $\exp\left(-\int \frac{2}{x}\,\mathrm{d}x\right) = \mathrm{e}^{-2\ln|x|} = \dfrac{1}{x^2}$. Multiplying through by it,

$$\frac{1}{x^2}\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{2}{x^3}\,u = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{x^2}\right) = -\frac{2}{x^2}.$$

Integrating, $\dfrac{u}{x^2} = \dfrac{2}{x} + C$ for a constant C, so the general solution is

$$y^2 = 2x + Cx^2.$$

(The original equation needs $x \neq 0$ and $y \neq 0$, so each solution lives on an interval of x not containing 0 on which $2x + Cx^2 > 0$, with y of one sign throughout.)

The three curves

Each of the three points has $x > 0$ and $y > 0$, so each solution curve is the part of $y^2 = 2x + Cx^2$ with $y > 0$ on the interval of positive x containing the point.

Through $(1,1)$**:** $1 = 2 + C$, so $C = -1$ and $y^2 = 2x - x^2$. Completing the square,

$$(x-1)^2 + y^2 = 1,$$

the circle with centre $(1,0)$ and radius 1. The solution curve is the upper semicircle $y = \sqrt{2x - x^2}$ for $0 < x < 2$.

Through $(2,2)$**:** $4 = 4 + 4C$, so $C = 0$ and $y^2 = 2x$: the parabola with vertex at the origin, axis along the x-axis. The solution curve is $y = \sqrt{2x}$ for $x > 0$.

Through $(4,4)$**:** $16 = 8 + 16C$, so $C = \frac{1}{2}$ and $y^2 = 2x + \frac{1}{2}x^2$. Completing the square, $\frac{1}{2}(x+2)^2 - y^2 = 2$, that is

$$\frac{(x+2)^2}{4} - \frac{y^2}{2} = 1,$$

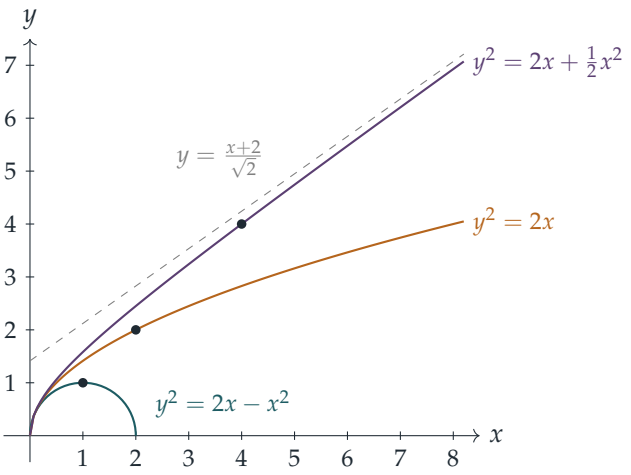
a hyperbola with centre $(-2,0)$, vertices $(0,0)$ and $(-4,0)$, and asymptotes $y = \pm\frac{1}{\sqrt{2}}(x+2)$. The solution curve is the upper half of the right-hand branch, $y = \sqrt{2x + \frac{1}{2}x^2}$ for $x > 0$.

> **Ask yourself** Why not include the left-hand branch of the hyperbola, or the lower halves?

A solution of the differential equation must be differentiable on an interval and satisfy the equation there, and the equation is undefined when $x = 0$ or $y = 0$. Starting at $(4,4)$ and moving left, the curve reaches $(0,0)$, where both fail, so it cannot be continued through. The left branch (with $x \leq -4$) and the lower halves satisfy $y^2 = 2x + \frac{1}{2}x^2$ but are different solutions, not the one through $(4,4)$.

Sketch

All three curves start at the origin with a vertical tangent (near $x = 0$ each behaves like $y \approx \sqrt{2x}$), and at the given points their gradients are $\frac{1}{1} - \frac{1}{1} = 0$, $\frac{2}{2} - \frac{1}{2} = \frac{1}{2}$ and $\frac{4}{4} - \frac{1}{4} = \frac{3}{4}$. For larger x the circle comes back down to $(2,0)$, the parabola grows like $\sqrt{x}$, and the hyperbola approaches its asymptote $y = \frac{1}{\sqrt{2}}(x+2)$. Since $2x - x^2 < 2x < 2x + \frac{1}{2}x^2$ for $x > 0$, the circle lies below the parabola, which lies below the hyperbola.



Checks

- Substitute back: if $y^2 = 2x + Cx^2$ then $2yy' = 2 + 2Cx$, so $y' = \frac{1+Cx}{y}$, while $\frac{y}{x} - \frac{1}{y} = \frac{y^2-x}{xy} = \frac{x+Cx^2}{xy} = \frac{1+Cx}{y}$. They agree.
- Stepping the original equation numerically (Runge–Kutta) from $(4, 4)$ to $x = 8$ gives $y \approx 6.92820$, matching $\sqrt{16+32} = \sqrt{48}$; similar runs from $(1, 1)$ and $(2, 2)$ match the semicircle and parabola to about 10^{-12}.

Common slips

- Getting the integrating factor as x^2 instead of x^{-2}: the coefficient of u is $-\frac{2}{x}$, so the exponent is $-2\ln x$.
- Forgetting the factor 2 from $\frac{du}{dx} = 2y\frac{dy}{dx}$, which leads to $y^2 = x + Cx^2$; that fails the back-substitution check.
- Sketching the whole circle, both halves of the parabola or both branches of the hyperbola as "the solution curve".

Beyond STEP

The trick here works for any *Bernoulli equation* $y' = f(x)y + g(x)y^n$: the substitution $u = y^{1-n}$ always makes it linear (here $n = -1$, so $u = y^2$). The care needed at $x = 0$ and $y = 0$ is a first glimpse of the existence and uniqueness theorem: through each point where the right-hand side is well behaved there is exactly one solution, and the family of curves can only meet at points where it isn't, as all of ours do at the origin. Boyce and DiPrima, *Elementary Differential Equations*, covers Bernoulli equations and this theorem.

WORKED SOLUTION

STEP III 2004, Question 9

Mechanics • Moments and Friction

STEP III 2004, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q9

The key idea

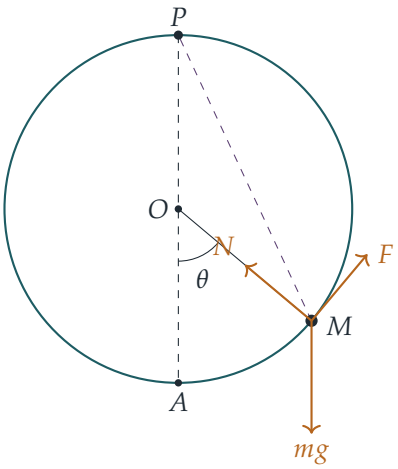
The hoop's weight acts through O, which is directly below the axis, so it has no moment about P. The only other force on the hoop (apart from the reaction at the axis) is the contact force from the mouse, so the hoop stays put exactly when that contact force has zero moment about P, that is, when its line of action passes through P. Everything follows from asking when the mouse's contact force can point along MP: once for the speed, and once for the friction limit.

Solution

The model

Treat the mouse as a particle of mass m and the hoop as a rigid ring (its mass will turn out not to matter) that turns freely about a smooth horizontal axis through P. Let $\theta = \angle AOM$, measured from the downward vertical, with the mouse running in the direction of increasing θ. While the hoop is at rest the mouse moves on a fixed circle of radius a about O at constant speed u.

The forces on the mouse are its weight mg, the normal reaction N from the hoop towards O, and friction F along the tangent, which we take as positive in the direction of increasing θ (up the hoop).



Resolving for the mouse along the tangent (no tangential acceleration, as the speed is constant) and along the radius (acceleration u^2/a towards O):

$$F = mg\sin\theta,$$
$$N - mg\cos\theta = \frac{mu^2}{a}.$$

When is the hoop in equilibrium?

By Newton's third law the mouse pushes on the hoop with a force equal and opposite to the contact force (N and F) on the mouse. The hoop's weight acts at O, vertically below P, and the axis reaction acts at P, so neither has a moment about P. The hoop therefore stays in equilibrium with POA vertical if and only if the contact force has zero moment about P, that is, its line of action passes through P.

Triangle OPM is isosceles ($OP = OM = a$) with $\angle POM = \pi - \theta$, so $\angle OMP = \frac{1}{2}\theta$. The line MP thus makes an angle $\frac{1}{2}\theta$ with the radius MO, tilted towards the direction of increasing θ. So the contact force on the mouse points along MP exactly when

$$\frac{F}{N} = \tan\tfrac{1}{2}\theta. \qquad (*)$$

(Equivalently, taking moments about P directly, the moment of the mouse's force on the hoop is $a\big[N\sin\theta - F(1+\cos\theta)\big]$, and $\sin\theta/(1+\cos\theta) = \tan\frac{1}{2}\theta$.)

? **Ask yourself** How do we turn this into a condition on u?

Use the equations of motion. The contact force on the mouse is its mass times acceleration minus its weight:

$$\mathbf{R} = m\cdot\frac{u^2}{a}\,\widehat{MO} + mg\,\widehat{\text{up}}.$$

But $\overrightarrow{MP} = \overrightarrow{MO} + \overrightarrow{OP} = a\,\widehat{MO} + a\,\widehat{\text{up}}$, with equal coefficients. So $\mathbf{R}$ is parallel to MP,

for every θ, if and only if $u^2/a = g$, that is,

$$u = \sqrt{ag}.$$

Algebraically: with $u^2 = ag$ the equations of motion give $F = mg\sin\theta$ and $N = mg(1+\cos\theta)$, so

$$\frac{F}{N} = \frac{\sin\theta}{1+\cos\theta} = \frac{2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta}{2\cos^2\frac{1}{2}\theta} = \tan\tfrac{1}{2}\theta,$$

which is $(*)$. For a general speed the moment about P works out as $m\sin\theta\,(u^2 - ag)$, which vanishes at every position only when $u^2 = ag$. So the mouse can keep the hoop in equilibrium by running at speed $\sqrt{ag}$, as required, provided the hoop can actually supply this friction. We look at that next.

The friction limit

The mouse's feet grip without slipping only while $|F| \le \mu N$. With $u = \sqrt{ag}$, and on the lower part where $|\theta| < \frac{1}{2}\pi$ so that $\cos\frac{1}{2}\theta > 0$, this is

$$\tan\tfrac{1}{2}|\theta| \le \mu \quad\Longleftrightarrow\quad |\theta| \le 2\arctan\mu.$$

So everything works until $\angle AOM = 2\arctan\mu$. (This point is on the lower half of the hoop only if $\mu < 1$; if $\mu \ge 1$ the mouse never reaches it.)

Past this point no choice of speed helps. Equilibrium needs $F/N = \tan\frac{1}{2}\theta > \mu$, and friction can't give that. With any speed and any friction satisfying $|F| \le \mu N$, the moment about P of the mouse's force on the hoop is

$$a\big[N\sin\theta - F(1+\cos\theta)\big] \;\ge\; aN(1+\cos\theta)\big(\tan\tfrac{1}{2}\theta - \mu\big) > 0.$$

What happens

Once the mouse passes $\angle AOM = 2\arctan\mu$ it can no longer grip hard enough. Its feet slip, friction falls to its limiting value μN, and the mouse can't hold its speed: it begins to slide back. Its contact force on the hoop now passes to the side of P, so it has a nonzero moment, and the hoop is no longer in equilibrium. The hoop starts to rotate about P. The moment above is in the sense that the normal push of the mouse wins over the reduced friction, so the hoop swings with A moving towards the mouse's side, the way the mouse was running. From then on the motion is no longer the simple equilibrium picture: the hoop swings about P while the mouse slides on it.

Checks

- At $\theta = 0$ the mouse is at A, the contact force $N = 2mg$ is vertical and passes through P, and no friction is needed, as expected.
- The speed $\sqrt{ag}$ has units of speed, doesn't depend on μ or on the hoop's mass, and makes the normal reaction $mg(1+\cos\theta)$, positive everywhere on the lower half.

- For $\mu = 0.5$ the critical angle is $2\arctan 0.5 \approx 0.927$ rad $\approx 53.1^\circ$. There $\sin\theta = 0.8$ and $1 + \cos\theta = 1.6$, so $F/N = 0.5 = \mu$ exactly. A numerical check of the moment about P at random positions, with $u^2 = ag$, gave zero to within $10^{-15}mga$.

Common slips

- Taking moments about O instead of P. The hoop pivots about P, and the normal reaction (which passes through O) does have a moment about P.
- Forgetting the centripetal term in N, which gives $N = mg\cos\theta$ and the wrong friction condition.
- Claiming the mouse simply "falls off". On the lower half $N > 0$ throughout. What fails is the grip, not the contact.

Beyond STEP

The neat fact here is that a body is in equilibrium about a pivot exactly when the resultant of the other forces passes through the pivot. The "line of action through P" argument replaces a page of resolving. The second half is a first taste of coupled systems, where a slipping contact feeds one body's motion into another's. Analysing the motion after slipping needs the hoop's moment of inertia and the full equations of rigid-body rotation. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers both torque about a fixed axis and friction-limited motion in this style.

WORKED SOLUTION

STEP III 2004, Question 10

Mechanics • Elastic Strings and SHM

STEP III 2004, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

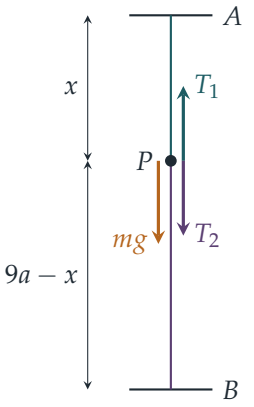
step.maths.org/questions/04-s3-q10

The key idea

Each string obeys Hooke's law only while it is taut, so the motion splits into stages according to which strings are taut. In each stage the acceleration is a constant multiple of $7a - x$, so the particle moves in simple harmonic motion about $x = 7a$, but with a different ω in each stage. We find the time and speed at the end of the first stage and use them to start the second.

Solution

We model P as a particle and the strings as light, obeying Hooke's law when stretched and exerting no force when slack. Take $x = AP$ measured downwards from A, so $PB = 9a - x$, and take downwards as positive.



The equations of motion

For $x > 6a$ the string AP is stretched by $x - 6a$, so its tension, pulling P up, is

$$T_1 = \frac{6mg(x-6a)}{6a} = \frac{mg(x-6a)}{a}.$$

The string BP is stretched by $(9a - x) - 2a = 7a - x$ when $x < 7a$, so its tension, pulling P down towards B, is

$$T_2 = \frac{6mg(7a-x)}{2a} = \frac{3mg(7a-x)}{a},$$

and $T_2 = 0$ for $x \geq 7a$, when BP is slack.

For $6a < x < 7a$, both strings are taut and Newton's second law downwards gives

$$\begin{aligned} m\ddot{x} &= mg - \frac{mg(x-6a)}{a} + \frac{3mg(7a-x)}{a} \\ &= \frac{mg}{a}(a - x + 6a + 21a - 3x) = \frac{mg}{a}(28a - 4x), \end{aligned}$$

so $\ddot{x} = \dfrac{4g}{a}(7a - x)$, as required.

For $7a < x < 9a$, only AP is taut, so

$$m\ddot{x} = mg - \frac{mg(x-6a)}{a} = \frac{mg}{a}(7a - x),$$

giving $\ddot{x} = \dfrac{g}{a}(7a - x)$, as required.

The time to reach B

Stage 1: $6a \leq x \leq 7a$. Put $y = x - 7a$. Then $\ddot{y} = -\omega_1^2 y$ with $\omega_1^2 = 4g/a$, which is simple harmonic motion about $x = 7a$. Starting from rest at $y = -a$ gives

$$x = 7a - a\cos\omega_1 t, \qquad \dot{x} = a\omega_1 \sin\omega_1 t.$$

Starting from rest at $x = 6a$, the particle accelerates downwards (since $\ddot{x} > 0$ there), so x increases from $6a$ and stays in this stage until $x = 7a$. That happens when $\cos\omega_1 t = 0$, first at $\omega_1 t = \pi/2$, so after time

$$t_1 = \frac{\pi}{2\omega_1} = \frac{\pi}{4}\sqrt{\frac{a}{g}},$$

with speed $a\omega_1 = 2\sqrt{ag}$, moving downwards.

Ask yourself The equation changes at $x = 7a$. How do we carry on?

The position and velocity are continuous at $x = 7a$, so the end of stage 1 gives the starting conditions for stage 2. We restart the clock there.

Stage 2: $7a \le x \le 9a$. Now $\ddot{y} = -\omega_2^2 y$ with $\omega_2^2 = g/a$, starting at the centre $y = 0$ with speed $2\sqrt{ag}$. So, with τ the time since $x = 7a$,

$$x = 7a + \frac{2\sqrt{ag}}{\omega_2}\sin\omega_2\tau = 7a + 2a\sin\omega_2\tau.$$

The amplitude is exactly $2a$, so the particle reaches $x = 9a$, which is the point B, when $\sin\omega_2\tau = 1$, first at $\omega_2\tau = \pi/2$. At that moment $\dot{x} = 2\sqrt{ag}\cos\omega_2\tau = 0$, so it arrives at B with zero speed. This stage takes

$$t_2 = \frac{\pi}{2\omega_2} = \frac{\pi}{2}\sqrt{\frac{a}{g}}.$$

Hence the time taken for the particle to reach B is

$$t_1 + t_2 = \frac{\pi}{4}\sqrt{\frac{a}{g}} + \frac{\pi}{2}\sqrt{\frac{a}{g}} = \frac{3\pi}{4}\sqrt{\frac{a}{g}}.$$

Checks

Energy confirms that P just reaches B at rest. Take the start, with BP stretched by a, and the arrival at B, with AP stretched by $3a$. The elastic energy goes from $\frac{6mg\,a^2}{2\cdot 2a} = \frac{3}{2}mga$ to $\frac{6mg\,(3a)^2}{2\cdot 6a} = \frac{9}{2}mga$. The increase is $3mga$, exactly the gravitational energy lost in falling $3a$, so the kinetic energy at B is zero. Numerically, with $a = 0.7$ and $g = 9.8$, stepping the full equation of motion from rest at $x = 6a$ gives $x = 7a$ at $t \approx 0.2099$, the lowest point $x = 9a$ at $t \approx 0.6297$, and $\frac{3\pi}{4}\sqrt{a/g} = 0.6297$.

Common slips

- Using T_2 for $x > 7a$: a string can't push, so once BP is shorter than $2a$ it exerts no force.
- Using ω_1 for the whole motion, or forgetting to restart the second SHM with the speed $2\sqrt{ag}$ inherited from the first.
- Calling the second stage a quarter of a period without checking that the amplitude $2a$ really takes P to B.

Beyond STEP

This is a piecewise-linear oscillator: the restoring force is linear but the stiffness changes at $x = 7a$. Each half has its own phase-plane ellipse, and the trajectory is made of arcs of two different ellipses joined where the string goes slack. Systems like this, from bouncing balls to impact oscillators, are a standard starting point in nonlinear dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats oscillations and energy methods in depth.

WORKED SOLUTION

STEP III 2004, Question 11

Mechanics • Two-Body Repulsion

STEP III 2004, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 10.6, *Two Bodies Become One*: https://beyondstep.keithmcnulty.org/ch10/two-body.html

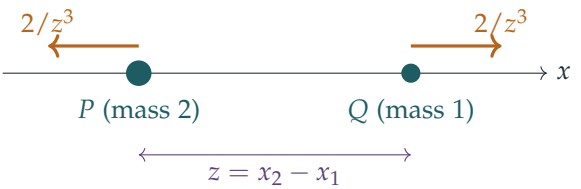
The key idea

Two particles pushing each other apart give two equations of motion, and two clever combinations of them do all the work. The *difference* $x_2 - x_1$ gives a single equation for the separation z, which the $v\,\mathrm{d}v/\mathrm{d}z$ trick integrates once (it is really energy conservation). The *weighted sum* $2x_1 + x_2$ is three times the centre of mass, and since the forces are internal it feels no net force at all.

Solution

The equation for z

Model P and Q as particles on a line, with x increasing from P towards Q (since $x_2 > x_1$). The repulsion pushes P in the negative direction and Q in the positive direction, each with magnitude $2/z^3$.



Newton's second law for each particle gives

$$2\ddot{x}_1 = -\frac{2}{z^3}, \qquad \ddot{x}_2 = \frac{2}{z^3}.$$

So $\ddot{x}_1 = -1/z^3$ and, subtracting,

$$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = \ddot{x}_2 - \ddot{x}_1 = \frac{2}{z^3} + \frac{1}{z^3} = \frac{3}{z^3},$$

as required.

Solving for z

First the initial conditions. At $t = 0$ we have $z = 1$. P moves towards Q, that is, in the positive direction, so $\dot{x}_1 = 1$, while $\dot{x}_2 = 0$. Hence $v = \dot{z} = 0 - 1 = -1$ at $t = 0$: the gap is closing.

With $\ddot{z} = v\,\mathrm{d}v/\mathrm{d}z$, the equation becomes

$$v\frac{\mathrm{d}v}{\mathrm{d}z} = \frac{3}{z^3} \quad \Longrightarrow \quad \tfrac{1}{2}v^2 = -\frac{3}{2z^2} + C.$$

At $z = 1$, $v^2 = 1$, so $\frac{1}{2} = -\frac{3}{2} + C$ and $C = 2$. Therefore

$$v^2 = 4 - \frac{3}{z^2}, \qquad \text{or} \qquad z^2\Big(\frac{\mathrm{d}z}{\mathrm{d}t}\Big)^2 = 4z^2 - 3.$$

The left-hand side is $\big(\frac{1}{2}\,\mathrm{d}(z^2)/\mathrm{d}t\big)^2$, which suggests putting $u = \sqrt{4z^2 - 3} \geq 0$. Then $u^2 = 4z^2 - 3$, so $u\,\dot{u} = 4z\dot{z}$, and the equation above says $(z\dot{z})^2 = u^2$, that is, $z\dot{z} = \pm u$. Hence

$$\dot{u} = \frac{4z\dot{z}}{u} = \pm 4.$$

Initially $u = 1$ and $z\dot{z} = -1 < 0$, so we take the minus sign: $u = 1 - 4t$.

> **?** **Ask yourself** What happens when u reaches 0, and does the sign of $\dot{z}$ stay negative?

No. At $t = \frac{1}{4}$ we get $u = 0$, so $z = \sqrt{3}/2$ and $v = 0$: the particles are momentarily at their closest. There $\ddot{z} = 3/z^3 > 0$, so z starts to increase, $\dot{z}$ becomes positive and the plus sign takes over, giving $u = 4t - 1$ for $t \geq \frac{1}{4}$ (continuous at $t = \frac{1}{4}$). In both cases $u^2 = (4t - 1)^2$, so for all $t \geq 0$

$$4z^2 - 3 = 16t^2 - 8t + 1 \quad \Longrightarrow \quad z^2 = 4t^2 - 2t + 1,$$

and since $z > 0$,

$$z = \sqrt{4t^2 - 2t + 1}.$$

This is the required result. (Note $4t^2 - 2t + 1 = 4(t - \frac{1}{4})^2 + \frac{3}{4} > 0$, so the square root is always defined.)

Finding x_1 and x_2

From the two equations of motion,

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}(2x_1 + x_2) = 2\ddot{x}_1 + \ddot{x}_2 = -\frac{2}{z^3} + \frac{2}{z^3} = 0.$$

This is conservation of momentum: the forces are equal and opposite. So $2x_1 + x_2$ is linear in t. At $t = 0$ it equals $2 \cdot 0 + 1 = 1$, and its rate of change is $2\dot{x}_1 + \dot{x}_2 = 2 \cdot 1 + 0 = 2$. Hence

$$2x_1 + x_2 = 2t + 1, \qquad x_2 - x_1 = \sqrt{4t^2 - 2t + 1}.$$

Subtracting the second from the first gives $3x_1$, and adding twice the second to the first gives $3x_2$:

$$x_1 = \frac{2t + 1 - \sqrt{4t^2 - 2t + 1}}{3}, \qquad x_2 = \frac{2t + 1 + 2\sqrt{4t^2 - 2t + 1}}{3}.$$

Checks

- At $t = 0$ these give $x_1 = 0$ and $x_2 = 1$. Differentiating, $\dot{x}_1(0) = \frac{1}{3}\big(2 - \frac{1}{2} \cdot (-2)\big) = 1$ and $\dot{x}_2(0) = \frac{1}{3}(2 - 2) = 0$, as required.
- A quicker route to z: with $w = z^2$, $\ddot{w} = 2\dot{z}^2 + 2z\ddot{z} = 2\big(4 - 3/z^2\big) + 6/z^2 = 8$, so $w = 4t^2 - 2t + 1$ from $w(0) = 1$ and $\dot{w}(0) = 2z\dot{z} = -2$. This sidesteps the sign worry altogether.
- As $t \to \infty$, $z \approx 2t - \frac{1}{2}$, so $x_1 \to \frac{1}{2}$ and $\dot{x}_2 \to 2$: P ends up at rest and Q carries all the momentum ($1 \times 2 = 2 \times 1$) and all the energy. The potential energy is $1/z^2$, so the energy at the start is $1 + 1 = 2$ and at the end it is $\frac{1}{2} \cdot 1 \cdot 2^2 = 2$.
- Stepping the original equations numerically (Runge–Kutta, step 10^{-4}) up to $t = 5$ matches the formulae for x_1, x_2 and z to within 10^{-11}, with closest approach $z = 0.8660\ldots = \sqrt{3}/2$.

Common slips

- Taking $\dot{z}(0) = +1$. P is moving towards Q, so the gap is shrinking.
- Writing $\mathrm{d}z/\mathrm{d}t = -\sqrt{4 - 3/z^2}$ for all time and missing the turning point at $t = \frac{1}{4}$.
- Forgetting the mass 2 in P's equation, which gives $\ddot{z} = 4/z^3$.

Beyond STEP

This is the two-body problem in miniature: the centre of mass moves uniformly, and the relative coordinate behaves like a single particle of *reduced mass* $m_1 m_2/(m_1 + m_2) = \frac{2}{3}$ feeling the same force, since $\frac{2}{3}\ddot{z} = 2/z^3$. The same split reduces planetary motion to a single orbit. The inverse-cube force is special too: z^2 turns out to be exactly quadratic in t, a hidden symmetry of the $1/z^2$ potential studied in conformal mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats reduced mass and two-body motion carefully.

WORKED SOLUTION

STEP III 2004, Question 12

Probability and Statistics • Indicators and Derangements

STEP III 2004, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q12

The key idea

Writing a count as a sum of indicator variables is one of the most useful tricks in probability. Expectations of indicators are just probabilities, so $\mathrm{E}(N)$ comes for free; the variance needs the covariances between pairs, and these are small but not zero, because the shirts are not assigned independently. Once we know N has mean and variance both equal to 1, the Poisson distribution with parameter 1 is the natural model to try.

Solution

The single indicator C_1

Every one of the $m!$ assignments of shirts is equally likely, so shirt 1 is equally likely to go to any of the m players. Hence $\mathrm{P}(C_1 = 1) = \frac{1}{m}$, and since C_1 takes only the values 0 and 1,

$$\mathrm{E}(C_1) = 1 \cdot \mathrm{P}(C_1 = 1) = \frac{1}{m},$$

as required. Because $C_1^2 = C_1$ (as $0^2 = 0$ and $1^2 = 1$), we also have $\mathrm{E}(C_1^2) = \frac{1}{m}$, so

$$\mathrm{Var}(C_1) = \frac{1}{m} - \frac{1}{m^2} = \frac{m-1}{m^2}.$$

The covariance of C_1 and C_2

Here we need $m \geq 2$. The product C_1C_2 equals 1 exactly when players 1 and 2 both have their own shirts, and 0 otherwise. Given that player 1 has shirt 1, the remaining $m-1$ shirts are shared out at random among the other $m-1$ players, so player 2 gets shirt 2 with probability $\frac{1}{m-1}$. Therefore

$$\mathrm{E}(C_1C_2) = \mathrm{P}(C_1 = 1,\, C_2 = 1) = \frac{1}{m} \cdot \frac{1}{m-1},$$

and

$$\mathrm{Cov}(C_1, C_2) = \mathrm{E}(C_1C_2) - \mathrm{E}(C_1)\,\mathrm{E}(C_2) = \frac{1}{m(m-1)} - \frac{1}{m^2}$$
$$= \frac{m-(m-1)}{m^2(m-1)} = \frac{1}{m^2(m-1)}.$$

It is positive: knowing that player 1 has the right shirt removes one wrong shirt from the pool for player 2, which slightly raises their chance.

Ask yourself Why isn't the covariance zero? Aren't the players' shirts chosen separately?

They are not. Shirts are assigned one to each player, so what player 1 gets affects what is left for everyone else. The C_i are *dependent*, and that is exactly why the covariance terms must be included in $\mathrm{Var}(N)$.

The mean and variance of N

By symmetry, every C_i has the same distribution as C_1, and every pair C_i, C_j with $i \neq j$ has the same joint distribution as C_1, C_2 (relabelling players and shirts together doesn't change the random assignment). By linearity of expectation, which needs no independence,

$$\mathrm{E}(N) = \sum_{i=1}^{m} \mathrm{E}(C_i) = m \cdot \frac{1}{m} = 1.$$

For the variance of a sum we add all the variances and all the covariances over ordered pairs $i \neq j$, of which there are $m(m-1)$:

$$\mathrm{Var}(N) = \sum_{i=1}^{m} \mathrm{Var}(C_i) + \sum_{i \neq j} \mathrm{Cov}(C_i, C_j)$$
$$= m \cdot \frac{m-1}{m^2} + m(m-1) \cdot \frac{1}{m^2(m-1)}$$
$$= \frac{m-1}{m} + \frac{1}{m} = 1.$$

So $\mathrm{E}(N) = \mathrm{Var}(N) = 1$, as required. (For $m = 1$ we would have $N = 1$ always, so $\mathrm{Var}(N) = 0$; the question implicitly assumes $m \geq 2$, as the covariance already required.)

Which approximation?

A Normal approximation is poor whatever the value of m. N takes only the whole-number values $0, 1, 2, \ldots$, and with mean 1 and standard deviation 1 almost all its probability sits on the three or four values $0, 1, 2, 3$. The distribution is cut off at 0, just one standard deviation below the mean, so it is strongly skewed to the right; a $\mathrm{N}(1, 1)$ distribution is symmetric and puts noticeable probability (about 7% even with a continuity correction) below $-\frac{1}{2}$, where N can never be. Crucially, increasing m doesn't

help: the mean and variance stay at 1 for every m, so the shape never spreads out into a bell.

A Poisson approximation, on the other hand, is plausible. N counts the "successes" among a large number m of events, each with a small probability $\frac{1}{m}$, and these events are only weakly dependent (the covariance $\frac{1}{m^2(m-1)}$ is tiny). That is the setting in which counts are approximately Poisson. Moreover, a Poisson variable has its mean equal to its variance, and N has exactly this property, so $\mathrm{Po}(1)$ is the natural choice.

The case $m = 4$

There are $4! = 24$ equally likely assignments. Writing an assignment as the list of shirts worn by players $1, 2, 3, 4$, the ones in which nobody has their own shirt are

$$\begin{array}{llllll} 2143, & 2341, & 2413, & 3142, & 3412, \\ 3421, & 4123, & 4312, & 4321. & & \end{array}$$

(To list them systematically: player 1 has shirt 2, 3 or 4, and in each case there are exactly three ways to finish, so $3 \times 3 = 9$.) Hence

$$\mathrm{P}(N = 0) = \frac{9}{24} = \frac{3}{8} = 0.375.$$

The Poisson approximation with parameter 1 gives $\mathrm{P}(N = 0) \approx \mathrm{e}^{-1}$. Using $\mathrm{e} \approx 2.72$,

$$\mathrm{e}^{-1} \approx \frac{100}{272} = \frac{25}{68}.$$

The relative error, measured against the true value, is

$$\frac{\frac{3}{8} - \frac{25}{68}}{\frac{3}{8}} = 1 - \frac{25}{68} \cdot \frac{8}{3} = 1 - \frac{200}{204} = \frac{1}{51} \approx 0.0196,$$

which is about 2%, as claimed. (If instead we measure the error relative to the approximation, we get $\frac{1}{50}$, exactly 2%; either way the agreement is remarkably good for such a small m.)

Checks

- For $m = 2$ the only possibilities are both right ($N = 2$) or both wrong ($N = 0$), each with probability $\frac{1}{2}$: mean 1, variance 1. Good.
- A computer enumerating all permutations for $m = 2, \ldots, 7$ confirmed $\mathrm{E}(C_1) = \frac{1}{m}$, $\mathrm{Var}(C_1) = \frac{m-1}{m^2}$, $\mathrm{Cov}(C_1, C_2) = \frac{1}{m^2(m-1)}$ and $\mathrm{E}(N) = \mathrm{Var}(N) = 1$ exactly, and found 9 derangements of 4 objects, so $\mathrm{P}(N = 0) = 0.375$ against $1/2.72 \approx 0.3676$.
- Note that $N = m - 1$ is impossible: if all but one player have the right shirt, so does the last. This is another sign that N is very far from Normal.

Common slips

- Treating the C_i as independent, which gives $\mathrm{Var}(N) = \frac{m-1}{m}$ instead of 1.
- Counting only $\binom{m}{2}$ covariance terms instead of $m(m-1)$, i.e. forgetting the factor 2 in $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.
- Listing the derangements carelessly: a systematic count (3×3) avoids missing one.

Beyond STEP

This is the classical "matching" or derangement problem. Using inclusion–exclusion one can show that $\mathrm{P}(N=0) = \sum_{k=0}^{m} \frac{(-1)^k}{k!}$, which tends to e^{-1} extremely fast, and in fact N converges in distribution to $\mathrm{Po}(1)$ as $m \to \infty$. The broader principle, that a count of many rare, weakly dependent events is nearly Poisson, is made precise by the Chen–Stein method. Blitzstein and Hwang, *Introduction to Probability*, uses indicator variables and the matching problem as recurring examples.

WORKED SOLUTION

STEP III 2004, Question 13

Probability and Statistics • Geometric Distribution

STEP III 2004, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q13

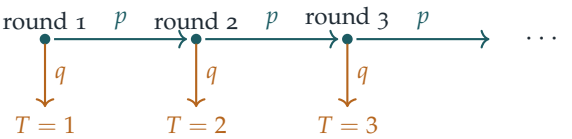
The key idea

Forget the competition for a moment and follow one competitor. The round in which he drops out is a geometric random variable, and the three competitors' drop-out rounds are independent. Every part is then a question about three independent geometric variables. The grand prize turns out to depend only on whether the *latest* drop-out round is shared, so the final part is the complement of the first two parts put together.

Solution

Write $q = 1 - p$, and assume $0 < p < 1$. For competitor j ($j = 1, 2, 3$) let T_j be the round in which he fails. He fails round k exactly when he gets through rounds $1, \ldots, k-1$ and then fails round k, so

$$\mathrm{P}(T_j = k) = p^{k-1}q, \qquad k = 1, 2, 3, \ldots$$



Since each round is passed or failed independently, and the competitors act independently, T_1, T_2, T_3 are independent. Note also that

$$\mathrm{P}(T_j \leq r - 1) = 1 - \mathrm{P}(\text{passes rounds } 1, \ldots, r-1) = 1 - p^{r-1}.$$

Part (i)

All three drop out in round k with probability $(p^{k-1}q)^3$, by independence. Summing the geometric series over k (common ratio $p^3 < 1$),

$$\mathrm{P}(T_1 = T_2 = T_3) = \sum_{k=1}^{\infty} q^3 p^{3(k-1)} = \frac{q^3}{1-p^3} = \frac{(1-p)^2}{1+p+p^2}.$$

Part (ii)

There are 3 choices for the competitor who drops out early. For a given choice, the other two both fail in round r with probability $(p^{r-1}q)^2$, and the early one fails in some round before r with probability $1 - p^{r-1}$. These events are disjoint for different choices, so

$$\mathrm{P}(\text{two in round } r,\ \text{third earlier}) = 3p^{2r-2}q^2\left(1-p^{r-1}\right).$$

Part (iii)

? **Ask yourself** Exactly when is the grand prize awarded, in terms of T_1, T_2, T_3?

Suppose the largest of the T_j is achieved by just one competitor, say $T_3 > T_1, T_2$, and let $m = \max(T_1, T_2)$. At round m competitor 3 is still in and gets through. Every other competitor still in at that round has $T_j = m$, so fails it. So competitor 3 wins the prize at round m.

Conversely, suppose the largest value is shared, say $T_2 = T_3 = M \geq T_1$. Before round M, both 2 and 3 are still in and both get through each round, so no round has a single survivor. At round M everyone remaining fails, so there is no survivor at all. So the prize is not awarded.

Hence the prize is *not* awarded exactly when the top two drop-out rounds are equal: either all three are equal (part (i)), or two are equal at some round r and the third is earlier. The second case needs $r \geq 2$, and it is the event of part (ii). Summing part (ii) over $r \geq 2$ and writing $s = r - 1$,

$$\begin{aligned}\sum_{r=2}^{\infty} 3q^2p^{2r-2}\left(1-p^{r-1}\right) &= 3q^2\sum_{s=1}^{\infty}\left(p^{2s}-p^{3s}\right) = 3q^2\left(\frac{p^2}{1-p^2}-\frac{p^3}{1-p^3}\right)\\ &= 3q^2\,\frac{p^2(1-p^3)-p^3(1-p^2)}{(1-p^2)(1-p^3)} = 3q^2\,\frac{p^2(1-p)}{(1-p^2)(1-p^3)}\\ &= \frac{3p^2q^2}{(1+p)(1-p^3)}.\end{aligned}$$

Adding part (i) and using $1 - p^3 = q(1 + p + p^2)$,

$$\mathrm{P}(\text{no prize}) = \frac{q^3(1+p)+3p^2q^2}{(1+p)(1-p^3)} = \frac{q^2(1+2p^2)}{(1+p)(1-p^3)} = \frac{(1-p)(1+2p^2)}{(1+p)(1+p+p^2)}.$$

Since $(1+p)(1+p+p^2) = 1+2p+2p^2+p^3$ and $(1-p)(1+2p^2) = 1-p+2p^2-2p^3$, their difference is $3p+3p^3$. Therefore

$$\mathrm{P}(\text{grand prize awarded}) = \frac{3p(1+p^2)}{(1+p)(1+p+p^2)}.$$

Checks

- As $p \to 0$ everyone fails round 1, and the answer to (iii) tends to 0. As $p \to 1$ ties become unlikely, and the answer tends to $\frac{3\cdot 2}{2\cdot 3} = 1$.
- Directly, the prize is awarded when one T_j beats the other two, so its probability is $3\sum_k p^{k-1}q\,(1-p^{k-1})^2$. Exact summation with $p = \frac{1}{2}$ gives $\frac{5}{7} \approx 0.714$, matching $\frac{3\cdot\frac{1}{2}\cdot\frac{5}{4}}{\frac{3}{2}\cdot\frac{7}{4}} = \frac{5}{7}$.
- A simulation of the competition itself (with 2×10^5 runs at $p = 0.6$) gave the prize in 77.9% of runs, against the formula's 0.7806.

Common slips

- Forgetting the factor 3 in part (ii) for the choice of which competitor drops out early.
- Thinking the prize needs the last two to drop out in different rounds. If two drop out together in round m and the third gets through it, he still wins. Only a tie at the *top* prevents the prize.

Beyond STEP

The question is about the maximum of independent geometric random variables and whether it is unique. For n competitors, the probability of a tie at the top does not settle down to a limit as $n \to \infty$. It oscillates very slightly with $\log n$, a well-known surprise in the analysis of algorithms such as leader election. The tools here, conditioning on one variable and summing geometric series, are developed in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 2004, Question 14

Probability and Statistics • Hypothesis Testing Errors

STEP III 2004, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q14

The key idea

Both hypotheses give $\overline{X}$ the same centre μ; they differ only in how spread out $\overline{X}$ is. So the evidence against H_0 is $\overline{X}$ landing *far from* μ, on either side, and that is exactly what a symmetric two-sided rejection region detects. After that, everything reduces to standardising $\overline{X}$, once under each hypothesis.

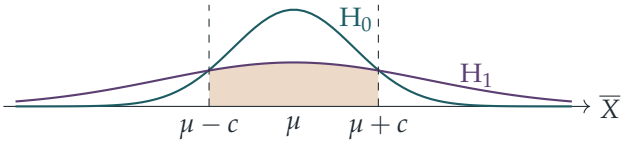
Solution

Why a two-sided test

Since $X \sim N(\mu, \sigma^2)$ and the sample values are independent, the sample mean satisfies

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Under either hypothesis $\overline{X}$ is centred at the known value μ; the only difference is that under H_1 its standard deviation $\sigma_1/\sqrt{n}$ is larger than $\sigma_0/\sqrt{n}$. Values of $\overline{X}$ close to μ are relatively more likely under H_0, while values far from μ are relatively more likely under H_1 (the wider curve has the fatter tails; see the figure). A large value of $\overline{X}$ is no more evidence for H_1 than an equally small one, because both distributions are symmetric about μ. So the natural test rejects H_0 when $|\overline{X} - \mu|$ is large, in either direction, and by symmetry the acceptance interval should be centred on μ: accept H_0 when $\mu - c < \overline{X} < \mu + c$. A one-sided test would be wrong here, as it would ignore half of the evidence for the larger spread.



The shaded area (accepting H_0 although H_1 is true) is β; the teal area outside the dashed

lines is α.

Finding c

Under H_0, $Z = \dfrac{\sqrt{n}\,(\overline{X} - \mu)}{\sigma_0}$ is standard normal. We accept H_1 when $|\overline{X} - \mu| \geq c$, that is when $|Z| \geq c\sqrt{n}/\sigma_0$. By the symmetry of the standard normal,

$$\alpha = \mathrm{P}\left(|Z| \geq \frac{c\sqrt{n}}{\sigma_0}\right) = 2\left(1 - \Phi\left(\frac{c\sqrt{n}}{\sigma_0}\right)\right), \qquad \text{so} \qquad \Phi\left(\frac{c\sqrt{n}}{\sigma_0}\right) = 1 - \tfrac{1}{2}\alpha.$$

Since Φ is strictly increasing, this says $c\sqrt{n}/\sigma_0 = z_\alpha$, and therefore

$$c = \frac{z_\alpha\,\sigma_0}{\sqrt{n}}.$$

β does not depend on n

Under H_1, $W = \dfrac{\sqrt{n}\,(\overline{X} - \mu)}{\sigma_1}$ is standard normal, and

$$\begin{aligned}\beta = \mathrm{P}(|\overline{X} - \mu| < c) &= \mathrm{P}\left(|W| < \frac{c\sqrt{n}}{\sigma_1}\right) = \mathrm{P}\left(|W| < \frac{z_\alpha \sigma_0}{\sigma_1}\right) \\ &= 2\,\Phi\left(\frac{z_\alpha \sigma_0}{\sigma_1}\right) - 1,\end{aligned}$$

where we substituted $c\sqrt{n} = z_\alpha \sigma_0$ from the previous part. The $\sqrt{n}$ in c cancels the $\sqrt{n}$ from standardising, so β depends only on α and σ_0/σ_1, not on n.

> **Ask yourself** Surely a bigger sample should make the test better?

Usually it does, because the two sampling distributions pull apart as n grows. Here they never do: both are centred at μ, and increasing n shrinks *both* standard deviations by the same factor $\sqrt{n}$. The whole picture simply rescales, so the areas α and β are unchanged. The sample mean is a poor statistic for testing a variance; it throws away the information about spread that the individual values contain.

The minimum ratio σ_1/σ_0

Write $r = \sigma_1/\sigma_0 > 1$.

Condition $\alpha < 0.05$. This means $\Phi(z_\alpha) = 1 - \frac{1}{2}\alpha > 0.975 \approx \Phi(1.960)$, so, as Φ is increasing, $z_\alpha > 1.960$.

Condition $\beta < 0.05$. This means $2\Phi(z_\alpha/r) - 1 < 0.05$, that is $\Phi(z_\alpha/r) < 0.525 \approx \Phi(0.063)$, so $z_\alpha/r < 0.063$, i.e.

$$r > \frac{z_\alpha}{0.063}.$$

Combining, we need $r > z_\alpha/0.063 > 1.960/0.063$. Conversely, if $r > 1.960/0.063$ we can choose z_α just above 1.960 with $z_\alpha < 0.063\,r$, and then both conditions hold. Hence

the ratio must exceed

$$\frac{1.960}{0.063} \approx 31.1,$$

so the minimum value of σ_1/σ_0 is approximately 31. The standard deviations must differ by a factor of about thirty before the sample mean can tell them apart this reliably, whatever the sample size.

Checks

- A simulation with $\mu = 3$, $\sigma_0 = 1$, $\sigma_1 = 2.5$, $\alpha = 0.1$ and $n = 1, 4, 9, 25$ gave rejection rates under H_0 of about 0.10 every time, and acceptance rates under H_1 of about 0.489 for every n, matching $2\Phi(1.645/2.5) - 1 \approx 0.489$.
- With $r = 1.960/0.063$ and $\alpha = 0.05$ exactly, $\beta = 2\Phi(0.063) - 1 \approx 0.050$, on the boundary as expected. Using exact inverse values of Φ gives $r \approx 31.3$, so the answer 31 is robust to the rounding in the given table values.
- Sanity: if $\sigma_1 = \sigma_0$ then $\beta = 2\Phi(z_\alpha) - 1 = 1 - \alpha$, as it must be when the hypotheses coincide.

Common slips

- Using a one-sided region such as $\overline{X} > \mu + c$: under H_1 the mean doesn't move, so there is no reason to favour one tail.
- Writing $\alpha = 1 - \Phi(c\sqrt{n}/\sigma_0)$, forgetting that the rejection region has two tails.
- Getting the inequality for r the wrong way round: small β needs z_α/r *small*, so r large.

Beyond STEP

This question shows the trade-off between the two error types and why the choice of test statistic matters. The Neyman–Pearson lemma says the most powerful test compares the likelihoods under the two hypotheses; here that leads to the statistic $\sum(X_i - \mu)^2$, whose χ^2_n distribution does separate the hypotheses as n grows, unlike $\overline{X}$. For a readable introduction to these ideas, see Blitzstein and Hwang, *Introduction to Probability*, or Grimmett and Welsh, *Probability: An Introduction*.

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